The Spectre and Chair44 Tilings and the $SU(2)$ Ladder: Exact Computations on Internal Space, Boundary Growth and Quantum Geometry

Brent Hartshorn and Tristan Romanov

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Abstract

Exact computations on three connected subjects are joined here, and every number in them is generated and checked when the paper is built.

Part I. The planar Spectre is an aperiodic monotile and Tsiokos's three-dimensional Chair44 is claimed to be one; both have hierarchical tilings, and they are compared here by exact computation. The Spectre inflates by the unit $\lambda^2 = \sqrt{15} + 4$ of $\mathbb{Z}[\sqrt{15}]$. Its supertile perimeter is $2F_{3k+5}+4$ at every order k, proved by a finite exact certificate, so it grows by the unit $2+\sqrt{5}$ of $\mathbb{Q}(\sqrt{5})$, faster than the linear inflation. The boundary of the limiting supertile is a fractal of box dimension $\log(2+\sqrt{5})/\log\lambda$, and the tail of the common-ancestor levels of neighbouring tiles decays with ratio tending exactly to $(2+\sqrt{5})/\lambda^2$. In the coordinates of the rank-four vertex module $\mathbb{Z}[\zeta_{12}]$, one level of the inflation is a 16×16 integer matrix on four key points, with expanding eigenvalues $\pm\lambda$, contracting $\pm\lambda^{-1}$ and marginal ± 1, and the internal plane, the projection along the expanding plane of each parity of level, is given exactly; the window is the attractor of contracting similarities and has positive area, and the seams where its pieces meet have box dimension $\log(5+2\sqrt{6})/\log(4+\sqrt{15})$, a lower bound for the dimension of its boundary. The Chair44 inflates by the integer 2 on $\mathbb{Z}^3$, with no Euclidean internal space (its internal space is 2-adic) and no twist; its centre set, that of the classical three-dimensional chair, has an explicitly computed pure-point diffraction measure. The main result is that every lattice-registered tiling obeying the 44 contact rules lies in the substitution hull, and that its symmetry group is conjugate to one of 1, C_2 (a half-turn about a face diagonal), C_4 and D_4. Both statements use Kamenetsky's unique-hierarchy criterion, whose finite certificate is checked here independently, and they transfer to every tiling by the Chair44 solid through Tsiokos's registration theorem, which has been audited but not refereed. Orientation sums and contact matrices separate the two tilings. Four finite-patch computations then test whether the Spectre's substitution hierarchy can serve as a clock: positional clustering shows no arrow, the radius needed to recognize an ancestor grows about threefold per level without making one, the look-back law is shared with a quadtree, and the parity flip of the construction shows in the outlines of supertiles and not in the tiles.

Part II. The exactly solved two-plaquette $SU(2)$ lattice gauge theory has the spin networks of the theta graph as its gauge-invariant states, and the electric energy of a link is the squared area of a surface that cuts it. In the ground state the mean-square area of the surface between the plaquettes is exactly $2/(3g^8)$ at strong coupling and $3/(\sqrt{10}\,g^2)$ at weak coupling. Counting spin-$\frac{1}{2}$ punctures against the Bekenstein–Hawking entropy gives $\gamma_I = \ln 2/(\pi\sqrt{3})$, a property of that count and not of the area spectrum, and the smallest vertex with a volume of its own, four spin-$\frac{1}{2}$ legs, has triple-product spectrum $\pm\sqrt{3}/4$. A third plaquette tests the strip formula $2\sqrt{2-\cos(\pi/4)} \approx 2.274$ for the weak-coupling gap. Four entries of the OpenAI math release are then read in the same language: the Weyl blocks

of S^4, $\mathbb{CP}^2$ and $S^2 \times S^2$ as spin-2 objects of $SU(2)_+ \times SU(2)_-$, recovering $\chi = 2, 3, 4$ and $\tau = 0, 1, 0$ from their metrics; the Penrose inequality and the Kerr horizon algebra against the smallest quantised area, as arithmetic and heuristics and not as theorems; Catalan counts of planar maps against the invariants of $SU(2)_k$; and the finite statements of a four-dimensional lattice gauge-fixing paper. This is not a model of quantum gravity: the graph is fixed and the Hamiltonian is a gauge-theory one.

Part III. The families of the release that the two parts meet are read from its catalogue by `oai_tree.py`, and the graph of their connections is drawn with the places where this paper meets them.

204 registered checks, 182 of them in the quantum gravity models, pass in the build together with the figure checks, and a failure stops it; they test that stated formulas agree with one another and with independent computation, and do not re-prove any theorem. The material on the OpenAI release is fork-level and is not an official OpenAI document.

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1 Overview

This paper joins two computations that were written separately and share a method. Every number, matrix and table in both is produced when the paper is built, from code that checks it: in Part I, exact integer and algebraic arithmetic and the finite certificates behind it; in Part II, the results that thirteen sympy models register, where each term of each displayed row is a library equation solved for the row's quantity, substituted back and required to give an identity. A failed check stops the build. What is written by hand is the argument.

Part I compares the planar Spectre with Tsiokos's Chair44: inflation, internal space, boundary growth, the hull of the Chair44 and the symmetry groups of its tilings, and four tests of whether a substitution hierarchy can serve as a clock. Part II reads the exactly solved two-plaquette $SU(2)$ lattice gauge theory as the kinematics of loop quantum gravity, and takes up four entries of the OpenAI math release in that language: Einstein four-manifolds, the Penrose inequality, strong cosmic censorship near Kerr, and random planar maps, with the finite statements of a four-dimensional lattice gauge-fixing paper. Part III reads from the release's own files how the families that the two parts meet, and the entries that the fork's Yang–Mills note names, are connected, and draws the graph. The material on the OpenAI release (Parts II and III) is fork-level and is not an official OpenAI document.

The parts do not use one another's results, and nothing in either is derived from the other. They touch at one place, which is a remark and not a result: the Weyl tensor. The discussion of Part I (Section 12) proposes that a flat two-dimensional initial surface would have no Weyl curvature, which first appears with the third spatial dimension. Part II (Section 21) computes, for three model Einstein four-manifolds, the two trace-free Weyl blocks as spin-2 objects of the two factors of $\mathrm{Spin}(4) = SU(2)_+ \times SU(2)_-$. The first is a Lorentzian reading about 2+1 and 3+1 dimensions and the second a Euclidean computation in four, and no map between them is given. Part III shows a second, indirect meeting: the fork's own notes tie both parts to entry 215 of the release, the two-dimensional $O(N)$ lattice models proved by block renormalization, Part I through a comparison of that renormalization with a substitution hierarchy and Part II through the scaling forms of Section 26. That is a statement about the notes and not a result of either part.

Part I
The Spectre and Chair44 tilings

2 Introduction

A tiling that exists only by inflation carries an arithmetic: the number by which it inflates, the field that number generates, and what that field leaves behind. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss [1, 2] inflates by a unit of a real quadratic field, and the Chair44 of Tsiokos [3] by the integer 2. The Spectre's vertices lie in $\mathbb{Z}[\zeta_{12}]$, a module of rank four, and one level of the inflation, acting on four key points of a supertile, is an integer matrix with eigenvalues $\pm\lambda$, $\pm\lambda^{-1}$ and ± 1: projecting along the expanding plane, which alternates with the

parity of the level, gives a two-dimensional internal space in which the vertices fill a bounded window. An integer inflation $m > 1$ acts as mI, nothing contracts, and the internal space is m-adic. This paper computes that difference exactly.

The Spectre is analysed in Sections 3–5 (substitution matrix, spectrum, marginal charges), 6 (perimeter growth) and 7 (four-dimensional lift), the Chair44 in Sections 8 and 9, with a proof, resting on one quoted input and two finite computations, that every Chair44 tiling lies in the substitution hull and a classification of the symmetry groups of its tilings in Section 8.5, and the two are compared in Section 10. Section 11 tests whether the Spectre's substitution hierarchy can serve as a clock.

What is not claimed matters as much. The inputs are these. For the Spectre: the published substitution rule of [2], that is, the matrix (1) together with the placement transforms of the construction, from which every placed patch is built. For the Chair44: Tsiokos's solid and its eight substitution children [3]. The einstein property of the Chair44 rests on Tsiokos's claimed proof [3], quoted; what Section 8.5 adds is that, given his registration theorem, no tiling by the solid is periodic. The hull theorem of Section 8.5 is proved for tilings obeying the contact rules, using Kamenetsky's unique-hierarchy criterion [11] with its finite certificate checked independently (Lemma 1), and for the solid it rests in addition on Tsiokos's registration theorem [3], quoted; that proof has been audited by the second author [6] but has not been refereed. That the Spectre diffraction is pure point is a published theorem [7, 8], not re-proved here, and the centre set of the Chair44 is that of the classical three-dimensional chair [12, 14]. Section 13 lists the rest.

3 Exact Spectral Theory of the Spectre Monotile Substitution

Figure 1 summarises the substitution that this section analyses.

The substitution acts on nine species (Γ, Δ, Θ, Λ, Ξ, Π, Σ, Φ, Ψ), with Γ carrying the Mystic pair $\Gamma_1 + \Gamma_2$, $\Gamma_2 = \mathrm{Tile}(b, a)$. The matrix is transcribed from the published rule [2]; every statement of Sections 3–5 that does not mention placed tiles is derived from it alone.

$$M = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 2 & 2 & 1 & 2 & 2 \\ 0 & 0 & 1 & 1 & 2 & 2 & 0 & 2 & 3 \end{pmatrix} \tag{1}$$

Fact 1 (Inflation factor and its arithmetic)**.** *The Perron eigenvalue is $\lambda^2 = \sqrt{15} + 4$, the fundamental unit of $\mathbb{Z}[\sqrt{15}]$: $4^2 - 15 \cdot 1^2 = 1$ is the least Pell solution. The linear inflation factor is*

$$\lambda = \sqrt{\sqrt{15} + 4} = \tfrac{1}{2}(\sqrt{6} + \sqrt{10}) = 2.8058837\ldots, \qquad (\lambda - \lambda^{-1})^2 = 6, \qquad (\lambda + \lambda^{-1})^2 = 10,$$

with minimal polynomial $x^4 - 8x^2 + 1$ and splitting field $\mathbb{Q}(\sqrt{6}, \sqrt{10})$. The two channel identities are proved in $\mathbb{Z}[\sqrt{15}]$ without evaluating a square root: each is $\lambda^2 + \lambda^{-2} \mp 2$, and the $\sqrt{15}$ parts cancel. Tile counts, and with them supertile areas, therefore grow by the factor λ^2 per level.

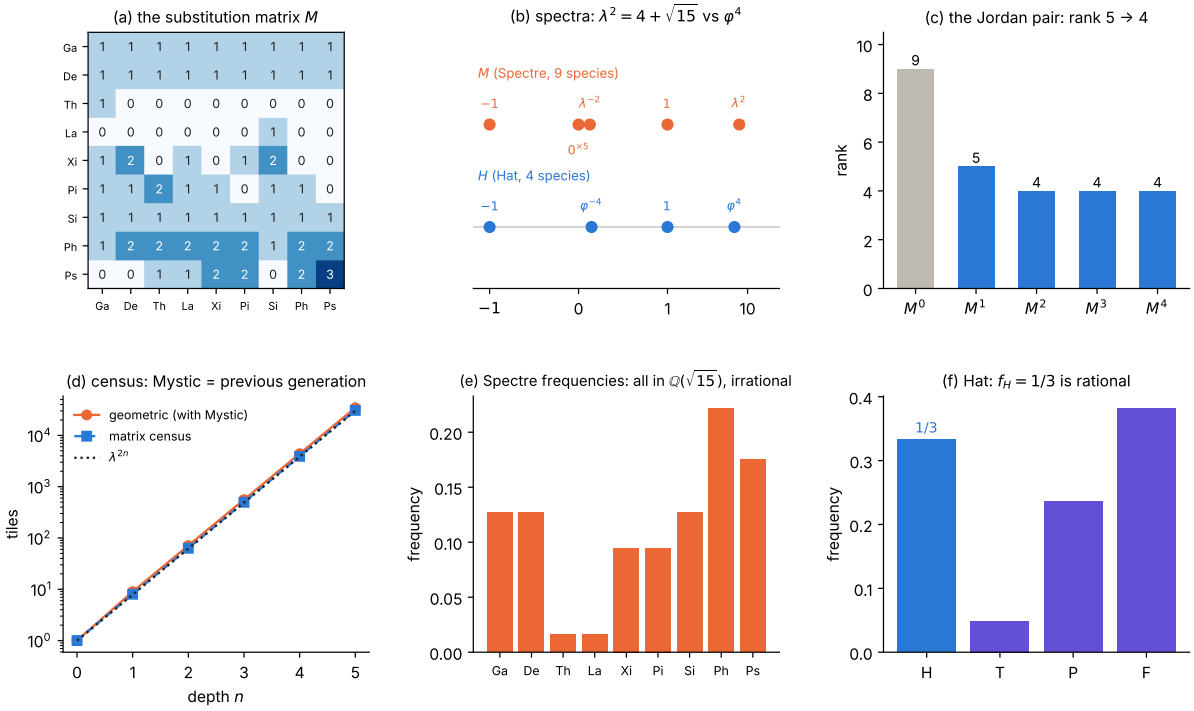


Figure 1: The substitution. (a) The matrix M. (b) Its spectrum on a symmetric logarithmic axis, against that of the Hat matrix: the Pisot pair $\lambda^{\pm 2}$ with $\lambda^2 = 4 + \sqrt{15}$, and $\varphi^{\pm 4}$. (c) The ranks of M^n drop from 5 to 4 and stay: the Jordan pair. (d) Tiles by depth, with and without the Mystic, against λ^{2n}. (e,f) The species frequencies: irrational for the Spectre, with the Hat's $f_H = 1/3$ rational.

Fact 2 (Transfer spectrum: a Jordan pair among the zero modes)**.** $\chi_M(x) = x^5(x-1)(x+1)(x^2-8x+1)$, *and the zero sector carries structure the spectrum alone does not show:*

$$\operatorname{rank} M = 5, \qquad \operatorname{rank} M^n = 4 \quad (n \geq 2).$$

The zero eigenvalue has algebraic multiplicity five and geometric multiplicity four, so M is not diagonalizable: its Jordan blocks at 0 have sizes $1, 1, 1, 2$. The kernel of M has dimension four and that of M^2 dimension five, so one census direction survives one substitution and is erased by the second. The remaining spectrum is the Pisot-unit pair $\{\lambda^2, \lambda^{-2}\}$ and two marginal eigenvalues ± 1. Relative to the Perron growth, the Pisot conjugate decays by $\lambda^{-4} = (4-\sqrt{15})^2 \approx 0.0161332$ per step and the marginal modes by λ^{-2}; so species frequencies converge at the rate λ^{-2} whenever a marginal charge of Section 5 is nonzero on the seed (as it is for the seed Δ), and at the rate λ^{-4} otherwise. The zero modes are gone after two steps.

4 Matrix Iteration Versus Placed Patch Counts

A patch can be counted two ways and the difference matters, so it is settled here before anything uses it.

The *matrix census* is what iterating (1) produces: nine species, one Γ entry, seeded with a single Δ (the seed of Table 1; seeding with Γ gives 7, 55, 433, ...). The *geometric census* is

what a placed patch contains, and a placed patch carries the Mystic as well, so it is larger. The two are not interchangeable.

Fact 3 (The Mystic is the previous generation)**.** *Row* Γ *of* M *is all ones, so* $N_\Gamma(n)$ *is the entire census at depth* $n-1$*; the Mystic is a second copy of it. Hence*

$$T_{\text{geom}}(n) = T(n) + T(n-1),$$

and both sequences obey $T(n{+}1) = 8T(n) - T(n{-}1)$*, the recurrence whose characteristic polynomial is the quadratic factor of* χ_M*. The recurrence holds for every seed and every* n*: the row of ones satisfies* $\mathbf{1}^\top(M^2 - 8M + I) = 0$*, checked entry by entry, so* $T(n{+}2) - 8T(n{+}1) + T(n) = \mathbf{1}^\top M^n(M^2 - 8M + I)e = 0$ *for every seed vector* e*.*

depth	matrix census	geometric census	Mystic
0	1	1	0
1	8	9	1
2	63	71	8
3	496	559	63
4	3905	4401	496

Table 1: The two censuses. The fourth column is the third minus the second, and is also the second column one row up — which is Fact 3. Charges are linear forms on the nine-species vector and are evaluated against the matrix column.

5 Conserved and Alternating Marginal Charges

Fact 4 (The marginal charges)**.** *The left eigenvectors for the marginal eigenvalues* ± 1*, in the species order of* (1)*, are*

$$Q_+ = (-3, -1, 0, 0, 1, 1, -2, 1, 2), \qquad Q_- = (-1, -1, 2, 0, 1, -1, 0, 1, 0),$$

satisfying $Q_+M = Q_+$ *and* $Q_-M = -Q_-$ *exactly over* $\mathbb{Z}$*. Evaluated on the matrix census of Section 4 at depths one to four,*

$$Q_+ = -1, -1, -1, -1, \qquad Q_- = +1, -1, +1, -1.$$

Q_+ *is conserved;* Q_- *alternates.*

Proposition 1 (Q_- and the parity of the construction)**.** *The construction applies a mirror at every level, so all tiles of a depth-n patch have the same handedness and the common determinant of their placements is* $(-1)^n$*; transcribing the published placement chain into* $\mathbb{Q}(\sqrt{3})$ *affine arithmetic, every determinant is exactly* ± 1*, as it must be. Since* Q_- *is a left eigenvector for* -1*,* $Q_-(n) = (-1)^n Q_-(e)$ *for a seed* e*. For the seed* Δ*,* $Q_-(\Delta) = -1$ *and the determinant equals* $-Q_-(n)$ *at every depth. This is a coincidence of two alternating sequences and not a way of reading parity from species counts:* Q_- *vanishes on single tiles of the species* Λ*,* Σ *and* Ψ*, and hence on the supertiles of those species at every depth.*

Fact 5 (The generational ledger)**.** *Row* Θ *of* M *is the indicator of* Γ *and row* Λ *the indicator of* Σ*, so*

$$N_\Theta(n) = N_\Gamma(n-1), \qquad N_\Lambda(n) = N_\Sigma(n-1)$$

hold exactly, not asymptotically: two species are a verbatim copy of an earlier generation.

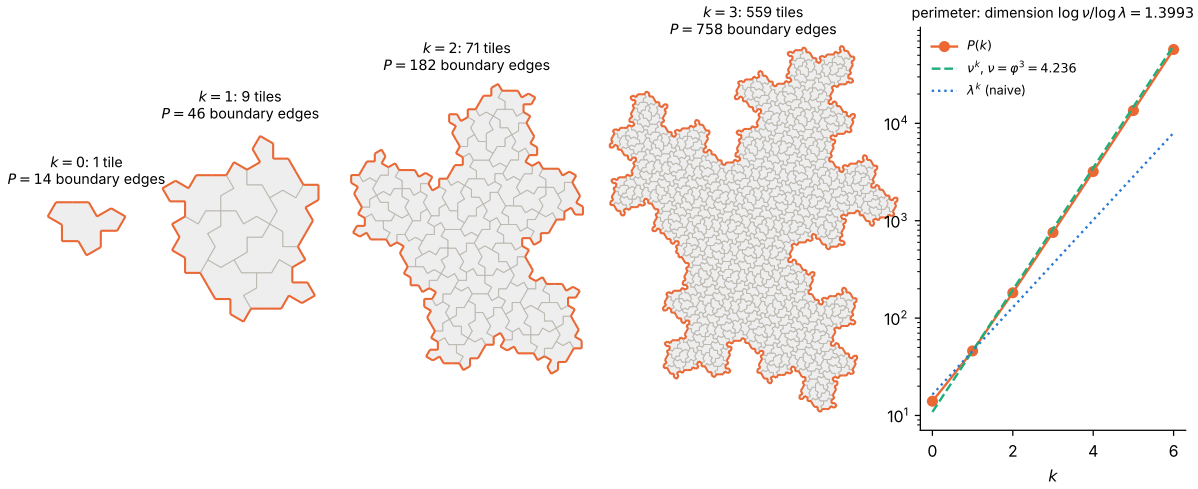


Figure 2: Supertiles of order $k = 0, \ldots, 3$ with their boundary edges (orange), and the perimeter $P(k)$ against ν^k with $\nu = \varphi^3$ and against the naive λ^k. The edge counts drawn are counted on the geometry; by Theorem 1 they equal the recurrence of this section.

5.1 The twist of the Spectre supertile

Fact 6 (The twist of the supertile)**.** *The four key points of a supertile (its quadrilateral) are moved by one level of the construction by a conjugate-linear map $F(w) = D\,\overline{w}$ on $\mathbb{C}^4$, and $B = D\overline{D}$ is complex linear. The quadrilaterals are not similar at any finite level: their corners lie in $\mathbb{Q}(\sqrt{3}, i)$, so squared side lengths lie in $\mathbb{Q}(\sqrt{3})$, which does not contain λ^2. They become similar in the limit: along the dominant eigenline of B, consecutive quadrilaterals are related by the ratio λ and a rotation of $-\theta_S$ on even levels and $+\theta_S$ on odd ones, with a relative correction that decays by λ^{-2} per level, where*

$$\tan\theta_S = \frac{5\sqrt{3}-6}{13}, \qquad \cos 2\theta_S = \frac{4+3\sqrt{3}}{10}, \qquad \theta_S = 11.565051\ldots^\circ .$$

The two-level map B has characteristic polynomial $(x-1)^2(x^2-8x+1)$, which carries the quadratic factor of χ_M, and its dominant eigenvalue is the real number λ^2: over two levels the limiting quadrilateral is dilated and not rotated.

The $\sqrt{15}$ enters the limiting one-level factor only through its modulus λ: the doubled phase $e^{-2i\theta_S} = \frac{1}{10}\big(4+3\sqrt{3}+i(3-4\sqrt{3})\big)$ lies in $\mathbb{Q}(\sqrt{3}, i)$, the field of the hexagonal tile coordinates, while $e^{i\theta_S}$ itself does not, since $\cos\theta_S$ has minimal polynomial $400x^4 - 560x^2 + 169$ over $\mathbb{Q}$, of degree four. The alternation of the sign is the mirror applied at each level. The twist is irrational as an angle: $\cos 2\theta_S$ has minimal polynomial $100x^2 - 80x - 11$, so $2\cos 2\theta_S$ is not an algebraic integer and $e^{2i\theta_S}$ is not a root of unity. This is the contrast drawn with the Chair44 in Section 9.

6 The other unit: a boundary operator in $\mathbb{Q}(\sqrt{5})$

Figure 2 shows the supertiles whose perimeter is counted here.

Everything above grades the substitution by area. The perimeter is graded by something else, and the two do not live in the same field.

Let $P(k)$ be the number of unit tile edges on the boundary of an order-k supertile. Reading the child multiset off the substitution table — the eight placement slots are the same for every species, and slot 7 is Γ in every rule, while Γ alone has a null slot and therefore seven children — and subtracting each glued edge twice gives, with Q the Γ perimeter and G the number of edges shared between two *different* children,

$$P(k{+}1) = 7P(k) + Q(k) - 2G(k{+}1), \qquad Q(k{+}1) = 6P(k) + Q(k) - 2G_\Gamma(k{+}1). \tag{2}$$

The coefficients are read from the rule, not fitted to counts, and (2) holds at every order (Theorem 1). Two facts make it small enough to solve. First, all eight non-Γ species have the *same* outline at every order, since they place the same child outlines in the same slots; this collapses a nine-type problem to two. Second, the glue decomposes by slot pair into thirteen contact sequences. Ten of them satisfy the recurrence of P from the first order on; the contacts of slots 1–4, 1–7 and 4–7 satisfy it only from $k = 5$ on, their first term being exceptional, so their characteristic polynomial carries an extra factor x. What closes the system is a boundary substitution in the form of a finite set of contact types.

Theorem 1. *For every $k \geq 0$, $P(k{+}3) = 5P(k{+}2) - 3P(k{+}1) - P(k)$ and $P(k) = 2F_{3k+5} + 4$, where F_n is the Fibonacci number; and $Q(k{+}3) = 5Q(k{+}2) - 3Q(k{+}1) - Q(k)$ for every $k \geq 1$.*

Proof. The slot transforms of the construction are $x \mapsto A_i x + M_i q_k$, with fixed A_i and fixed integer 4×16 matrices M_i, where $q_k \in \mathbb{Z}^{16}$ is the quadrilateral of order k and $q_{k+1} = \Phi q_k$. So two placed order-k outlines have a relative placement $x \mapsto Ax + Lq_k$ with L independent of k. The length $S(k)$ of their common boundary is the sum, over the pairs of their children, of the same quantity one order down, provided the two have disjoint interiors. Two placements are identified when Lq_k agrees for all k; this is decided on $q_0, \ldots, \Phi^5 q_0$, because the minimal polynomial of q_0 is $(x^2 - 1)(x^4 - 8x^2 + 1)$. Starting from the sibling pairs, the child pairs close on 93 types. Every other child type is separated at every order: by a radius bound for $k \leq 60$, and by the exact eigen-decomposition of q_k for $k > 60$, with a margin of at least 0.097 in units of the quadrilateral's diagonal. On the 93 types the interior overlap vanishes at order 0, hence at every order by the same sum. So the supertiles are genuine patches; this is proved here, not assumed. The state $(P, Q, S_1, \ldots, S_{93})$ then obeys $V(k+1) = \mathcal{B}V(k)$ with a fixed 95×95 integer matrix $\mathcal{B}$. The sequence $e^\top \mathcal{B}^k p(\mathcal{B}) V(0)$, with $p(x) = x^3 - 5x^2 + 3x + 1$, vanishes for $k < 95$, hence for all k by Cayley–Hamilton; the same argument gives the recurrence of Q. The closed form follows because $F_{n+6} = 4F_{n+3} + F_n$ and the seeds agree. The computation is `boundary_certificate/check_spectre_boundary_proof.py`. It works from an independent exact transcription of the construction and does not use the rest of the build. □

Fact 7. *For every $k \geq 3$, $P(k) = 5P(k-1) - 3P(k-2) - P(k-3)$ (Theorem 1), with $P = 14, 46, 182, 758, 3198, 13534$ and $P(6) = 57318$ also counted exactly (Fact 8).*

The seed of that recurrence is counted from placed tiles, and the count used a rounding key: two edges were identified when their endpoint coordinates agreed to four decimals. A key of that kind can fail in both directions, by merging edges that differ below the tolerance and splitting edges that agree above it, and nothing inside a floating-point computation bounds the failure — it can only be removed.

Fact 8. *The same sequence, recounted with edges identified by* equality *of exact $\mathbb{Q}(\sqrt{3})$ endpoint coordinates and from an independent transcription of the placement chain, agrees at every order*

computed, $0 \le k \le 6$. *The rounding key is therefore not load-bearing for any of the boundary results.*

The factorisation of the characteristic polynomial χ_∂ of the perimeter recurrence (Fact 10 below) is not left as a spectral remark. Its quadratic factor applied to the sequence annihilates the growing part and leaves behind the constant belonging to the marginal eigenvalue. By Theorem 1 the identity holds for the perimeters at every order:

Fact 9. $P(k+2)-4P(k+1)-P(k) = -16$ *for all* $k \ge 0$. *Together with the seed it is equivalent to the recurrence of Fact 7, and it is the* $+1$ *eigenvalue made arithmetic — the boundary's analogue of* Q_+.

Fact 10. *The companion matrix of that recurrence — the boundary transfer operator — has characteristic polynomial* $\chi_\partial(x) = (x-1)(x^2-4x-1)$. *The marginal eigenvalue* $+1$ *is a conserved boundary quantity, of the same kind as the* ± 1 *of Section 5.*

Fact 11. *The Perron root is* $\nu = 2+\sqrt{5} = \varphi^3 = 4.23606797750$.

So, for every order (Theorem 1), the area of a supertile grows asymptotically by $\lambda^2 = 4+\sqrt{15}$ per level and its perimeter by $2+\sqrt{5}$. This is a $\mathbb{Q}(\sqrt{15})$ / $\mathbb{Q}(\sqrt{5})$ divide between area and perimeter of the *same* tile. The supertile boundary is golden although the tiling is not.

Since $P(k) \sim c\,\nu^k$ unit edges bound a supertile of diameter of order λ^k, and since $\nu > \lambda$ the perimeter outgrows the linear scale with exponent $\log\nu/\log\lambda = 1.39925321391$. That exponent is the box dimension of the limiting outline.

Theorem 2. *For each parity* p, *the rescaled supertiles* $\lambda^{-k}S_k$ *with* $k \equiv p$ *modulo* 2 *converge in the Hausdorff metric to a compact set* K_p *that is the closure of its interior. The box dimension of* ∂K_p *exists and equals* $\log\nu/\log\lambda$.

Proof. Rescaled by λ^{-k}, the placements of Theorem 1 have translations $M_i q_{k-1}/\lambda^k$, and $q_k = \lambda^k a + (-\lambda)^k b + O(1)$. So on each parity they converge geometrically to similarities of ratio $1/\lambda$, and $\lambda^{-k}S_k$ converges to the attractor K_p of the limiting graph-directed system.

The areas of $\lambda^{-k}S_k$ converge to a positive limit, and area is upper semicontinuous under Hausdorff limits. The children matrix $\left(\begin{smallmatrix}7&1\\6&1\end{smallmatrix}\right)$ has Perron root λ^2. So K_p has positive area and its pieces overlap in measure zero.

A grandchild of S (slot 3 of slot 7) is separated at every order from every contact it could have outside its supertile, with a margin of 0.267 quadrilateral diagonals. Take a Spectre tiling of the plane whose tiles group, for every k, into the construction's order-k supertiles, with every two adjacent supertiles placed as in one of the types of Theorem 1. A limit of the supertiles S_k, re-centred at tiles whose distance from the outline grows without bound, is such a tiling: those distances do grow, since the area of S_k grows by λ^2 and its perimeter only by ν, and inside each S_k adjacent supertiles are placed as in one of the finitely many types (this is the hierarchical structure of [2, Thms 2.2, 5.1]). In that tiling a ball of that relative radius about the grandchild therefore lies in the supertile. So K_p has interior and, since its pieces are dense, it is the closure of its interior.

For the upper bound, every point of ∂K_p is within any ε of the outline of $\lambda^{-k}S_k$ for large k. That outline is covered by the depth-m pieces in contact with a piece of a neighbouring supertile, and there are at most $C\nu^m$ of them: the contact matrix on the 93 types of Theorem 1 has characteristic polynomial $x^{84}(x-1)^3(x+1)^4(x^2-4x-1)$, with ν a simple dominant root.

For the lower bound, 20 of the types share an edge at every order $k \geq 1$, and they contain an irreducible block of spectral radius ν that sibling pairs reach. So at least $c\nu^m$ pairs of depth-m pieces of two sibling limit tiles meet, each in a point on the boundaries of both siblings. Among the sibling types that reach the block are pairs of two non-Mystic children, which are similar copies of K_{p+1}; applied on both parities, the bound transfers to ∂K_p. A box of side λ^{-m} meets boundedly many pieces, because the pieces have area of order λ^{-2m} and overlap in measure zero.

The finite inputs are checked by `boundary_certificate/check_spectre_fractal.py`. □

The theorem does not use the self-similar Spectre-type tilings of [8], whose tiles are fractal by construction. The limiting Spectre supertile is a fractal in its own right: it has interior, and its boundary has box dimension strictly between 1 and 2. The growth is also visible at every finite order: $P(k+3) \geq 4P(k+2)+30$ for every k, while $\lambda^2 < 16$ hence $\lambda < 4$, so $P(k+1)/P(k) > 4 > \lambda$ for every $k \geq 2$, and a perimeter argument assuming $P(k) \sim \lambda^k$ is wrong at every order.

6.1 What the golden appearance is not

φ^3 next to the Hat's φ^4 invites the conjecture that the Spectre's boundary combinatorics *are* the Hat's. They are not.

Fact 12. *x^2-4x-1 divides neither χ_M nor $\chi_H(x) = (x-1)(x+1)(x^2-7x+1)$, the characteristic polynomial of the four-metatile substitution of the Hat [1].*

Fact 13. *φ^n has minimal polynomial $x^2 - L_n x + (-1)^n$ with L_n Lucas ($n \geq 1$). So $x^2 - 4x - 1$ is the minimal polynomial of φ^3 and $x^2 - 7x + 1$ that of φ^4: any two powers of φ have Lucas quadratics as minimal polynomials, and the resemblance carries no information beyond $\mathbb{Q}(\sqrt{5})$.*

The asymmetry is worth recording: the Spectre's own unit is not golden, since χ_M contributes the trace 8 and 8 is not a Lucas number. The three operators sit at traces 8, 7 and 4 with norms $+1$, $+1$ and -1.

What survives is weaker and true. The ratio $\mu = \lambda^2/\nu$ has minimal polynomial $x^4 + 32x^3 - 46x^2 - 32x + 1$ — degree four, not two — so it lies in the compositum $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ and in neither quadratic field alone. Both λ^2 and ν are Spectre quantities; the Hat does not enter μ.

6.2 A closed form, and the directions of the boundary edges

The recurrence has a closed form in Fibonacci numbers.

Fact 14. *The supertile perimeter is $P(k) = 2F_{3k+5}+4$ for every $k \geq 0$, where F_n is the Fibonacci number (Theorem 1).*

It follows from the recurrence by induction, because $F_{n+6} = 4F_{n+3} + F_n$ makes $2F_{3k+5}+4$ satisfy the same recurrence and the seeds agree. It says three things that the recurrence did not. The growth $\nu = \varphi^3$ is one Spectre level advancing the Fibonacci index by three, which is how the golden unit arises. The marginal eigenvalue $+1$ is the constant 4, so $(P(k) - 4)/2$ is every third Fibonacci number. And the remaining root of the characteristic polynomial $(x-1)(x^2-4x-1)$, the conjugate of φ^3, is $2 - \sqrt{5} = -\varphi^{-3}$, which is negative, so that component alternates in sign with the parity of the level. That is the same parity that flips the determinant of the placement matrix, but any quantity alternating with level parity does so, and this is weak evidence of a link to the chirality channel and is not claimed as one.

The boundary edges also have directions. The fourteen edges of the Spectre point along multiples of 30°. Let h_j, $j \in \mathbb{Z}/6$, be half the number of boundary edges of an order-k supertile whose direction is $j \cdot 30°$ modulo 180°, so that $\sum_j h_j = P(k)/2$. Splitting the six classes by parity gives, counted on the placed tiling through $k = 6$,

$$\underbrace{h_0 + h_2 + h_4}_{\text{even}} = F_{3k+4} + 1, \qquad \underbrace{h_1 + h_3 + h_5}_{\text{odd}} = F_{3k+3} + 1, \tag{3}$$

whose sum doubled is $P(k)$. The difference of the two totals is exactly F_{3k+2}, which is the amplitude of the $\mathbb{Z}/2$ character in Table 2. The remaining characters of $\mathbb{Z}/6 = \mathbb{Z}/2 \times \mathbb{Z}/3$, the $\mathbb{Z}/3$ and faithful ones (each up to complex conjugation), stay bounded: their largest amplitude over $k \le 6$ is 3.61. The group $\mathbb{Z}/6$ is the one that acts on the hexagonal directions of the tiling, and the data identify the group, not the action.

Table 2: Boundary edges of an order-k supertile by direction modulo 180°. Even and odd are the class totals of (3); $|\hat{h}|$ is the largest amplitude of the $\mathbb{Z}/3$ and faithful characters of the folded histogram. Counted on the placed tiling (`chair44_spectre_link.py`); the $k = 6$ row took 66 s. The $P(k)$ column is the recurrence sequence and agrees with the exact count of Fact 8.

k	$P(k)$	even	odd	even − odd	$\lvert\hat{h}\rvert$
0	14	4	3	1	1.00
1	46	14	9	5	1.00
2	182	56	35	21	2.65
3	758	234	145	89	2.65
4	3198	988	611	377	3.61
5	13534	4182	2585	1597	3.61
6	57318	17712	10947	6765	1.73

Counted, not derived. The closed form and the recurrence are proved for every order (Theorem 1); the split (3) is a count and is not proved beyond $k = 6$. What is proved is the arithmetic that ties it to the closed form: $F_{3k+4} + F_{3k+3} + 2 = P(k)/2$. The boundedness of the other characters is an observation. The directions are counted with the fork's own tile builder, so this is a count and not an independent second route.

The fine decomposition. Resolved by species and edge index and counted on placed tiles for $1 \le k \le 6$ (`spectre_golden.py`; the single tile at $k = 0$ is dropped, and with it the edge types that occur only there), the boundary has 40 edge types, and only 22 of them satisfy the recurrence of Theorem 1. Of the other 18, 14 are bounded and of period two, such as $1, 2, 1, 2, 1, 2$, and the remaining four grow without obeying it or its obvious order-four extension: the edges 3 and 4 of Π both give $1, 7, 22, 91, 378, 1599$ and those of Ψ both give $0, 5, 31, 141, 607, 2581$, and their ratios still approach ν. The operator of Theorem 1 is exact for the aggregates; the operator on the fine types is larger, and its spectrum below ν is not determined by these orders.

6.3 The balanced exponent

A boundary that grows by ν per level while the area grows by λ^2 raises the question of how to weigh the two against each other, and the answer is a single exponent.

Proposition 2. *Interfaces at coarseness $1 \le c \le N$ in a depth-N patch, that is, unit edges shared by two different supertiles of order $N-c$, number $\frac{1}{2}\big((T(c)-N_\Gamma(c))P(N-c)+N_\Gamma(c)Q(N-c)-P(N)\big)$, with Q the Mystic perimeter of (2) and $N_\Gamma(c)=T(c-1)$; so, since P and Q grow like ν^k (Theorem 1), they grow in c by $\mu=\lambda^2/\nu$. If each such unit edge is thickened to a neighbourhood of width $\delta_c=\delta_0\lambda^{-sc}$ and cross-section δ_c^2, as for a curve in $\mathbb{R}^3$, the exponent making every level carry comparable volume is*

$$s^* = 1 - \frac{\log \nu}{\log \lambda^2} = 0.300373393044. \tag{4}$$

Proof. The depth-N patch is the order-N supertile, of perimeter $P(N)$. It is the union of $T(c)$ supertiles of order $N-c$ with disjoint interiors (Theorem 1), counting a Mystic as one, and $N_\Gamma(c)=T(c-1)$ of them are Mystics (Fact 3). Summing their perimeters counts every interface edge twice and every edge of the outline once, which gives the count. It grows in c by λ^2/ν because $T(c)\sim a\lambda^{2c}$ and $P(N-c), Q(N-c)\sim b\,\nu^{N-c}$. With the B_c interface edges at coarseness c, each of cross-section δ_c^2 the volume at level c is of order $\mu^c\lambda^{-2sc}$, which is independent of c exactly when $\lambda^{2s}=\mu$, that is, at $s=s^*$. □

The natural guess $s=2$, one factor of the unit per level, is wrong, and Section 6 says why: every interface edge is a *unit* tile edge whatever level it sits at, so length does not scale with level and only the population does. Two cautions attach to (4). It is exact but *not algebraic*. The ratio $\log\nu/\log\lambda^2$ is irrational, since $\nu^p=\lambda^{2q}$ with integers $p,q\neq 0$ would put an element of $\mathbb{Q}(\sqrt{5})\setminus\mathbb{Q}$ equal to one of $\mathbb{Q}(\sqrt{15})\setminus\mathbb{Q}$; and an irrational ratio of logarithms of algebraic numbers is transcendental by the Gelfond–Schneider theorem [15]. The thickening that defines s^* is a modelling choice and not a property of the tiling; given it, s^* is exact, since ν is (Theorem 1). And s^* had to be derived rather than fitted: because the interface count depends on $T(c)$ and on $P(N-c)$ together, a single-exponent fit needs c large and $N-c$ large at once, which no finite patch provides. Fitting one anyway to the measured counts gives a growth of 2.66 per level at depth 4 and 2.41 at depth 5, against $\mu\approx 1.859$. The closed form of Proposition 2, with its Mystic term, matches those counts exactly at both depths, so it reproduces the same fitted values (`spectre_boundary.py`), which is how we know the drift was the estimator and not the geometry.

7 The Spectre monotile in four dimensions

Every edge of the Spectre tiling is a unit vector at a multiple of 30°. Following Chéritat and Ma [10], who lift the tilings by the deformable tiles Tile(a,b) of the Hat family to four dimensions, the even multiples are taken to lie in a first plane and the odd multiples in a second. Write $\omega=e^{i\pi/3}$ and $z=e^{i\pi/6}$. An even edge is $p+q\omega$ and an odd edge is $z(r+s\omega)$ with integers, so walking along edges assigns every vertex a point (p,q,r,s) of $\mathbb{Z}^4$, and the projection $(u,v)\mapsto au+bv$ is the deformation of the tile.

Fact 15 (The integer lift)**.** *Every vertex of a Spectre patch is a sum of unit vectors at multiples of 30°, so it lies in $\mathbb{Z}[\zeta_{12}]$, which has the basis $1,\omega,z,z\omega$; its coordinates (p,q,r,s) in that basis are its lift, which is therefore well defined. On the level-3 patch (3734 vertices) the vertex differences have rank 4, and the physical position is $a(p+q\omega)+bz(r+s\omega)$, which is the tiling itself at $a=b=1$ and gives a homogeneous quadratic form in (a,b) for the area of a tile.*

Fact 16 (The inflation as an integer matrix)**.** *The supertile recursion moves four key points of the supertile by rotations through multiples of* $60°$*, a mirror and translations built from the points themselves; on* $\mathbb{Z}[\zeta_{12}]$ *each is an integer map, so one level is a* 16×16 *integer matrix* Φ *on the four key points. It reproduces the floating-point recursion of the construction for three levels (largest difference* 3×10^{-14}*), and an independent integer transcription reproduces it exactly for six levels; and*

$$\chi_\Phi(x) = (x-1)^4(x+1)^4\,(x^4-8x^2+1)^2, \qquad x^4-8x^2+1 = \big(x^2-(4+\sqrt{15})\big)\big(x^2-(4-\sqrt{15})\big),$$

with roots $\pm\lambda$*,* $\pm\lambda^{-1}$ *and* $\lambda^2 = \sqrt{15}+4$*. The key-point vector has minimal polynomial* $(x^2-1)(x^4-8x^2+1)$*, and on the part of its cyclic subspace where* x^4-8x^2+1 *vanishes* Φ *acts by the companion matrix*

$$\begin{bmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

of determinant 1 *and trace* 0*.*

The unit $4+\sqrt{15}$ has norm one, which is why the block is unimodular: two eigenvalues expand ($\pm\lambda$) and two contract ($\pm\lambda^{-1}$). The negative roots come from the mirror applied at every level; over two levels Φ^2 acts on the expanding pair as the real number λ^2. The Chair44, by contrast, inflates by $2\,I_3$ on $\mathbb{Z}^3$, of determinant 8: every eigenvalue expands and nothing contracts, so there is no Euclidean internal space; the internal space of the chair is 2-adic (Section 8).

Fact 17 (The internal plane)**.** *A linear functional of* (p,q,r,s) *has bounded range on the supertile corners exactly when it kills the expanding part (on which* Φ^2 *acts as* $4+\sqrt{15}$*) of every iterate of the key-point vector. Each level mirrors the frame, so the odd iterates are mirrored first; the expanding parts then span two dimensions and the annihilator is the plane*

$$\begin{bmatrix} \frac{5}{11}+\frac{3\sqrt{15}}{11} & \frac{3\sqrt{15}}{11}+\frac{16}{11} & -\frac{5\sqrt{15}}{11}-\frac{12}{11} & 0 \\ \frac{3}{11}+\frac{4\sqrt{15}}{11} & \frac{3}{11}+\frac{4\sqrt{15}}{11} & -\frac{3\sqrt{15}}{11}-\frac{5}{11} & 1 \end{bmatrix}$$

(rows are functionals on (p,q,r,s)*), with entries in* $\mathbb{Q}(\sqrt{15})$*. It agrees with a plane fitted to the lifted vertices to principal cosines above* $1-10^{-6}$*. It is bounded on the patches of one parity of level and its mirror image on the other. The extent of the image, each level taken with its bounded plane, is 9.77 at level* 3 *and 9.84 at level* 4*, while the physical extent grows by a factor 2.89 per level (the inflation factor is 2.8059). The Galois conjugate* $z \mapsto z^5$*, the obvious candidate for the internal space, is not bounded: its extent is 50 at level* 3*.*

The plane is bounded for one frame and its image under the mirror for the next, which is why a single fitted plane cannot serve every level. The alternation is exact, and it makes the window the attractor of contracting maps. The slot maps of the construction use only rotations by multiples of $60°$ and the mirror. Under that group, $\mathbb{Z}[\zeta_{12}] \otimes \mathbb{R}$ is two copies of one irreducible plane, so it has a one-parameter family of invariant planes, not only the two complex embeddings. The conjugate $z \mapsto z^5$ is equivariant for the $30°$ rotation as well, and it is not the internal space: the images of the supertile key points under it grow like λ^k.

Theorem 3. *On levels of parity* p*, the expanding parts of the eight slot translations span a plane* E_p*, and every slot map sends* E_p *to* E_{p+1}*. Projected along* E_p*, the marginal* (± 1) *parts of the*

translations are removed by a fixed shift of the internal origin. What remains is $\lambda^{-k}\gamma_{i,p}$. The window W of the tile positions, read in the frame of each tile, is the attractor of the graph-directed system

$$Z_p(S) = \bigcup_i \bar{A}_i^{-1}\Big(\gamma_{i,p} + \lambda^{-1}\bigcup_{S'} Z_{p+1}(S')\Big)$$

of similarities of ratio $1/\lambda$. Each $Z_p(S)$ has positive area, and its pieces overlap in measure zero. The system is dual to that of Theorem 2, with shape matrix $\left(\begin{smallmatrix}7 & 6\\ 1 & 1\end{smallmatrix}\right)$. Its contact matrix, on 1192 types, has characteristic polynomial $(x-1)^{35}(x+1)^{35}x^{1114}(x^4-10x^2+1)^2$, and spectral radius $\sqrt{2}+\sqrt{3}$. Let the seam $\Sigma_p(S)$ be the union of the intersections of distinct pieces of $Z_p(S)$. Then

$$\dim_B \Sigma_p(S) = \frac{\log(\sqrt{2}+\sqrt{3})}{\log\lambda} = \frac{\log(5+2\sqrt{6})}{\log(4+\sqrt{15})} = 1.110977\ldots,$$

and the lower box dimension of $\partial Z_p(S)$ is at least this value, for every S and p.

Proof. The plane, intertwining and coboundary statements are linear algebra on the eigencomponents of q_0. They are checked to sixty digits, together with a metric in which every internal map is orthogonal (the even products form a group of order 6). The bottom-up expansion of a tile's internal coordinate is then the stated system.

Two distinct pieces of the same depth m descend from a pair of sibling pieces. If they lie within $\delta\lambda^{-m}$ of each other ($\delta = 10^{-20}$, the margin of the certificate), so do their ancestors at every depth, so none of those pairs is pruned and the relative placement of the two pieces is one of the 1192 types. No type is the identity (the nearest candidate is at distance 4.11), so a piece is determined by its type relative to another, and a disc of radius $\delta\lambda^{-m}/2$ meets at most 1193 pieces of depth m. Give $Z_p(S)$ its natural measure ρ, in which a piece of depth m and type q has mass $\lambda^{-2m}w_q/w_{(p,S)}$, with w the Perron vector of the shape matrix. Then $\rho(B(x,r)) \le Cr^2$, so $Z_p(S)$ has positive area by the mass distribution principle. The vector a of areas satisfies $a \le \lambda^{-2}Na$, with N the shape matrix; pairing with the positive left Perron vector of N, of root λ^2, forces equality, so the pieces overlap in measure zero.

For the upper bound on the seam, a meeting point of two pieces of depth one lies, at each depth m, in a pair of meeting pieces whose type is live. There are $O(m(\sqrt{2}+\sqrt{3})^m)$ such pairs, since the dominant root is double, and each lies in a disc of radius of order λ^{-m}. For the lower bound, 142 types lie on infinite paths of the contact graph, so their pieces meet at every depth. They contain an irreducible block carrying x^4-10x^2+1, reached from 16 sibling types, so at least $c(\sqrt{2}+\sqrt{3})^m$ pairs of pieces of depth m meet. A disc of radius λ^{-m} meets boundedly many pieces of depth m, because they have area of order λ^{-2m} and overlap in measure zero, so a fixed fraction of that many discs is needed to cover the seam. A meeting point x of two pieces A and B lies on ∂A: every neighbourhood of x contains a whole piece of B of some depth, of positive area, which would otherwise meet A in positive measure. The pieces in the sibling types that reach the block are similar copies of $Z_{p'}(S')$ for all four choices of S' and p', so the lower bound holds for every $\partial Z_p(S)$. The computation is `boundary_certificate/spectre_window.py`. □

The theorem bounds the dimension of ∂W from below and not from above, because a point of ∂W need not lie near a seam. An upper bound needs the pieces of the windows that neighbour W in a covering of internal space, the role played in Theorem 2 by the neighbouring supertiles of a tiling. Baake, Gähler, Mazáč and Sadun show that the window of the self-similar member CASPr of the family and its lattice translates tile internal space [8]. The Spectre is mutually

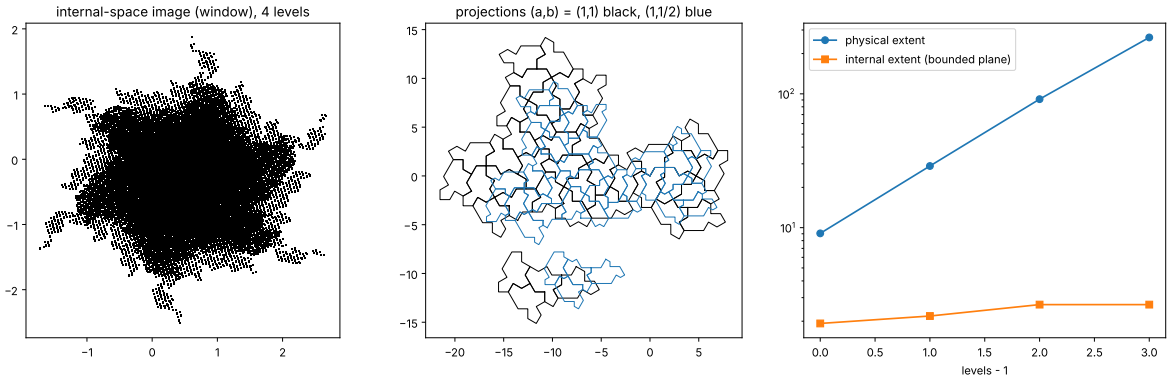


Figure 3: The lift of the Spectre. (a) The internal image of the vertices of a level-4 patch through the exact bounded plane: the window. (b) The projections $(a, b) = (1, 1)$ (black) and $(1, \frac{1}{2})$ (blue) of the same lifted tiles. (c) The physical extent of the lifted patch, which grows with the level, and the internal extent, which does not.

locally derivable from a reprojection of CASPr with the same lattice. Identifying W with their window, and showing that the neighbour placements of that tiling lie among the 1192 types, would give $\dim_B \partial W = 1.110977\ldots$; this is not done here. They obtain the same value for CASPr as a Hausdorff dimension, extracted from its orbit separation dimension, and remark that whether the dimension is the same everywhere along the boundary deserves a closer analysis [8]. Theorem 3 derives the value for the seams of the Spectre's own window from the substitution, as a box dimension. The image in the fitted plane is bounded, and its boundary cell count (a level-5 patch, holes from finite sampling filled; Figure 3 shows level 4) grows by $2^{1.26}$ and $2^{1.16}$ at the successive halvings of the cell size from 0.2, against 2^1 for a smooth curve, while the area grows by $2^{1.87}$. The boundary is not a smooth curve, since by Theorem 3 its lower box dimension exceeds 1; a box count at these resolutions gives about 1.2, a crude estimate, against the exact 1.111 of the seams. It is not the dimension $\log \nu / \log \lambda$ of the supertile outline in the physical plane (Section 6). Fractal windows of this kind are the Rauzy-type windows that Baake, Gähler, Mazáč and Sadun obtain [7, 8], who prove that the Spectre tilings are mutually locally derivable from model sets of a 4:2 cut-and-project scheme with such windows; the analogous statement for the Hat family is [9]. Whether anything like this exists for a three-dimensional tile is the question raised by Kamenetsky [11], who finds that the obvious four-dimensional analogues of the Chair44 fail a closure test; the integer inflation of the chair leaves nothing to project.

8 From Plane to Bulk: Tsiokos's Chair44

Everything above is two-dimensional; this section asks which of it depends on that. Tsiokos's Chair44 [3] is a candidate three-dimensional einstein, claimed to fill $\mathbb{R}^3$ only aperiodically. The tile is the seven-cube chair (a $2 \times 2 \times 2$ block less one corner) with each of its 24 unit faces carrying eight small pyramids of signed heights. Without the pyramids, the substitution of eight chairs on the doubled chair is the classical three-dimensional chair substitution [14, 12]; the decorations select the frames of the children. Its einstein property and strong aperiodicity rest on Tsiokos's claimed proof, which we quote and have not refereed. Section 8.5 re-proves its combinatorial half: no tiling obeying the contact rules has a nonzero period, and each has

a finite symmetry group. The geometric half (R), that every tiling by the solid obeys those rules, and the existence of a tiling by the solid are quoted, so aperiodicity is proved here only conditionally on (R). What is recomputed is the combinatorial atlas: `symchair44.py` generates the tile in exact rational arithmetic with no external data and agrees vertex for vertex with the published solid (`symchair44_tests.py`, `chair44_tests.py`); the Facts are in `chair44facts.py` and the supplement.

8.1 Combinatorial Reconstruction of the Chair44 Tile

The mesh has 2138 vertices, 6408 edges and 4272 triangles, is a closed manifold of Euler characteristic 2, and has volume exactly 7 for *every* base width w and height unit h, because the 192 signed heights sum to zero and the volume is $7 + \frac{4}{3}w^2h\sum_s \gamma_s$. The substitution places eight copies of the chair on the doubled chair, with the refinement $(G, t) \mapsto (GH,\ 2t + Gu)$. There are exactly $3^8 = 6561$ ways to cover the doubled chair by eight rotated chairs; gluing each to depth two and demanding exact complementarity ($\gamma_A + \gamma_B = 0$ for the pyramids on a shared face) leaves four free height classes for 6555 of them and twelve for six. Of those six, three use all 24 orientations and the first in sort order is the published substitution; the selection rule is a convention and is stated as one. The twelve classes are the heights $\pm1, \ldots, \pm12$. The contact atlas follows from the labels: of 2388 raw face contacts, 1194 are mirror images, 44 are legal, and 30 occur in the depth-three hierarchy, giving 372 site equations of rank 180 and a kernel of dimension 12, the twelve free classes. These counts reproduce the published atlas from the labels alone.

Table 3: The Spectre and the Chair44 side by side. d is the dimension, λ the linear inflation, ν the boundary growth per level, $D_b = \log\nu/\log\lambda$, $s^* = 1 - D_b/d$ and $\mu = \lambda^d/\nu$. The Spectre value of ν is proved (Theorem 1), and D_b is the box dimension of the boundary of the limiting supertile (Theorem 2). The last two rows are about what is claimed or quoted, not computed.

	Spectre	Chair44
d	2	3
λ	$\frac{1}{2}(\sqrt{6}+\sqrt{10}) \approx 2.806$	2
tile count per level	$\lambda^2 \approx 7.87$	$8 = \lambda^3$
boundary growth ν	$\varphi^3 \approx 4.236$	4
D_b	1.3993	2
s^*	0.3004	1/3
μ	1.8586	2
fractal excess $D_b - (d-1)$	+0.3993	0
area and perimeter fields	$\mathbb{Q}(\sqrt{15})$, $\mathbb{Q}(\sqrt{5})$	both $\mathbb{Q}$
species, orientations	9 species, 12 orientations	24 rotations
zero sector	not semisimple	semisimple
frequencies	irrational (Pisot)	rational, 1/24
symmetry groups of a tiling	not computed	1, C_2^{diag}, C_4, D_4 (Theorem 5)
strict chirality	yes, by the tile	yes, by the contacts
spectrum of the tiling	pure point [8]	2-adic, pure point, every atlas tiling (Corollary 2)

8.2 The boundary operator in three dimensions

The Chair44 super-tile of depth N has $24 \cdot 4^N$ boundary faces and $12(8^N - 4^N)$ shared faces, since 24 faces on each of 8^N tiles less two for every shared face leaves $24 \cdot 4^N$; the identity holds

for all N. The faces point equally in all six axis directions, 4^{N+1} each, since the chair has four unit faces in each of them; this is the opposite of the Spectre's direction histogram with its golden split. The interface counts obey the form that the law of Proposition 2 takes when there is no Mystic,

$$B_c = \tfrac{1}{2}\big(T(c)P(N-c) - P(N)\big) = 12 \cdot 4^N(2^c - 1), \qquad T(c) = 8^c,\ P(k) = 24 \cdot 4^k,$$

so the interface law is not an accident of the plane. What is different is the exponent. With $\lambda = 2$ and $\nu = 4$ the boundary grows by exactly λ^2, so $D_b = 2 = d - 1$: the supertile boundary of Tsiokos's Chair44 is an ordinary surface, with no fractal excess, where the Spectre's has $+0.3993$. Area and perimeter inflate in the same field, so there is no divide between them, and the balanced exponent is the rational $s^* = 1/3$. The separation of the area and perimeter fields observed for the Spectre in Section 6 does not occur here.

8.3 The orientation spectrum, and the 2-adic route

Let the 24 proper rotations be the species and let $M_{FF'}$ count the children of an F-oriented chair that are F'-oriented. The characteristic polynomial is $x^3(x-8)(x-2)^{20}$, the minimal polynomial is $x(x-8)(x-2)$, every row and column sums to 8, and $\operatorname{rank} M^n = 21$ for all $n \geq 1$. The zero sector is semisimple, there is no negative eigenvalue, and orientation frequencies converge to $1/24$ at the rate $(2/8)^n$ in the depth n. The Spectre's matrix is the opposite on each count: a Pisot unit, irrational frequencies and a zero sector with a Jordan pair. The consequence is a narrowing. The Pisot route to pure-point diffraction and the irrational-frequency witness of aperiodicity (Section 3) both need an inflation that is an irrational Pisot number, with conjugates to supply a Euclidean internal space, and frequencies that are irrational; the integer inflation 2, Pisot only vacuously, with rational frequencies supplies neither.

The replacement is 2-adic, and it is classical: the undecorated chair substitution, in any dimension, is a standard example of a lattice substitution with a modular coincidence, pure point diffractive with a 2-adic internal space [12, 13]; for the labels of the Chair44 it is recorded in [3, Thm 8.1]. The substitution acts on $24 \times 7 = 168$ labelled unit cells, is primitive (every entry of its 168×168 matrix is positive at power 3), and has a modular coincidence: at depth 3 there are 336 of 512 addresses, among them $(0, 0, 2)$, at which all 168 labels expand to the same label. By the theorem of Lee and Moody (2001) with Schlottmann (2000) [12, 16, 17], a primitive lattice substitution with a modular coincidence has fixed-point label classes that are regular model sets with a 2-adic internal space, so its diffraction is pure point and the fixed point is limit-periodic. That implication is a cited theorem and is not re-proved, and it concerns the fixed point; that every tiling lies in the substitution hull, which was left open upstream, is Theorem 4 below. What is added here is a recomputation of the finite part from our own generator, which agrees on the depth and the address, and a rate. A subset automaton on the label sets (only 209 are reachable, 41 of them with more than one label) counts the digit words of length M that have not yet fixed the label: $c(M) = 1, 8, 64, 176, 400, 848, 1744, \ldots$, that is $c(M+1) = 2c(M) + 48$ and $c(M) = 28 \cdot 2^M - 48$ for $M \geq 1$. This holds for every such M: the listed values give it for $M \leq 4$, and the vectors v_M of counts per automaton state satisfy $v_5 = 3v_4 - 2v_3$ exactly, so applying the linear transfer map gives $v_{M+2} = 3v_{M+1} - 2v_M$ for all $M \geq 3$, and $c(M)$ obeys the same recurrence. The undetermined fraction is $c(M)/8^M \sim 28 \cdot 4^{-M}$, and the transfer matrix on the non-singleton states has spectral radius exactly 2, so the set of addresses not yet determined, which is where the window boundaries lie, has box dimension $\log_2 2 = 1$ in the internal space $\mathbb{Z}_2^3$ of dimension 3. The two 2's are the same number as the

linear inflation and the subdominant orientation eigenvalue, and no link between them is claimed there; Section 9 finds the first-moment sum rule that both obey. The Spectre rests on a Pisot unit and an irrational window; Tsiokos's Chair44 rests on an integer inflation and a 2-adic one.

8.4 Why Three Dimensions Do Not Automatically Resolve Chirality

A common heuristic about parity rests on a sentence: a planar patch embedded in a three-dimensional bulk has its reflection realized as an ambient rotation, so parity that is forbidden on the surface is restored by embedding. The Chair44 does not touch that sentence. It does touch the suggestion that three dimensions are what restores parity.

Fact 18. *The undecorated seven-cube chair has* 6 *symmetries, 3 of them mirrors, and is achiral. The decorated Chair44 has only the identity, so it is chiral. Of the* 2388 *raw face contacts,* 1194 *place a mirror image against the root, and none is legal: all* 44 *legal contacts are proper rotations, so no copy of the tile touches its own mirror image anywhere in the hierarchy.*

The chirality is forced twice over: by the labels, which break the three mirror planes of the seed, and by the contacts, which admit no mirror pair. So, for this tile, passing from the plane to three dimensions does not by itself make mirror images available. Three caveats attach. The einstein property is Tsiokos's claim, and Section 8.5 proves aperiodicity only given (R). The forced chirality was tested on all six substitutions that leave twelve height classes and does not depend on the selection rule, but a decoration outside that family is not excluded. And one example is one example: a three-dimensional aperiodic tile whose legal contacts include a mirror image would show that forced chirality is not a feature of dimension three (`symchair44.legal_contacts` tests any labelled candidate). This section does not show that the Chair44 is an einstein monotile, or aperiodic, or that the pyramid decoration is the only one that works; it re-derives its atlas and finds it equal to the published one. A numerical scan of sliding and tilted placements of smoothed variants (`chair44_rigidity.py`) found no non-lattice contact, which is evidence for rigidity and not a proof of it.

8.5 Every tiling lies in the hull, and the symmetry groups of tilings

The 2-adic route above is stated for the fixed point of the substitution. Tsiokos left two questions open [3]: whether *every* tiling by the Chair44 lies in the hull of the substitution, and which symmetry groups its tilings have. The decorations force every tiling to be hierarchical: a short Berger-style account is due to Goodman-Strauss [4], simpler equivalent matching rules to Flicker [5], and a finite certificate for the unique hierarchy to Kamenetsky [11]. Flicker argues informally that no slide plane is possible and states that only 30 of the 44 contacts appear in an infinite tiling; Theorem 4 and Corollary 1 are the rigorous form. This subsection answers the two questions. The proofs are due to the second author [6]; they combine one level of the hierarchy with a finite computation on the eight unit cells around a lattice vertex.

Set-up. A chair *pose* is $x \mapsto gx + t$ with g one of the 24 proper rotations O and $t \in \mathbb{Z}^3$. An *atlas tiling* is a tiling of $\mathbb{R}^3$ by chairs in poses in which every face-adjacent pair has a relative pose among the 44 legal contacts of Section 8. Label each unit cell by the frame of the chair that covers it and by which of its seven cells it is; there are 168 labels, the 24×7 of Section 8. Write σ for the substitution, of inflation 2.

Definition 1. A *block* is the labelling of the eight unit cells around a lattice vertex. It is *legal* if it occurs in $\sigma^n(j)$ for some label j and some n, and *admissible* if the chairs it names are consistent and pairwise disjoint and every face-adjacent pair has a relative pose among the 44 legal contacts. Every block of an atlas tiling is admissible.

One input is taken from the literature and is not re-derived here; the other is proved below from a finite certificate that is checked independently.

(R) Every tiling by Q is, up to an isometry, an atlas tiling: it is registered, and every pair of tiles sharing a feature is in a relative pose among the legal contacts [3, Thm 5.6, Lemma 5.3]; every shared unit face carries features. This is the geometric half of [3]. Its written proof has been audited step by step by the second author, and all of its finite inputs recomputed in exact arithmetic, including each of the 5,273 collision witnesses (`chair44_independent/registration_audit.py`, `REGISTRATION_AUDIT.md`) [6]; no gap was found. Tsiokos's formal proof of it [3] has also been rebuilt from source in an independent build. The build succeeds, and the dependency report of the theorem lists only the three standard axioms of the proof assistant and named decision steps evaluated by its compiler, whose results are trusted as well as its kernel; the 162 reports printed by the rebuild agree, theorem by theorem and up to a namespace prefix, with the published log, and the log left by the build equals the published one byte for byte (`lean_rebuild/`). The 67 proof files of the build differ from the externally delivered proofs only in marked compile fixes, and no delivered statement is changed beyond notation (`lean_rebuild/discharge_audit_indep.py`, our re-implementation of the project's audit rule, which adds a comparison of every declaration's statement; the judgement that the statement differences are notation is ours). A build shows that the formal statement holds; that the formal statement says what (R) says is a reading of it, not a computation. It has not been refereed.

(H) Every atlas tiling T partitions uniquely into complete level-1 supertiles, which form an atlas tiling $D(T)$ with $T = \sigma(D(T))$. This is Kamenetsky's Proposition 7.1 [11] applied to the Chair44, and Tsiokos proves it for tilings by Q [3, Thms 6.4, 6.5]. It is proved here as Lemma 1.

Lemma 1 (Kamenetsky's criterion, checked)**.** *(H) holds.*

Proof. Call two chairs compatible if their interiors are disjoint and, if they share a unit face, their relative pose is one of the 44 legal contacts. An enclosure of a chair A is a set of pairwise compatible chairs, compatible with A, covering its 24 unit faces; it is live if the chairs of A and the enclosure can all have their faces covered again by pairwise compatible chairs. Kamenetsky's three finite conditions (C1)–(C3) are checked by `hierarchy_certificate/check_chair44_hierarchy.py`, written independently of [11] and [6]. (C1) The relative poses of touching children in the supertiles of all levels are 30. The poses of a level-1 supertile that are admissible next to a reference supertile, with all 64 child pairs compatible, are all aligned; they are exactly the doubles $(g, 2t)$ of the 44 legal contacts (g, t). The coarsening closure of the 30 is therefore the 44, and these 44 are the contacts of the atlas. (C2) The 44 are exactly the poses that obey the 135 face-to-face rules they induce: the face rules are tight, so a tiling obeying the face rules is an atlas tiling. (C3) A chair has 33 enclosures, 18 of them not live. For each of the 15 live ones exactly one level-1 supertile containing A is

compatible with A and the enclosure (U). Either A is that supertile's central child T_0 and all eight children are already present, or T_0 is a chair of the enclosure and every enclosure of T_0 compatible with A and the enclosure contains all eight children (Pr).

Let T be an atlas tiling and A a tile of T. Its enclosure E in T is one of the 33, and it is live because the rest of T covers the faces again; so (U) gives a unique supertile $S(A)$ compatible with $A \cup E$. Any supertile of T containing A is compatible with $A \cup E$, so it can only be $S(A)$. By (Pr), applied to the enclosure of T_0 in T, all eight children of $S(A)$ are tiles of T. If A' is another child, $S(A)$ is compatible with A' and its enclosure, all of whose chairs are tiles of T, so $S(A') = S(A)$. The supertiles $S(A)$ therefore partition T, and the partition is unique. Two touching supertiles of the partition have all their touching child pairs among the 44 contacts, so by (C1) their relative pose is $(g, 2t)$ with (g, t) legal. Halving gives an atlas tiling $D(T)$ with $T = \sigma(D(T))$. □

Lemma 2 (computation)**.** *There are 360,189 admissible blocks, in* 15,214 *orbits under O. For each of them, all* 27 *windows of size $2 \times 2 \times 2$ in its substitution, a $4 \times 4 \times 4$ block, are legal.*

The count is the second author's computation, with `chair44_vertex_blocks.py` and `chair44_hull_prune.py` on the atlas published with [3]. It is repeated by an independent implementation that derives the atlas from the pyramid geometry and, separately, from the eight children alone, and published as a certificate of O-orbit representatives with a stand-alone verifier [6]. `chair44_hull_admissible.py` repeats the count with the 44 contacts that `symchair44.py` derives from its own labels; this is independent code on the same eight children, which define the tile. A further implementation, written from the definitions during a referee review of this paper and sharing no code with the others, derives the 44 contacts from Tsiokos's pyramid data and reproduces 360,189 admissible blocks, 15,214 orbits and no illegal child window, checking every block and not only orbit representatives (`hierarchy_certificate/chair44_blocks_indep.py`, with its output). The lemma is not vacuous: enlarging the atlas by the five non-atlas contacts of a single violating block yields 360,213 admissible blocks and 24 failures. The count runs for minutes, so the build of this paper does not repeat it.

Theorem 4. *Every atlas tiling lies in the hull Ω_σ of the substitution. Hence, by (R), every tiling by Q lies, up to an isometry, in Ω_σ.*

Proof. Let P be a finite patch of an atlas tiling T. By (H) every $D^m(T)$ is an atlas tiling on a lattice $L_m = 2^m\mathbb{Z}^3 + o_m$, $L_{m+1} \subset L_m$. For 2^{n-1} larger than the diameter of P, P lies in $\sigma^n(\beta)$ for the block β of $D^n(T)$ at the vertex v of L_n nearest to P, and β is the window at offset $2^{-n}(v - v') + (1, 1, 1)$ of $\sigma(\beta')$ for the block β' of $D^{n+1}(T)$ at the vertex v' of L_{n+1} nearest to v. As β' is admissible, β is legal by Lemma 2, and so is P. No exhaustion of space by one supertile is used, so tilings with fault planes are covered. □

Corollary 1. *Only the* 30 *contacts that occur in supertiles appear in an atlas tiling; the other* 14 *legal contacts occur in none.*

Proof. The block at a corner of a shared unit face contains both cells across it, and the face-adjacent pairs in legal blocks realise exactly the 30 supertile contacts. □

Corollary 2. *Ω_σ, under translations of $\mathbb{R}^3$, is minimal and uniquely ergodic with pure-point dynamical spectrum, and it is the hull of a regular model set with internal space $\mathbb{Z}_2^3$. Every atlas tiling is pure-point diffractive; by (R), so is every tiling by Q.*

Proof. The substitution is primitive, so Ω_σ is minimal and uniquely ergodic. The modular coincidence of [3, Thm 8.1], recomputed above, and the theorem of Lee and Moody [12] give pure-point spectrum of the hull [17]. In a uniquely ergodic hull every element has the same diffraction, and Theorem 4 places every atlas tiling in Ω_σ. □

The space of all tilings by Q is not minimal: it is the union of the translates $g\Omega_\sigma$ over the isometries g. Nor is every element of Ω_σ a model set or limit-periodic; singular elements of a model-set hull need not be.

A second route to the finite part. The legal blocks depend only on the substitution, so they can be recomputed from the cell substitution of Section 8, which our generator builds from nothing (`chair44_hull.py`). The distinct $2\times2\times2$ windows of $\sigma^n(j)$ over all 168 labels number 168, 600, 1278, 1398, 1410, 1410 for $n = 1,\dots,6$, so 1410 blocks are legal. The set is closed, in that all 27 child windows of every one of its members lie in it, and it contains the windows of $\sigma(j)$; so it is the whole language and not merely the place where a count stopped. The central window $\Phi(w)$ of $\sigma(w)$ maps the legal blocks to themselves and has exactly 27 periodic blocks, all of them fixed. These are the 27 fixed seeds of [3], and their stabilisers in O have order 1 for 24 of them and 8 for 3. Both counts agree with the second author's.

Symmetry. Write $\mathrm{Sym}(T)$ for the group of isometries that map an atlas tiling T to itself, chairs with their frames; for a tiling by Q this is its geometric symmetry group, since Q has no self-isometry (Fact 18). A symmetry maps poses to poses, so its linear part is a quotient of two proper frames: it is proper, with integer translation part, and its kernel consists of periods, which vanish by (H). So $\mathrm{Sym}(T)$ embeds in O, as Tsiokos showed for tilings by Q, with $|\mathrm{Sym}(T)| \le 24$; he exhibited orders 1 and 8 [3].

Theorem 5. *For every atlas tiling T, $\mathrm{Sym}(T)$ is conjugate in O to exactly one of 1, C_2^{diag} (a half-turn about a face-diagonal axis), C_4 and D_4, and each occurs. By (R), the same holds for every tiling by Q.*

Lemma 3 (position of symmetry axes)**.** *Let $L_n = 2^n\mathbb{Z}^3 + o_n$ be the vertex lattice of the level-n supertiles of an atlas tiling T, so that $o_{n+1} \equiv o_n \pmod{2^n}$, and let $g \in \mathrm{Sym}(T)$ have linear part $R \neq I$. (1) If R is a half-turn or quarter-turn about a coordinate axis, the axis of g is a lattice line of every L_n. (2) If R is a half-turn about a face diagonal, the axis of g lies in a coordinate plane through vertices of every L_n and meets every L_n. (3) If $\mathrm{Sym}(T)$ is not cyclic, its fixed point is a vertex of every L_n. In cases (1) and (2) no single point of the axis need be a vertex of every L_n.*

Proof. The hierarchy is unique by (H), so g preserves every L_n. Write $g(x) = R(x-c)+c$ with c on the axis; since R preserves $2^n\mathbb{Z}^3$, g preserves L_n if and only if $(I-R)(c-o_n) \in 2^n\mathbb{Z}^3$. (1) Take the axis along e_3. For a half-turn $I-R = \mathrm{diag}(2,2,0)$; for a quarter-turn $I-R$ acts on the (e_1,e_2)-plane with determinant 2. Either way each $\delta_n^{(i)} = (c_i - o_{n,i}) \bmod 2^n$, $i = 1,2$, is 0 or 2^{n-1} (for $n=0$: 0 or $\frac12$). If $\delta_n^{(i)} = 2^{n-1}$ then, since $o_{n+1} \equiv o_n \pmod{2^n}$, $c_i - o_{n+1,i} \equiv 2^{n-1}$ or $3\cdot2^{n-1} \pmod{2^{n+1}}$, neither 0 nor 2^n, against the condition at level $n+1$. So every $\delta_n^{(i)} = 0$. (2) For $R(x,y,z) = (y,x,-z)$ the condition reads $(c_1 - o_{n,1}) - (c_2 - o_{n,2}) \in 2^n\mathbb{Z}$ and $2(c_3 - o_{n,3}) \in 2^n\mathbb{Z}$. The second gives $c_3 \equiv o_{n,3}$ as in (1); by the first, $c + s(1,1,0)$ lies in L_n for $s \equiv o_{n,1} - c_1$. (3) A non-cyclic subgroup of D_4 contains two half-turns about perpendicular coordinate axes, or

one about a coordinate axis and one about a perpendicular face diagonal; its fixed point is the intersection of the two axes, and by (1) and (2) each of its coordinates is $\equiv o_n \pmod{2^n}$. □

Proof of Theorem 5. No element of order three. Such an element is a rotation about a line in direction $(1, \pm 1, \pm 1)$. Along the line the three coordinates change at the same rate, so at some point all three lie in $(\frac{1}{100}, \frac{99}{100})$ modulo 1. That point is interior to a unit cell, hence to the chair covering it (for Q: farther from every cell face than the pyramid height $12 \cdot 10^{-4}$), so the rotation fixes that chair with its frame and is the identity. So $\mathrm{Sym}(T)$ is a 2-group, contained in a copy of D_4; up to conjugacy in O its candidates are 1, C_2 (coordinate axis), C_2^{diag}, C_4, the two Klein groups and D_4. By Theorem 4 every block below is legal.

Vertex-centred groups. Let $\mathrm{Sym}(T)$ be non-cyclic, with fixed point x, a vertex of every L_n by Lemma 3(3), and let w_n be the block of $D^n(T)$ at x. Then $w_n = \Phi(w_{n+1})$. The substitution commutes with O, so $\mathrm{Stab}(w_{n+1}) \le \mathrm{Stab}(w_n)$, and since the w_n cover balls of radius 2^n about x, $\mathrm{Sym}(T) = \bigcap_n \mathrm{Stab}(w_n)$; the stabilisers are constant from some N_0 on. A block β that occurs infinitely often among $w_{N_0}, w_{N_0+1}, \ldots$ satisfies $\beta = w_a = w_b$ with $N_0 \le a < b$, so it is Φ-periodic and $\mathrm{Stab}(\beta) = \mathrm{Sym}(T)$. The Φ-periodic legal blocks are the 27 fixed seeds, with stabilisers 1 and D_4; this is the count recomputed above. So a non-cyclic group is D_4, and the Klein groups do not occur.

Axial groups. Let $\mathrm{Sym}(T) = K$ be cyclic and non-trivial, with axis A in direction e, and let O_e be the rotations about e. By Lemma 3, $A \cap L_n = v_n + 2^n \mathbb{Z}e$; take v_n nearest to a fixed point p of A. Then $v_n - v_{n+1} = j_n 2^n e$ with $j_n \in \{-1, 0, 1\}$, and the blocks w_n of $D^n(T)$ at v_n satisfy $w_n = W_{j_n}(w_{n+1})$, where W_j is the window of $\sigma(\cdot)$ centred at je. The elements of O_e commute with every W_j, so $\mathrm{Stab}(w_n) \cap O_e$ decreases to K; the full stabilisers may be larger, since an element outside O_e acts about the moving point v_n. If $j_n = j$ for all $n \ge N$, then $y = v_N + j2^N e$ satisfies $y - v_n = j2^n e$ for $n \ge N$, so y is a vertex of every L_n and $\mathrm{Sym}(T)$ is 1 or D_4 by the previous case, not K. So the shifts are not eventually constant, and their tail lies in a strongly connected component carrying two distinct shifts of the graph G_K whose vertices are the legal blocks w with $\mathrm{Stab}(w) \cap O_e = K$ and whose edges $w \to W_j(w)$ are labelled j.

Conversely, take a cycle in G_K whose labels $j_0, \ldots, j_{q-1}$ are not all equal and unroll it periodically: $v_n = p - x_n e$ with $x_n = \sum_{m<n} j_m 2^m$. With $J_i = \sum_{m<q} j_{i+m} 2^m$ (indices mod q) and $y_i = J_i/(1 - 2^q)$, induction gives $x_n = y_0 - 2^n y_{n \bmod q}$; since $0 < |J_i| < 2^q - 1$, $|x_n| \le \eta 2^n + 1$ with $\eta = \max_i |J_i|/(2^q - 1) < 1$, so the blocks cover cubes of half-width $(1 - \eta)2^n - 1$ about p and define a tiling in the hull. Every element of K fixes A and stabilises every w_n about $v_n \in A$, so $K \le \mathrm{Sym}(T)$. A point $p + ze$ lies in every L_n only if z is the 2-adic number $-y_0$, which is not an integer; by Lemma 3(3) the symmetry group is cyclic about A, hence K.

A computation finds such components for C_2^{diag} (14 blocks on each face-diagonal axis) and C_4 (4 blocks). For a half-turn about a coordinate axis the 18 blocks of G_K, among them blocks whose full stabiliser is D_4 or a Klein group, form only components of constant shift. Explicit tilings built from such cycles show exactly the predicted group on large central regions; that is a consistency check, and the exclusion of further symmetries rests on the argument above. □

Remark 1. In particular $|\mathrm{Sym}(T)| \le 8$, and the bound is attained. No atlas tiling has a 3-fold axis, a Klein four-group of symmetries, or exactly a 2-fold coordinate axis.

The theorem is about tilings and does not conflict with Fact 18, which is about the tile: the decorated chair has only the identity, and a tiling by it can still be invariant under a rotation of its whole pattern. Nor does the bound of 8 conflict with the 24 orientations that occur

in a depth-two supertile (Fact 19): by Theorem 4 every tiling contains all of them, and it is symmetric under at most a third of the 24 rotations.

Scope. For atlas tilings, Theorem 4, Corollaries 1 and 2 and Theorem 5 rest on Lemma 1, on Lemma 2, and, for the corollary, on the theorem of Lee and Moody [12] with its consequence for the hull [17]. For tilings by Q they also rest on (R), quoted, audited by the second author, and not refereed. The admissible-block count, the axial-group analysis and the explicit patches are the second author's computations, repeated independently; the build of this paper recomputes the language of legal blocks, its closure, the fixed seeds and their stabilisers. Since $\mathrm{Sym}(T)$ embeds in O, no atlas tiling has a nonzero period, so by (R) no tiling by Q is periodic; that is the whole bearing of this section on strong aperiodicity. (R) itself and the existence of a tiling by Q are quoted, so the einstein property and strong aperiodicity of the Chair44 remain Tsiokos's claim.

8.6 Uniform frequencies hold in boxes, not along lines

Corollary 2 says that every finite patch has a frequency, reached uniformly by boxes of side L in every tiling. It is tempting to read this as a statement about any scan of a tiling, for instance the sum of an observable along a line of cells. It is not, and this subsection says what is true. Everything below is proved from, or is, an exact finite computation (`chair44_lines.py`) except the last proposition, which is conditional on the internal space of Corollary 2.

Box means. The matrix of the cell substitution on the 168 labels is primitive with every row and column sum 8, so each label has frequency $1/168$ in every tiling, and the box mean of a cell observable is its mean over the labels. We use two observables of a cell. The *height* p is the sum of the signed heights of the pyramids on the faces of the cell; it vanishes on five of the seven cells of the chair and is -36 and $+36$ on the other two (the two values cancel, as they must, since the 192 heights sum to zero). The *notch vector* $d = g(1,1,1) \in \{\pm 1\}^3$ belongs to the chair that covers the cell. Both have box mean exactly 0, in each component.

Proposition 3 (chair stacks)**.** *Six of the 168 labels, namely cell* 0 *of the chairs whose notch points along* $\pm(1,1,1)$ *(three along each sign), satisfy* $\sigma(j)_{000} = \sigma(j)_{111} = j$*. For each of them and every* n*, the chairs of* $\sigma^n(j)$ *form a stack in which each chair fills the notch of the next, and each of the seven cells of the chair gives a line of identical labels in direction* $(1,1,1)$*, of length* 2^n *on the main diagonal and* $2^n - 1$ *for the other six cells. On the line of one cell type* p *is* -36 *throughout, on that of another* $+36$*, and on each of the seven lines each component of* d *is* ± 1*.*

Proof. Label the cells (x, y, z) of the stack by the difference $e = (x - z,\ y - z)$. The lines of the stack are exactly the cells whose difference lies in a set T of seven values, and a cell of difference e has a label $\Lambda(e)$. We show that every cell of $\sigma^n(j)$ with difference in T has the label $\Lambda(e)$, by induction on n. For $n = 0$ the single cell has difference $(0,0)$ and label $j = \Lambda(0,0)$. A cell of $\sigma^{n+1}(j)$ is the child $(a,b,c) \in \{0,1\}^3$ of a cell of $\sigma^n(j)$ whose difference is $e' = \big((e_0 - a + c)/2, (e_1 - b + c)/2\big)$. So it suffices that, for every $e \in T$ and every child position compatible with e, the difference e' lies in T and $\sigma[\Lambda(e'), (a,b,c)] = \Lambda(e)$. That is 14 cases for each of the six labels, and they all hold (`chair44_lines.py`). The set T is $\{(0,0), \pm(1,0), \pm(0,1), \pm(1,1)\}$, and the cells of the cube $[0, 2^n)^3$ with difference e number $2^n - \max(0, e_0, e_1) + \min(0, e_0, e_1)$: that is 2^n on the main diagonal $e = (0,0)$ and $2^n - 1$ for the other six. □

The stack lies in every tiling. It occurs in $\sigma^n(j)$ for every n, and by Theorem 4 and minimality every legal patch occurs in every tiling. So in every Chair44 tiling and for every L there are

L consecutive cells on a line in direction $(1,1,1)$ on which the mean of p is exactly -36 (and others on which it is $+36$) and the mean of each component of d is ± 1, while the box means are 0. Averages along these lines do not converge to the box mean, and no uniformity in L can hold for them.

Corner lines. For $b \in \{0,1\}^3$ the cells tb of $\sigma^n(j)$, $0 \le t < 2^n$, are the n-th iterate of the constant-length-2 substitution $\tau_b(j) = \sigma[j,0]\,\sigma[j,b]$ on the same labels, since the child positions along the line are always 0 and b. Every label leads into one of its closed strongly connected classes, each of which is primitive and carries exact letter frequencies, so the mean of each cell observable along the line has an exact limit for each starting label. Table 4 gives the mean of the notch vector; the mean of p is 0 on all of them. The seven lines are the lines through the corner of the cube in the seven directions in $\{0,1\}^3$; they are also read off the label cube, with agreement.

Table 4: Corner lines of $\sigma^n(j)$: closed classes of τ_b and the exact limit of the mean of the notch vector. The box mean is $(0,0,0)$.

direction b	closed classes of τ_b	mean of d
$(0,0,1)$	one class of 4	$(1,1,0)$
$(0,1,0)$	one class of 4	$(1,0,1)$
$(0,1,1)$	one class of 4	$(1,0,0)$
$(1,0,0)$	one class of 4	$(0,1,1)$
$(1,0,1)$	one class of 4	$(0,1,0)$
$(1,1,0)$	one class of 4	$(0,0,1)$
$(1,1,1)$	6 fixed letters	$(-1,-1,-1)$ and $(1,1,1)$

Proposition 4. *Assume that the hull Ω of label configurations, under the action of $\mathbb{Z}^3$ by translations, is uniquely ergodic with invariant probability measure $\mathbb{P}$ and has maximal equicontinuous factor $\pi : \Omega \to \mathbb{Z}_2^3$ with $\pi_*\mathbb{P}$ the Haar measure (Corollary 2; for the action of $\mathbb{R}^3$ on tilings the factor is a solenoid, not $\mathbb{Z}_2^3$). Let $v \in \mathbb{Z}^3 \setminus 0$. Then the action of the translation S^v on Ω is not uniquely ergodic. There are a function f of a finite window of labels and a tiling T such that the averages $\frac{1}{L}\sum_{t<L} f(S^{tv}T)$ along the line through 0 in direction v converge to a number different from $\int f\, d\mathbb{P}$.*

Proof. The closure of $\{tv : t \in \mathbb{Z}\}$ in $\mathbb{Z}_2^3$ is $H = \mathbb{Z}_2 v$, a closed subgroup of infinite index, so its Haar measure is 0. For $x \in \Omega$ the set $K = \pi^{-1}(\pi(x) + H)$ is compact, non-empty and S^v-invariant, so it carries a S^v-ergodic probability measure $\mathbb{P}'$. Since $\mathbb{P}(K) = 0$, $\mathbb{P}' \neq \mathbb{P}$, and functions of a finite window of labels are dense in $C(\Omega)$, so some such f has $\int f\, d\mathbb{P}' \neq \int f\, d\mathbb{P}$. By Birkhoff's theorem the averages along the line converge to $\int f\, d\mathbb{P}'$ for $\mathbb{P}'$-almost every tiling. □

Proposition 3 and Table 4 are explicit instances for the directions $(1,1,1)$ and the others in $\{0,1\}^3$. The failure depends on the observable and on the line: the corner lines all have p-mean 0, and a coordinate-axis line can have bounded sums of d_z while another has sums of d_z growing like L. On the corner line in direction $(0,0,1)$, τ_b maps the four letters of its closed class onto two words uv and $u'v'$, each letter of the closed class goes to uv if it is a first letter of one of these two words and to $u'v'$ otherwise, and d_z is $+1$ on u, v and -1 on u', v'. So beyond the first level the line reads $uvu'v'uvu'v'\cdots$, d_z runs $+1,+1,-1,-1,\ldots$ and its sums stay in $\{0,1,2\}$.

On the corner line in direction $(1,0,0)$ every letter of the closed class has $d_z = 1$, so the sums equal L (`chair44_lines.py`). Nothing here concerns directions that are not lattice directions: the cells met by a line of irrational slope are not an orbit of a subgroup of $\mathbb{Z}^3$, and the argument does not apply.

9 Unfolding the Chair44 and Comparing it with the Spectre

Section 8 compared the two tilings by their combinatorics. This section brings the three-dimensional tile down to two dimensions in the two exact ways, unfolding its surface into a net and slicing and projecting the hierarchy, and then asks the same questions of both tilings. Each tile is reduced to a point (the centroid of a Spectre, the centre of the bounding cube of a chair) and an orientation vector ($e^{i\theta}$ for a Spectre, the direction $d = g(1,1,1)$ of the missing corner, the notch, for a chair). Every number is a Fact computed by `chair44_unfold.py`, `chair44_slice.py` and `chair44_findings.py`, with the Spectre sums on the patches of `spectre.py` through level 5, which has 34649 tiles.

9.1 The twist of the Tsiokos Chair44 supertile

Fact 6 computed the twist of the Spectre supertile. The same question for the Chair44 has a simpler answer.

Fact 19. *The eight chairs of the supertile fill the doubled chair cell for cell, so the inflation is a scaling by* 2 *with no rotation and no mirror: its twist is* 0. *The eight child frames are proper rotations through* $0, 90, 120, 180$ *degrees, two of each, and their mean matrix is exactly* $I/4$*, so the mean frame over the* 8^n *chairs of a level-n supertile is* $I/4^n$*. All* 24 *rotations of the cube occur already at depth two.*

The Spectre supertile is turned by $\theta_S = 11.565°$ and mirrored at every level (Fact 6), and θ_S is irrational as an angle; every child rotation of the chair is a rational multiple of π, so no irrational twist can arise from this substitution. The mean matrix $I/4$ is the statement $\sum_i H_i = 2I$: the eight child rotations add up to the linear inflation times the identity. This is a first-moment sum rule, and it is the only structure found here that links the two 2's noted at the end of Section 8; whether it is related to the repeated eigenvalue 2 of the orientation matrix is not shown.

9.2 Polyhedral Unfolding and Surface Net Construction

The surface of the chair is a sphere made of 24 unit squares, with 26 vertices and 48 edges, whose folds are 27 convex, 18 flat and 3 concave. Keeping the 23 edges of a spanning tree of its face-adjacency graph as hinges, cutting the other 25 edges, and folding every face into the plane gives a 24-omino. There are 261,434,325,600 spanning trees (the matrix-tree theorem, exact). Of 3000 random trees, 1253 give a valid net and 1747 overlap themselves; the valid nets have perimeters from 28 to 50, and 2 of them tile the plane by translation. The smallest perimeter found is a sampled value and not a minimum (any 24-omino has perimeter at least 20). The control is the same code on a single cube, over all 384 of its spanning trees: it returns exactly the 11 nets of the cube. Figure 4 shows three nets and the perimeters.

The supertile surface is the chair surface scaled by 2, so its net is the chair net with each square replaced by a 2×2 block: the area grows by 4 and the length by 2, which is the dimension $D_b = 2$ already found for the boundary (Section 8). Unfolding therefore does not produce a

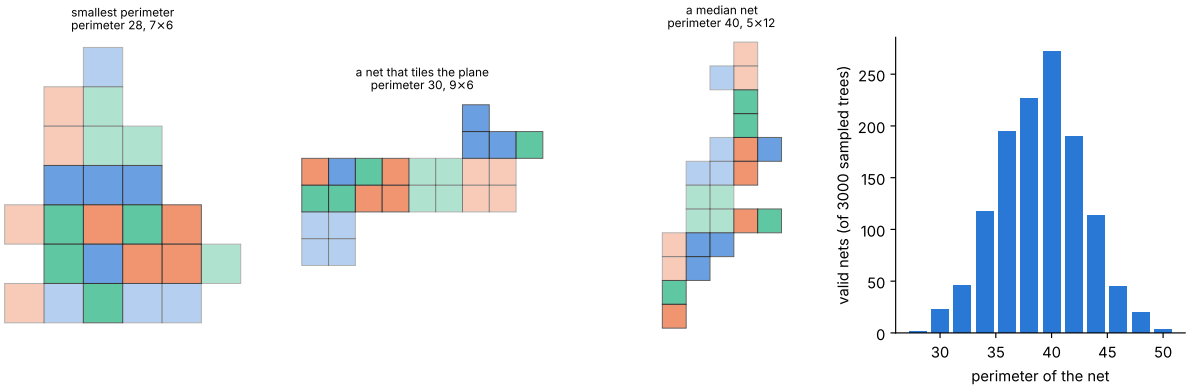


Figure 4: Unfolding the surface of the chair. Three nets (colour is the axis of the face normal, strong for the outward + side): the smallest perimeter among the sampled valid nets, one of the nets that tile the plane by translation, and a median one; and the perimeters of all the valid nets in the sample.

planar analogue of the Spectre's outline, whose perimeter grows with exponent 1.3993 at every order. That is a negative result, and the exact boundary count had already implied it.

9.3 Chair44: Slices, projections and the notch vector

A layer $z = c$ of the depth-n patch cuts each chair it meets into a square of 4 cells or an L-tromino of 3 (the cross-sections found are 3 and 4 cells), so a slice is a tiling of a planar region by polyominoes. The projection of the chair centres along z has, by the projection theorem, the $k_z = 0$ plane of the three-dimensional diffraction as its Fourier transform; this was checked against a three-dimensional FFT.

Fact 20. *The 8^n centres of the bounding cubes of the depth-n patch are distinct integer points on two cosets of $2\mathbb{Z}^3$ ($7, 56, 448, 3584$ and $1, 8, 64, 512$ points for $n = 1, \dots, 4$), and the notch vectors sum to $2^n(1, 1, 1)$. The interface length inside the layer $z = 0$ is $4^{n+1} - 4 \cdot 2^n$ for $n \le 4$ ($8, 48, 224, 960$).*

The sum of the notch vectors in Fact 20 is the sum rule of Fact 19 applied to the vector $(1, 1, 1)$, and it was also counted. It is far smaller than the sum of 8^n random signs, which is of order $8^{n/2}$: the intensity of the notch-weighted structure factor at the origin per tile is $3 \cdot 2^{-n}$ instead of order 1. The layer interface grows by a factor approaching 4 per level over the levels computed, again dimension 2. Figure 5 shows a slice, the $k_z = 0$ plane at level 5, the origin values, and the median intensity at random wave vectors.

9.4 The same questions asked of the Spectre

The orientation sum of the Chair44 grows by exactly the linear inflation, 2 per level. The same sum for the Spectre can be computed on the placed tiles.

Fact 21. *For the Chair44 the notch sum is $2^n(1, 1, 1)$ exactly (each component $4, 8, 16, 32$ at levels $2, \dots, 5$). For the Spectre, $|\sum_j e^{i\theta_j}|/\lambda^n = 1.4413, 1.4480, 1.4488, 1.4489$ at levels $2, \dots, 5$ (patches of $71, 559, 4401, 34649$ tiles), and the ratio of the last two sums is 2.8061 against $\lambda = 2.8059$. The*

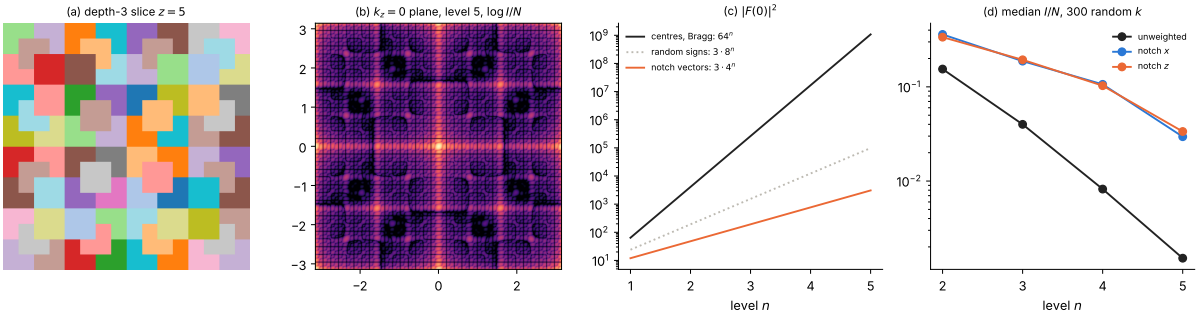


Figure 5: Slices and projection of the hierarchy. (a) A depth-3 layer, coloured by tile. (b) The $k_z = 0$ plane of the diffraction of the level-5 chair centres, which is the Fourier transform of their projection. (c) The squared structure factor at the origin: the centres (Bragg, 64^n), random signs ($3 \cdot 8^n$) and the notch vectors ($3 \cdot 4^n$). (d) The median of I/N at 300 random wave vectors, unweighted and weighted by a notch component.

first is exact and the second is a measurement over floating-point sums; Fact 23 in Section 10 explains it by an exact recursion.

In both tilings the orientation sum grows like the linear inflation. The tile counts grow like λ^{dn}, so $|S|^2/N \propto \lambda^{(2-d)n}$ is constant for the Spectre ($d = 2$; the values are $1.8136, 1.8303, 1.8324, 1.8327$) and 2^{-n} in each component for the Chair44 ($d = 3$). On this evidence the two tilings differ in that quantity through the dimension alone, and not through the details of the tile. This is an observation about one choice of orientation weight. For the Chair44 it is the sum rule; for the Spectre, whose inflation reverses orientation, a sum rule of the form $\sum H_i = \lambda I$ is not available in the same way, and the growth comes instead from the two-level recursion of Fact 23.

Where the power goes is asked with a deliberately loose statistic: the median of $I(k)/N$ over 300 fixed random wave vectors, with k in the box $[0, 2\pi/\ell]^d$ and ℓ the tile scale (the square root of the tile area for the Spectre, $7^{1/3}$ for the chair), against the number of tiles. It falls in both, as $N^{-0.34}$ for the Spectre centroids and $N^{-0.72}$ for the Chair44 centres, and as $N^{-0.26}$ and $N^{-0.39}$ with orientation weights: the intensity is concentrated on few wave vectors, as in a Bragg spectrum, and faster for the chair, whose centres lie on two cosets of $2\mathbb{Z}^3$ (Fact 20); the cause of the difference in rate is not analysed. Figure 6 and Table 5 gather the comparison.

Table 5: The two tilings under the same questions (Facts 19 and 21). The first four rows are exact; the sum and the slopes of the Spectre are measurements on placed patches.

	Spectre	Chair44
supertile twist	$11.565°$, mirrored each level	0, no mirror
orientations	12 (steps of $30°$)	24 rotations
supertile outline, dimension	1.3993	2
tile count per level	$\lambda^2 \approx 7.87$	8
orientation sum	$\approx 1.4489\,\lambda^n$	$2^n(1,1,1)$ exactly
$\lvert S\rvert^2/N$	constant, ≈ 1.83	2^{-n} per component
median I/N, slope in N	-0.34 (weighted -0.26)	-0.72 (weighted -0.39)

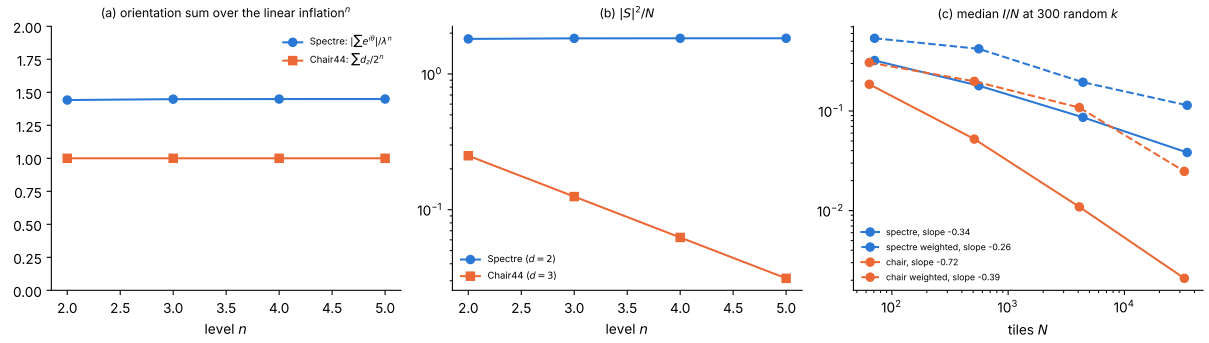


Figure 6: The same questions for both tilings. (a) The orientation sum over the linear inflation to the n: constant for both. (b) $|S|^2/N$, the intensity at the origin per tile of the orientation weights: constant for the Spectre, 2^{-n} for the chair. (c) The median of $I(k)/N$ at 300 random wave vectors against the number of tiles, unweighted and weighted.

9.5 Scope, Limitations, and Epistemic Scope

The nets are a sample of a space of 261,434,325,600 trees, and nothing is claimed about the best net. The point and the orientation vector of each tile are choices, and the sum rule is a statement about the notch vector, not about every function of the orientation. The Spectre patches of `spectre.py` are mirror-type placements; the sum is unchanged by the mirror, but the fixed random wave vectors are not. The median statistic compares two dimensions at one scaling of k and is noisy. None of this decides whether the diffraction measure of either tiling is pure point: for the Spectre that is a published theorem [7, 8], and the exact analysis of the Chair44 centre set is in Section 10.

10 Scattering Diffraction, Orientations, and Adjacency Relations

This section asks four exact questions of both tilings. The tools are in the repository of curved-tile programs spectre-monotile-py; each Fact is computed there and in `monotilefacts.py`.

10.1 The diffraction of the Chair44 centres

Take one point per chair, the centre of its bounding cube, and call the set at depth n $X_n \subset \mathbb{Z}^3$.

Fact 22 (Exact diffraction of the centre set)**.** *$X_n = 2X_{n-1} \cup (B_n \setminus B'_n)$, where B_n is the set of odd points of the cube $[1, 2^{n+1} - 1]^3$ and B'_n the odd points with every coordinate above 2^n (the corner octant); the recursion reproduces the placed patch for $n \le 4$. The structure factor is a finite sum of products of geometric series and equals the direct sum over the centres (relative difference 1.1×10^{-15}). At the dyadic wave vectors $k = 2\pi m/2^J$, m not all even, its limit per centre, $a(k) = \lim_n 8^{-n} \sum_{x \in X_n} e^{ik\cdot x}$ (so $a(0) = 1$, and the Bragg intensity is $|a(k)|^2$ times the squared density $1/49$), is the exact amplitude*

$$a(k) = \tfrac{7}{8}\, 8^{1-J}\big(\pm 1 + \tfrac{1}{7}\big),$$

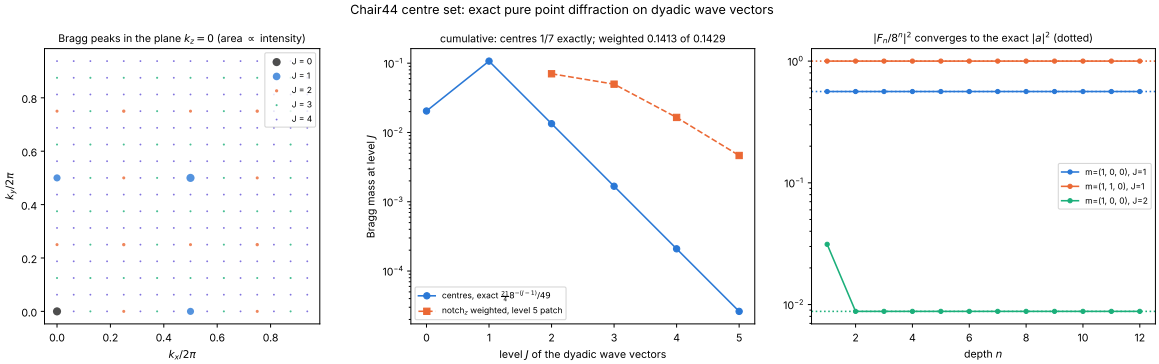


Figure 7: Bragg structure of the Chair44 centre set. The exact amplitudes at the dyadic wave vectors by level, the level masses $\frac{21}{4}8^{1-J}$ and their running sum approaching the density, and the finite-patch intensity converging to it.

with sign $-$ when $m_1 + m_2 + m_3$ is odd, so at $J = 1$, $a = -\frac{3}{4}$ on an axis, $a = -\frac{3}{4}$ on the body diagonal and $a = 1$ on a face diagonal; the last says only that every centre has $x + y$ and $y + z$ even. The squared amplitudes at level J sum to $\frac{21}{4}8^{1-J}$, and

$$\frac{1}{49}\Big(1 + \sum_{J\geq 1} \tfrac{21}{4}8^{1-J}\Big) = \frac{1}{7} = \textit{density of the centres.}$$

Wave vectors that are not dyadic carry no amplitude.

The Parseval sum is complete: all the mass of the density is in the Bragg peaks, so the diffraction measure of the centre set is pure point, and the 2-adic route of Section 8 is made explicit for this set. The centre set depends only on the undecorated chair substitution, so its pure-point diffraction also follows from [12, 13]; what is added is an explicit worked example. The mass of the notch-weighted measure is only numerically consistent with this. That the diffraction of the Spectre is pure point is a published theorem [7, 8] and is not re-proved here; the contrast is only in the route, a cut-and-project scheme with an internal space for the Spectre and a dyadic Parseval sum for the chair.

10.2 Orientation sums and their channels

Section 9 measured the sum of the orientation vectors of both tilings. The chair's is exactly 2^n in each component. For the Spectre it can now be written as a recursion.

Fact 23 (The orientation recursion)**.** *Let S_n be the vector over the nine tile labels of the sums of $e^{i\theta}$ over the tiles of the depth-n supertile of each label. Then $S_{n+1} = C\,\overline{S_n}$ with C the 9×9 matrix of the slot rotations of the construction; the conjugate appears because the inflation reverses orientation. Two levels give $C\overline{C}$ with leading eigenvalue $\lambda^2 = \sqrt{15} + 4$, and the sums over the placed tiles agree with the recursion (largest relative difference 2×10^{-15}, to level 5). Hence the entry S_n^{Δ} of the Δ supertile, the orientation sum of the depth-n supertile itself, satisfies $|S_n^{\Delta}| \sim c\,\lambda^n$ for a constant $c > 0$, along even and odd n alike.*

Numerically, from the recursion to eighty digits, c^2 is the root of $400x^4 - 1200x^3 + 840x^2 - 180x + 9$ equal to

$$c^2 = \frac{15 + 5\sqrt{3} + 3\sqrt{5} + 3\sqrt{15}}{20} = 2.0994,$$

a closed form that is identified, not proved; it is not part of the Fact. It could be proved exactly from the dominant eigenvector of $C\overline{C}$ over $\mathbb{Q}(\sqrt{3}, \sqrt{5}, i)$.
The constant is the ratio 1.4489 measured in Section 9. Splitting the orientation sums by the irreducible representations of the rotation group of the cube for the chair, and by the harmonic m of the orientation angle for the Spectre, gives the growth exponent of each channel.

Fact 24 (Orientation channels)**.** *The* 24*-dimensional orientation matrix of the Chair44 has characteristic polynomial* $x^3(x-8)(x-2)^{20}$*: the trivial channel has eigenvalue* 8 *and every non-trivial channel is exactly* 2 *times a projector. With the exponent* $h = \ln \varrho / \ln \lambda_{\mathrm{lin}}$ *of the leading eigenvalue* ϱ *in the linear inflation* λ_{lin}*, the trivial channel has* $h = 3$ *and all others* $h = 1$*. For the Spectre the harmonics* $m \in \mathbb{Z}/12$ *give* $h = 2$ *for* $m = 0, 6$*,* $h = 1$ *for* $m = \pm 1$ *and* $m = \pm 5$ *(the Galois partners of* ± 1 *under* $z \mapsto z^5$*), and* $h \leq 0$ *for* $m = \pm 2, \pm 3, \pm 4$*. The sums for* $m = \pm 2, \pm 3$ *stay bounded, while those for* $m = \pm 4$ *grow linearly in* n *on the orders computed, so there* $h = 0$ *with a polynomial factor.*

The intensity at the origin per tile of a channel with exponent h in dimension d scales as $\lambda_{\mathrm{lin}}^{(2h-d)n}$. The chair's orientation weights therefore have $|S|^2/N \propto 2^{-n}$ in the non-trivial channels ($h = 1$, $d = 3$) and the Spectre's has an exponent 0 for $m = \pm 1$ ($h = 1$, $d = 2$), which is the difference drawn in Figure 6.

10.3 Which faces meet which

Fact 25 (Contact matrices)**.** *Count how often native edge* a *of one tile meets native edge* b *of a neighbour. For the Spectre (level-*4 *patch,* 29,208 *contacts) the* 14×14 *symmetric matrix has* 62 *nonzero entries, rank* 14*, is bipartite, connected, and has a spectrum symmetric about zero. For the Chair44 (depth-*3 *patch,* 5,376 *contacts, equal to* $12(8^3 - 4^3)$*) the* 24×24 *matrix of native unit faces has* 93 *nonzero entries, rank* 24*, is not bipartite, and has 2 connected components.*

The Spectre's edges fall into two classes that never meet within a class, and the chair's faces do not. The boundary operators differ in the same way as the inflations: the supertile perimeter of the Spectre obeys the recurrence with characteristic polynomial $(x-1)(x^2-4x-1)$ of Section 6, and the surface of the chair obeys $S_N = 24 \cdot 4^N$, with characteristic polynomial $x - 4$.

10.4 The Hat and the Turtle see their lattice; the Spectre does not

Computation, not Fact. The Hat and the Turtle, unlike the Spectre at $(a, b) = (1, 1)$, tile on a hexagonal lattice $\mathbb{C}/\mathbb{Z}[\zeta_3]$, and that can be read in reciprocal space. For the vertex set of a level-5 patch at unit density, with a Gaussian window of width 9, a fraction 0.86 of the Bragg intensity above threshold of the Hat lies within 12% of a point of the dual lattice (from 16 peaks) and 1.00 for the Turtle (from 12 peaks), against 0.00 and 0.00 for the same lattice rotated or rescaled. The Spectre has strong peaks as well, but only a fraction 0.00 of them is on such a lattice (control 0.06, 36 peaks). This is the reciprocal-space form of the statement that Tile$(1, 1)$ is not on the lattice. It is one patch and one window, no exponent is claimed, and it does not bear on the pure-point theorem. For the Spectre the control fraction exceeds the measured one, so this statistic cannot by itself show the absence of a lattice; that statement rests on the vertices spanning the rank-four module $\mathbb{Z}[\zeta_{12}]$ (Fact 15).

11 What the substitution hierarchy supplies

Whether the level index of a substitution can be read as an arrow, a record, a look-back or a clock is a question that any physical reading of these tilings would need answered 12. Four readings are tested on finite patches against controls. The programs are in the `spectre` repository, built on `hierarchy_cache.py`, which walks the substitution tree and keeps each tile's address; the numbers are those they print at build time, and distances are in tile scales ℓ.

Clustering. Barbour, Koslowski and Mercati [18] use a scale-invariant clustering measure of equal-mass points, $C_S = \ell_{\mathrm{rms}}/\ell_{\mathrm{mhl}}$, to identify an arrow of time in the gravitational N-body problem. For the tile centroids of the level-n patch C_S is 0.360, 0.539, 0.604, 0.626 for $n = 1, \ldots, 4$, but the rise is the footprint: against a triangular lattice cut to the same outline the excess is 0.028 at $n = 1$ and at most 0.001 for $n = 2, 3, 4$. Positional clustering carries no arrow.

Records. How large a neighbourhood must a tile see to name its level-k ancestor? By leave-one-out prediction from the radius-r neighbourhood, the level-1 ancestor is determined at 3ℓ (error 0.000, against 0.151 at 2ℓ), while the level-2 ancestor is still wrong in 0.446 of cases at 4ℓ and 0.072 at 8ℓ; at full size the radius grows by about a factor of three per level. Finer levels are free and coarser ones cost a patch three times larger, an asymmetry that is an arrow only if an observer's patch grows in one direction, which the tiling does not supply.

Look-back. Let h be the level of the lowest common ancestor of two tiles at distance d (Figure 8). If depth were look-back time, $\langle h \rangle$ would grow as $\log_\lambda d$; it does, with slope 0.92 (0.89 for a quadtree with $\lambda = 2$ through the same code), below 1 because h is capped. For tiles 1 to 2ℓ apart, $\Pr(h \geq k+1)/\Pr(h \geq k)$ is 0.483, 0.518 for $k = 2, 3$ in the Spectre and 0.525, 0.504 for $k = 3, 4$ in the quadtree. The outline dimension $\ln \nu / \ln \lambda = 1.3993$ would give $\nu/\lambda^2 = 0.538$ and the quadtree $1/2$; the data do not separate them, but an exact count does. For tiles sharing an edge, the pairs whose lowest common ancestor has level j are the edges between different children of the level-j supertiles. Their number in a level-N supertile is $c_O G_O(j) + c_G G_G(j)$, where c_O, c_G count its level-j supertiles of the two shapes and $G_O(j) = \frac{1}{2}\big(7P(j-1) + Q(j-1) - P(j)\big)$, $G_G(j) = \frac{1}{2}\big(6P(j-1) + Q(j-1) - Q(j)\big)$ are the glue of (2) for each shape, which Theorem 1 gives exactly; the Mystic is taken as one tile, so the four edges between its halves have $h = 0$ (taking them at $h = 1$ changes only the first ratio below, to 0.368). As $N \to \infty$, $\Pr(h \geq k+1)/\Pr(h \geq k)$ is 0.394, 0.498, 0.528, 0.536 for $k = 1, \ldots, 4$ and tends to ν/λ^2. So the tail ratio is ν/λ^2, not $1/2$. The dictionary holds on average and is shared by any self-similar hierarchy, and spatial neighbours can be several levels apart in it.

Parity. The common determinant of the placements in a depth-n patch alternates in sign with n (Proposition 1), so the construction mirrors at every level. Were that a clock, reflection-odd statistics would alternate with level (Figure 9). For tile pairs they do not: with θ_o, θ_t the orientations of two tiles o and t, after both are reflected to a common handedness, and ϕ the direction from o to t, the mean of $\sin(\theta_o + \theta_t - 2\phi)$, verified to change sign under a mirror image, for pairs 1 to 3ℓ apart is 0.25 when the common ancestor is the level-1 supertile and -0.12, -0.14, -0.15 for levels $2, 3, 4$, and the swirl $\langle \sin(\theta - \phi) \rangle$ about a level-k supertile centre, with ϕ now the direction of the tile from the centre, is negative at every level (-0.43, -0.21, -0.15

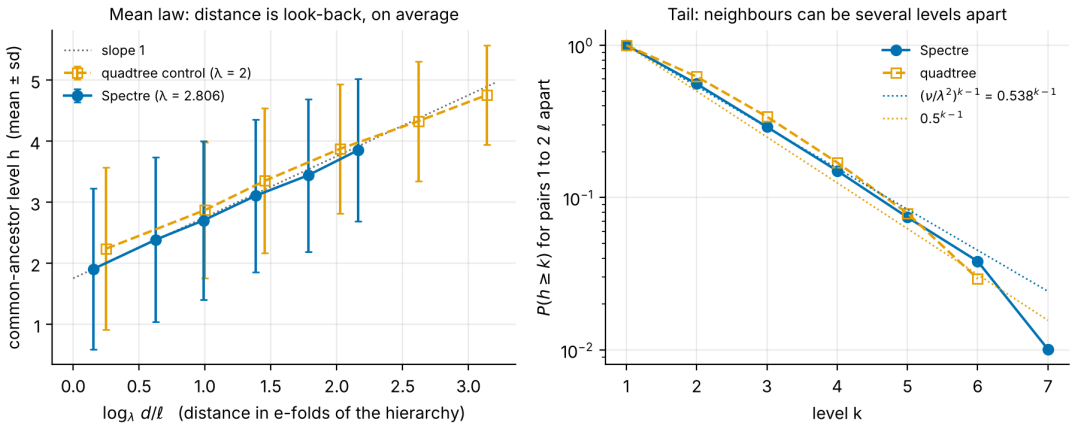


Figure 8: Left: mean common-ancestor level against $\log_\lambda d/\ell$ (with standard deviation), Spectre and quadtree control. Right: the tail $\Pr(h \geq k)$ for pairs 1 to 2ℓ apart, with 0.538^{k-1} and 0.5^{k-1}.

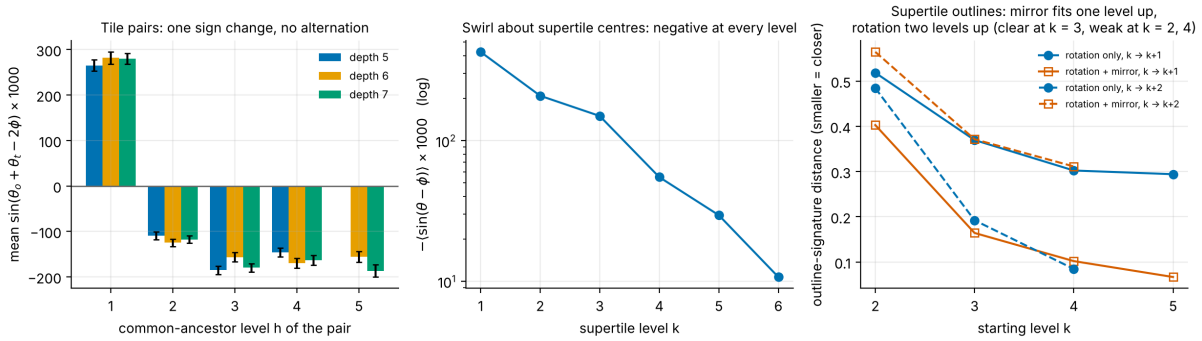


Figure 9: (a) The reflection-odd pair statistic by common-ancestor level, depths 5 to 7. (b) The swirl of tile orientations about supertile centres. (c) Outline-signature distance between same-type supertiles one and two levels apart, by rotation only and with reflection.

for $k = 1, 2, 3$). In outlines the flip is visible: a level-k supertile outline is nearer to the mirror image of the same-type level-$(k+1)$ one than to a rotation in 24 of 24 comparisons, and nearer to a rotation of the level-$(k+2)$ one in 15 of 15, though the rotation distance is below 0.8 times the reflection distance only for $k = 3$ (0.19 against 0.37; for $k = 2$ the two are within that margin). The alternation lives in the shapes of supertiles and not in the tiles; the mirror image of a Spectre tiling is a Spectre tiling, so it belongs to the substitution as written and is not a dynamics.

What follows. Positional clustering shows no excess over its control beyond the first level; the record asymmetry and the look-back dictionary are real but generic; the parity alternation comes from the rule and not from the tile data. None supplies a rule for what propagates, and none supplies a physical clock; what that would mean physically is discussed in 12. (Figures for the clustering and record computations are in `paper/figures/spectre_hier_clustering` and `spectre_hier_records`.)

12 Discussion: The Big Bang Pancake Conjecture

This conjecture about the initial state of the universe is stated so that it can be argued with: the Big Bang was a flat two-dimensional singularity, connected to the planar Spectre monotile [2], and the emergence of three-dimensional space is connected to Tsiokos's Chair44 [3]. Two classical background facts, a reading through the inflation arithmetic of the two tilings, and a proposal on the Weyl tensor are given. Nothing is derived; what would support or retire the reading is stated.

Conjecture 1 (The initial surface)**.** *The Big Bang was not a point-like singularity but a flat two-dimensional one. That flat surface is connected to the Spectre monotile, and the emergence of three-dimensional space from it is connected to the Chair44.*

Two background facts are kept apart from the conjecture. The first is classical and exact. An infinite uniform flat sheet of surface density σ produces a gravitational field $g = 2\pi G\sigma$ that does not depend on the distance from the sheet, because the mass in view grows as r^2 exactly as the force per unit mass falls as r^{-2} (equivalently, by Gauss's law on a pillbox). Fields of parallel sheets are uniform and superpose, so sheets of equal and opposite density cancel outside, and two equal sheets cancel between them. The second concerns the setting. Ghost-free bimetric gravity of Hassan and Rosen [21] gives dynamics to a second metric, and a pre-geometric origin for it has been proposed [22]; a picture in which the initial state consists of two sheets, one for each metric, and the Big Bang is their separation is the kind of reading that such a setting invites. No such model is built here, and no claim is made about the dual-sheet cosmologies themselves. The earlier observation that the area law $S_{\mathrm{rel}} = \delta A/4G$ is consistent with the s-wave dynamics of the Nariai horizon [23], the converse of the result of Dorau and Much [24], is the setting in which the question arose.

What the computations of this paper offer is a precise form of the contrast the conjecture needs. A planar tile with a unimodular inflation, here the Spectre, has an internal space of dimension two and a window; a tile with an integer inflation, here the Chair44, has none and is 2-adic. The conjecture would then say that the surface is of the first kind and the bulk of the second, and that the passage from one to the other is the emergence of the third dimension. That is a reading and not a result: no map from tile data to a physical quantity is given, the dichotomy is not derived from any dynamics, and the passage between the two kinds of inflation is not constructed. The hierarchy section of 8.5 tried the substitution hierarchy as that quantity, and it does not supply the map. The reading would be supported by a physical quantity computed from the tile data that distinguishes the two cases, and retired by showing that the dichotomy is an artefact of the two examples, for instance by a three-dimensional tile with a unimodular inflation. A planar counterexample of the other kind already exists: the hexagonal monotile of Socolar and Taylor [19] inflates by 2, and its tilings are model sets with a 2-adic internal space [20]. Its matching rules are carried by decorations, or by a shape that is not a connected region, so the dichotomy can survive it only as a statement about connected tiles whose shape alone forces aperiodicity; this narrowing is part of the reading, not a consequence of it.

Remark 2 (A proposal on the Weyl tensor). In three spacetime dimensions the Weyl tensor vanishes identically, the Riemann tensor is fixed by the Ricci tensor, and vacuum general relativity has no local gravitational degrees of freedom: gravitational waves need a third spatial dimension. That is classical. The *proposal*, the first author's and not a derived result, is that this is the form

in which a flat two-dimensional initial surface would have vanishing Weyl curvature (compare Penrose's Weyl curvature hypothesis, that the Weyl tensor vanishes at the initial singularity): a 2+1 substrate has no Weyl tensor to set to zero, and the Weyl degrees of freedom would first appear with the third spatial dimension, the step the conjecture assigns to the Chair44. Nothing here derives it. No map from tile data to a metric is given; it is not shown that a 2+1 substrate can acquire 3+1 Weyl degrees of freedom as space emerges; and the hierarchy section of 8.5 leaves open what the hierarchy depth is: if it is the third spatial dimension, a separate time is needed, and if it is look-back time, gravitational waves need a propagation rule that the tiling does not supply.

13 Conclusion

The two tilings separate cleanly in the quantities computed: the Spectre by a unit of $\mathbb{Z}[\sqrt{15}]$, a perimeter growing by a unit of $\mathbb{Q}(\sqrt{5})$ at every order, and an exact internal plane; the Chair44 by inflation 2 on $\mathbb{Z}^3$, no Euclidean internal space, no twist, an explicit pure-point centre set, notch sums $2^n(1,1,1)$ and face contacts that are not bipartite. The main new result is the hull theorem and the classification of the symmetry groups of Chair44 tilings in Section 8.5; for tilings by the solid it is conditional on one quoted, unrefereed input, Tsiokos's registration theorem (R), and it rests on the finite computations of Lemmas 1 and 2. The hierarchy tests of Section 11 are negative or generic.

What the Facts do not reach is as follows. The Spectre results take the published substitution rule (matrix and placement transforms) as input, and the Chair44 results Tsiokos's solid and substitution. The einstein property and aperiodicity of the Chair44 rest on Tsiokos's claimed proof, which we quote and have not refereed [3]; Section 8.5 proves the absence of periods only given his registration theorem. The hull theorem and the symmetry classification of Section 8.5 are proved for tilings obeying the 44 contact rules, from Kamenetsky's unique-hierarchy criterion (Lemma 1, its finite certificate checked independently) and from the count of admissible blocks of Lemma 2, which the build of this paper does not repeat but which three further implementations, described after the lemma, reproduce; for tilings by the solid they rest in addition on Tsiokos's registration theorem, quoted, audited by the second author (its written steps checked, all of its finite inputs recomputed), and not refereed. The counts of legal blocks and fixed seeds are recomputed. That the Spectre diffraction is pure point is a published theorem and is quoted. The perimeter recurrence of the Spectre, its closed form, the growth bound and the boundary exponent are proved for every order by the finite certificate of Theorem 1, which is independent code. The box dimension of the limiting boundary, Theorem 2, uses in addition a Spectre tiling of the whole plane with the construction's hierarchy, obtained in its proof as a limit of supertiles. The twist of the Spectre supertile is a limiting statement. The closed form of the orientation constant is identified and not proved, the window theorem (Theorem 3) gives the dimension of the seams and only a lower bound for the boundary, whose upper bound would need a covering of internal space by windows that is not proved here, the weighted diffraction of the chair is only numerically consistent with pure point, and the Hat and Turtle diffraction is one patch with one window. The boundary operator governs the perimeter in the aggregate at every order; resolved by species and edge index, on the orders counted, a few components obey neither it nor its obvious extensions, while their ratios still approach ν, and the full fine operator is not claimed. The hierarchy results are numerics on finite patches, with controls, and are not Facts.

One further problem remains open: a four-dimensional analogue of the chair, Kamenetsky's

Problem 9.1 [11]. An independent implementation of his closure test, which reproduces his three-dimensional count (exactly 3 of the 2187 homochiral assignments are coarsening-closed with at most 300 contact poses, Kamenetsky's cap, with $30 \to 44 \to 44$ contacts), was run on 12,784 further four-dimensional assignments. Half of them extend the three-dimensional marking on a facet, and half are random. The test stops at 400 contact poses, and every one of them reached that cap, so none is coarsening-closed with fewer than 400 poses. A simulated-annealing search on Kamenetsky's statistic $|\Sigma(P_0)|$ made 12,132 further evaluations. It got from his random-start plateau of 1566 down to 273, which is still above the value 158 of his structured families, and it found no closed assignment; the question stays open. The search code and its logs are in `nd_search/`; they are not part of this paper's build and were not rerun for it.

Part II
Quantum geometry on the $SU(2)$ ladder

14 Introduction to the quantum geometry part

Loop quantum gravity describes quantum geometry on the Hilbert space of spin networks: graphs whose links carry spins of $SU(2)$, and whose area and volume operators have discrete spectra [42, 49]. The Hamiltonian formulation of $SU(2)$ lattice gauge theory [45] lives on the same space on a fixed graph, and the square of the electric field on a link is the Casimir $j(j+1)$ of its spin, the number whose square root the area operator takes. The fork from which this part is generated (the repository `openai_math_yang_mills`, called the fork below) has already solved two small cases of that lattice theory exactly: one plaquette, and two plaquettes sharing a link, whose gauge-invariant states are spin networks on the theta graph, with seven links and two trivalent vertices. This part reads the second system, the *ladder*, as the kinematics of loop quantum gravity, writes the dictionary down, and checks it. Four questions organise it.

- **What does the geometry of the ladder's ground state look like?** The surface that separates the two plaquettes cuts only the shared link, so its mean-square area is the ground-state value of $c_2(J)$. It is $2/(3g^8)$ at strong coupling and $3/(\sqrt{10}\,g^2)$ at weak coupling, both exact, and both checked against numerical ground states (Section 18).
- **What does counting punctures fix?** For a horizon pierced by N links of spin $\frac{1}{2}$, the entropy $S = A/4\ell_P^2$ is matched by the count of invariant states if the Immirzi parameter is $\ln 2/(\pi\sqrt{3})$. We derive this from the library's own entropy and Planck length, and keep it as a property of that count, not of the area spectrum (Section 20).
- **What is the smallest vertex with a volume of its own?** Four spin-$\frac{1}{2}$ legs have a two-dimensional space of intertwiners, and the triple-product operator on it has spectrum $\pm\sqrt{3}/4$: one volume, two orientations (Section 17).
- **Which gravity entry of the release speaks this part's language?** Entry 348, on Einstein four-manifolds, is a statement about the curvature operator of $\mathrm{Spin}(4) = SU(2)_+ \times SU(2)_-$, whose Weyl blocks are spin-2 objects of the two $SU(2)$s. We compute those blocks for S^4, $\mathbb{CP}^2$ and $S^2 \times S^2$ from their metrics, check the topology and the numbers the papers use, and state what corresponds to a spin network and what does not (Section 21).

Every term of every displayed row is the source equation solved for the row's quantity, substituted back and required to give an identity, and every number is checked: 182 checks over the quantum gravity model, 204 over the 13 models that this build runs, and 91 more over the figures pass in the build, and a failure stops it. The checks test that stated formulas agree with one another and with independent computation (Weyl's integration formula against numerical Haar quadrature, exact perturbation theory against numerical ground states, explicit matrices against exact eigenvalues); they do not re-prove any theorem or establish any physics.

What this part is not. It is not a model of quantum gravity. The graph is fixed, the Hamiltonian is that of a gauge theory and not the Hamiltonian constraint of gravity, nothing is invariant under diffeomorphisms, there is no volume operator beyond the triple product on four-valent intertwiner spaces of Section 17, and there is no semiclassical or continuum limit. What the ladder supplies is a Hilbert space of spin networks and an operator that measures the flux through a link, and what is checked is what that gives.

15 The library, conjoined

Every statement about quantum gravity ends in the same few constants: G, $\hbar$, c, k_B, and the Planck length built from the first three. Equations that state the same quantity can be set equal once each is solved for it. Figure 10 does this for 7 quantities at once, among them the gravitational constant, which appears in Newton's law, the Schwarzschild radius, the Bekenstein–Hawking entropy and the Planck length. The equations are Mass-energy equivalence, Planck relation, Kinetic energy, Thermal energy, Mass gap in physical units, Newton's second law, Newton's law of gravitation, Coulomb's law, Hooke's law, Schwarzschild radius, Bekenstein-Hawking entropy, Planck length, Escape velocity, Boltzmann entropy, de Broglie relation, Compton wavelength, Euclidean correlator decay.

16 Quantum geometry in the library

The knowledge graph gains five equations for the geometry of a spin network, and four quantities to read them with (spin, Casimir eigenvalue, Immirzi parameter, puncture count): the Casimir, the area spectrum, the area of N spin-$\frac{1}{2}$ punctures, the horizon area, and the leading count of puncture microstates. Figure 11 conjoins them with the library's entropy and Planck length. The equations are LQG area spectrum, Area of N spin-1/2 punctures, Horizon area, Bekenstein-Hawking entropy, Spin Casimir, Spin-1/2 puncture microstates, Boltzmann entropy.

The area of a surface is a sum over the links that pierce it. A single link of spin j adds an amount that is a function of the Casimir alone.
The area spectrum of a spin-network surface [42, 49].

$$\begin{aligned} A(\{j_i\}) &= 8\pi\gamma_I\,\ell_P^2 \sum_i \sqrt{j_i(j_i+1)}, \qquad A_{\min} = 4\sqrt{3}\,\pi\gamma_I\ell_P^2 \;\; (j=\tfrac{1}{2}) \\ \sqrt{j(j+1)} &= \left(j+\tfrac{1}{2}\right) - \frac{1}{8j} + O(j^{-2}), \qquad \sqrt{(j+\tfrac{1}{2})(j+\tfrac{3}{2})} - \sqrt{j(j+1)} \;\xrightarrow{j\to\infty}\; \tfrac{1}{2} \end{aligned} \tag{5}$$

6 checks passed (`models/qg_spin_network.py`).

Two properties of the spectrum matter below. It has a smallest nonzero value, and it is not equally spaced: the step from one allowed spin to the next is 0.548 in units of $8\pi\gamma_I\ell_P^2$ from $j=\frac{1}{2}$

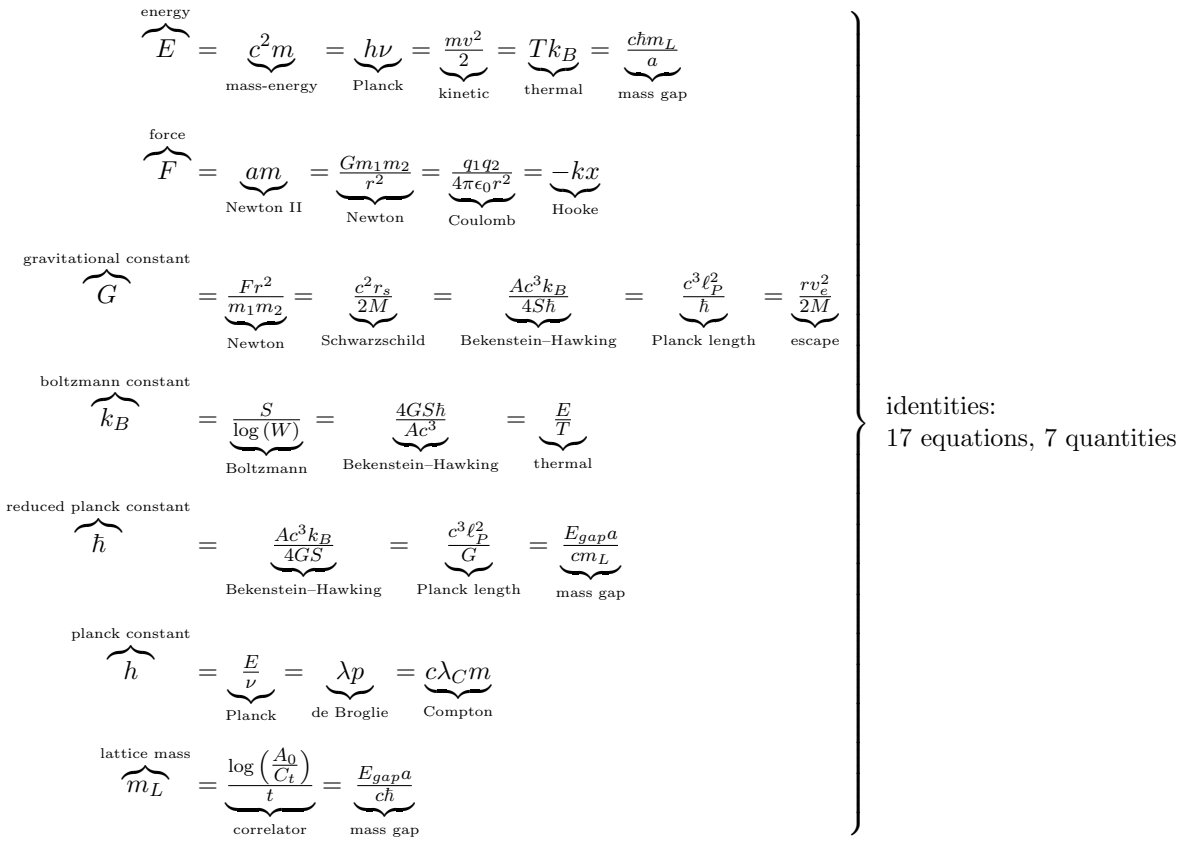


Figure 10: The physics of the library, conjoined. Each row is one quantity, overbraced with its name; each term is a library equation solved for it, underbraced with the equation's name. The letters are those of the equations, so an equals sign says that each equation states the same quantity, not that its other letters take one common value.

$$\overbrace{A}^{\text{area}} = \underbrace{8\pi\sqrt{c_2}\ell_P^2\gamma_I}_{\text{LQG area}} = \underbrace{4\sqrt{3}\pi N\ell_P^2\gamma_I}_{N\text{ punctures}} = \underbrace{4\pi r_s^2}_{\text{horizon}} = \underbrace{\frac{4GS\hbar}{c^3 k_B}}_{\text{Bekenstein–Hawking}}$$

$$\overbrace{c_2}^{\text{casimir eigenvalue}} = \underbrace{j\,(j+1)}_{\text{Casimir}} = \underbrace{\frac{A^2}{64\pi^2\ell_P^4\gamma_I^2}}_{\text{LQG area}}$$

$$\overbrace{W}^{\text{microstate count}} = \underbrace{2^N}_{2^N\text{ states}} = \underbrace{e^{\frac{S}{k_B}}}_{\text{Boltzmann}}$$

$$\overbrace{S}^{\text{entropy}} = \underbrace{\frac{Ac^3k_B}{4G\hbar}}_{\text{Bekenstein–Hawking}} = \underbrace{k_B\log(W)}_{\text{Boltzmann}}$$

$$\overbrace{A_{\min}}^{\text{one puncture}} = \underbrace{8\pi\gamma_I\ell_P^2\sqrt{\tfrac{3}{4}}}_{\text{LQG area, } c_2=3/4} = \underbrace{4\sqrt{3}\,\pi\gamma_I\ell_P^2}_{N\,=\,1\text{ puncture}}$$

$$\overbrace{\gamma_I}^{\text{spin-1/2 count only}} = \underbrace{\frac{\ln 2}{\pi\sqrt{3}}}_{S/k_B\,=\,\sqrt{3}\pi\gamma_I N\,=\,N\ln 2} \approx 0.1274$$

identities:
7 equations, 6 quantities
correspondences:
4 links

$$\underbrace{c_2 = j(j+1)}_{\text{Casimir}} \overset{\text{defines}}{\longrightarrow} \underbrace{A = 8\pi\gamma_I\ell_P^2\sqrt{c_2}}_{\text{LQG area}}$$

$$\underbrace{A = 4\sqrt{3}\pi N\gamma_I\ell_P^2}_{N\text{ punctures}} \overset{\text{specialises}}{\longrightarrow} \underbrace{A = 8\pi\gamma_I\ell_P^2\sqrt{c_2}}_{\text{LQG area}}$$

$$\underbrace{A = 8\pi\gamma_I\ell_P^2\sqrt{c_2}}_{\text{LQG area}} \overset{\text{same field}}{\longleftrightarrow} \underbrace{S = \frac{k_Bc^3A}{4G\hbar}}_{\text{Bekenstein–Hawking}}$$

$$\underbrace{W = 2^N}_{2^N\text{ states}} \overset{\text{same field}}{\longleftrightarrow} \underbrace{S = k_B\log W}_{\text{Boltzmann}}$$

Figure 11: Quantum geometry in the library, conjoined. Each identity row is one quantity (the area of a surface, the Casimir it is a function of, the number of microstates, the entropy) stated by several library equations, each solved for it. The equals sign says that each equation states the same quantity, not that the other letters take a common value; in the microstate row the 2^N is the leading exponential of the exact count, which is a Catalan number (Section 17). The two rows below are exact numbers, rederived from the library's own entropy and Planck length: G, $\hbar$ and c cancel, leaving $S/k_B = \sqrt{3}\pi\gamma_I N$. The arrows are links of the library, not equations: the Casimir defines the area, the N-puncture area specialises it, and the area spectrum and the entropy state the same area only once a counting fixes γ_I, which depends on which spins are counted (Section 20).

to 1, 0.522 from 1 to $\frac{3}{2}$, and falls to $\frac{1}{2}$ (Figure 13a). The Immirzi parameter γ_I is a free number in the spectrum.

The same Casimir is the electric energy of a link in the lattice gauge theory, which is what lets the ladder serve as a model. The next display states that exactly, link by link.
The link Casimir as a squared area, and the ladder's electric energy [45].

$$E_l^2 = c_2(j_l) = \Big(\frac{A_l}{8\pi\gamma_I \ell_P^2}\Big)^2$$
$$\sum_{l=1}^{7} E_l^2 = 3c_2(j_a) + 3c_2(j_b) + c_2(J) = \sum_{l=1}^{7}\Big(\frac{A_l}{8\pi\gamma_I \ell_P^2}\Big)^2 \tag{6}$$

3 checks passed (`models/qg_spin_network.py`).

This is a relation between two operators on one Hilbert space. It does not make the ladder's dynamics gravitational: the Hamiltonian constraint of loop quantum gravity is a different operator, and nothing in the ladder's Hamiltonian is built from areas in that way.

17 Intertwiners and the quantum tetrahedron

A vertex of a spin network carries an intertwiner: an invariant vector in the tensor product of the spins that meet there. The dimension of that space is the Haar integral of the product of characters, which Weyl's integration formula makes an exact integer. At a trivalent vertex it is 1 or 0 according to the triangle rule, which is the rule that selects the ladder's states and is why they are labelled by link spins alone. At a four-valent vertex it counts the intermediate spins, and from there on a vertex has quantum numbers of its own.
Intertwiner spaces: the dimension of the invariants of a vertex [48].

$$\dim \mathrm{Inv}\Big(\bigotimes_{i=1}^{n} V_{j_i}\Big) = \int_{SU(2)} \prod_{i=1}^{n} \chi_{j_i}(U)\, dU = \big[z^0\big]\big(1-z^2\big)\prod_i \chi_{j_i}(z)$$
$$n=3:\quad \dim = \begin{cases} 1 & \triangle(j_1,j_2,j_3) \\ 0 & \text{otherwise} \end{cases} \qquad n=4:\quad \dim = \#\{J:\ \triangle(j_1,j_2,J),\ \triangle(J,j_3,j_4)\} \tag{7}$$
$$j_i=\tfrac{1}{2},\ n=N:\quad \dim = C_{N/2} = \frac{1}{N/2+1}\binom{N}{N/2}\ :\ 1,\ 2,\ 5,\ 14,\ 42,\ \ldots$$

6 checks passed (`models/qg_spin_network.py`).

For four spin-$\frac{1}{2}$ legs the space is two-dimensional, and on it the triple product of three of the spins is a Hermitian operator that commutes with the total spin. It is the squared volume of the quantum tetrahedron [48] up to a constant, and its spectrum is computed here from explicit 16×16 matrices.
The four-valent vertex of four spin-1/2 as a quantum tetrahedron [48].

$$Q = \epsilon_{abc}\, J_a^{(1)} J_b^{(2)} J_c^{(3)}\Big|_{\mathrm{Inv}(V_{1/2}^{\otimes 4})}, \qquad \dim \mathrm{Inv} = 2, \qquad \mathrm{spec}\, Q = \Big\{-\tfrac{\sqrt{3}}{4}, +\tfrac{\sqrt{3}}{4}\Big\}, \qquad V \ \propto\ \sqrt{|Q|} \tag{8}$$

4 checks passed (`models/qg_spin_network.py`).

The prefactor that turns Q into a volume is not computed: the statement made is that the smallest vertex with quantum numbers of its own has a single nonzero volume squared, $\sqrt{3}/4$

in the units of Q, with two orientations. The same operator is built for any four spins. In the coupled basis it is $i(x_K - x_{K'})$ times an orthogonal matrix of $6j$ symbols (the spectrum does not depend on a phase that depends on one intermediate spin alone), and it agrees with the explicit-matrix construction to 4×10^{-15} for eight sets of spins.
Triple-product (volume) operator on a four-valent intertwiner space.

$$\begin{gathered} Q = \epsilon_{abc} J_{1a} J_{2b} J_{3c} = i\,[\,J_1 \cdot J_2,\ J_2 \cdot J_3\,] \ \text{ on } \ \mathrm{Inv}(V_{j_1} \otimes V_{j_2} \otimes V_{j_3} \otimes V_{j_4}), \qquad [Q, J_{\mathrm{tot}}] = 0, \ \ Q^\dagger = Q, \ \ \operatorname{spec} Q = -\operatorname{spec} Q \\ \dim \mathrm{Inv} = \#\{K:\ \triangle(j_1, j_2, K),\ \triangle(K, j_3, j_4)\}, \qquad V \propto \sqrt{|Q|} \end{gathered} \tag{9}$$

3 checks passed (`models/qg_volume.py`).
Exact spectra of the triple product for four equal and mixed spins.

$$\begin{gathered} \operatorname{spec} Q: \quad (\tfrac12)^4:\ \pm\tfrac{\sqrt3}{4}; \quad (1)^4:\ 0, \pm\sqrt3; \quad (\tfrac32)^4:\ \pm\tfrac{3\sqrt3}{4},\ \pm\tfrac{3\sqrt{35}}{4}; \quad (1,1,\tfrac12,\tfrac12):\ \pm\tfrac{1}{\sqrt2} \\ \operatorname{Tr} Q = \operatorname{Tr} Q^3 = 0, \qquad \operatorname{Tr} Q^2 = \tfrac38,\ 6,\ \tfrac{171}{4},\ 198 \quad \left(j = \tfrac12, 1, \tfrac32, 2\right) \\ Q_{KK'} = i\,(x_K - x_{K'}) \sum_L U_{KL}\, y_L\, U_{K'L}, \ \ U_{KL} = \sqrt{(2K+1)(2L+1)} \begin{Bmatrix} j_1 & j_2 & K \\ j_3 & j_4 & L \end{Bmatrix} \end{gathered} \tag{10}$$

3 checks passed (`models/qg_volume.py`).

The spectra are symmetric under $Q \to -Q$, and vanish when a leg has spin 0, when the vertex is trivalent. For four spin-1 legs there is a state of zero volume, as the dimension, three, is odd. The largest eigenvalue, against the triple product of three vectors of length $\sqrt{j(j+1)}$ of the regular tetrahedron, rises from 0.794 at $j = \frac32$ to 0.921 at $j = 8$ and stays below one: a trend, not a proof of a semiclassical limit.
Largest volume of four equal spins against the regular tetrahedron.

$$\frac{\lambda_{\max}(Q)}{\frac{4}{3\sqrt3}\,[j(j+1)]^{3/2}} = 0.866,\ 0.796,\ 0.794,\ 0.808,\ 0.840,\ 0.867,\ 0.921 \ \ \left(j = \tfrac12, 1, \tfrac32, 2, 3, 4, 8\right) \ \text{increasing, below 1} \tag{11}$$

1 checks passed (`models/qg_volume.py`).

Where the four-valent vertex is. After gauge fixing, every vertex of the strip of the next section is trivalent, so the strip carries no volume. A four-valent lattice vertex first appears for three plaquettes in an L shape (a at the origin, b to its right, c above), at the corner that all three share: its four links are the one shared by a and b, the top of b, the one shared by a and c, and the right side of c. At strong coupling the ground state is the electric vacuum dressed by strings of spin $\frac12$ around plaquettes. Only a string around b together with one around c puts spin $\frac12$ on all four legs, with the two loops each contracting a pair of legs, and on that vector $Q^2 = \frac{3}{16}$. Its amplitude is the second-order value $4/(9g^8)$, so that $\langle Q^2 \rangle_0 = 1/(27g^{16})$ at leading order.
The four-valent vertex of the L-shaped cluster at strong coupling.

$$\begin{gathered} \langle Q^2 \rangle_0 = \frac{3}{16}\,|\psi_{bc}|^2 = \frac{3}{16}\Big(\frac{4}{9\,g^8}\Big)^2 = \frac{1}{27\,g^{16}}\left(1 + O(g^{-8})\right) \quad \text{(L shape, strong coupling)} \\ \psi_{bc} = 2\,\frac{(1/g^2)^2}{(3g^2/2)(3g^2)} = \frac{4}{9g^8}, \quad E_b = \tfrac32 g^2,\ E_{bc} = 3g^2 \end{gathered} \tag{12}$$

5 checks passed (`models/qg_volume.py`).

This estimate was first conditional: the amplitude $4/(9g^8)$ is the one two disjoint strings have, and it was not derived for the L shape. The Hamiltonian of the L-shaped cluster is now

built, and the estimate is tested against it. The basis labels the spins j_a, j_b, j_c, the shared links J_{ab}, J_{ac}, and an intermediate spin K at the four-valent vertex, in the channel $(J_{ab}J_{ac})K$, $(j_bj_c)K$. The plaquette operators are built vertex by vertex from explicit Clebsch–Gordan coefficients, with the recoupling at the four-valent vertex included, and the signs are not guessed: the relation $\chi_{1/2}(U_p)\Psi_s = \sum_{s'}\langle s'|W_p|s\rangle\Psi_{s'}$ is checked pointwise at random group elements for 45 pairs of a state and a plaquette, with the wave functions evaluated by direct tensor contraction.
L-shaped cluster: basis, energies and plaquette operators with a four-valent vertex.

$$
\begin{aligned}
&|j_a, j_b, j_c, J_{ab}, J_{ac}; K\rangle, \quad \triangle(j_a, j_b, J_{ab}),\ \triangle(j_a, j_c, J_{ac}), \quad K:\ (J_{ab}J_{ac})K,\ (j_bj_c)K \\
&H = \frac{g^2}{2}\big[2C(j_a) + 3C(j_b) + 3C(j_c) + C(J_{ab}) + C(J_{ac})\big] + \frac{1}{g^2}\big(6 - W_a - W_b - W_c\big) \\
&\langle s'|W_p|s\rangle = \prod_{l\in p}\sqrt{\tfrac{d_{j_l}}{d_{j'_l}}}\ \sum_{\mu}\ \pm\ \prod_{v\in p} F_v(\mu_v), \qquad F_v = \langle \iota'_v|\, R\otimes R\,|\iota_v\rangle, \quad R_\mu = C\big(j\, m, \tfrac{1}{2}\mu\,|\, j'\, m{+}\mu\big)
\end{aligned}
\tag{13}
$$

7 checks passed (`models/su2_lattice_lshape.py`).

The three plaquette operators are real, symmetric and commute in the interior of the truncation (and fail to at its boundary, as they should); with a plaquette trivial they reduce in absolute value to the ladder's squared-6j elements; the kinetic matrix of the harmonic limit has eigenvalues 4, $4 \pm \sqrt{2}$, those of the strip. The ground-state energy is $6/g^2 - 2/g^6 + O(g^{-10})$ at strong coupling, and the amplitude of the state with strings around b and c is $4/(9g^8)$ (ratio to the numerics tending to 1 at $g^2 = 8, 12, 16$).
The four-valent vertex of the L-shaped cluster: <Q^2> in the ground state.

$$
\begin{aligned}
&\langle Q^2\rangle_0 = \sum_s \big|\psi_s\big|^2\,\langle s|Q^2|s\rangle, \qquad Q = \epsilon_{abc}J_{ab,a}J_{ac,b}J_{bt,c} \\
&g^{16}\langle Q^2\rangle_0 \ \xrightarrow{g\to\infty}\ \frac{1}{27}\ \text{(confirmed, see the numbers below)}, \qquad \psi_{bc} = \frac{4}{9g^8}
\end{aligned}
\tag{14}
$$

5 checks passed (`models/su2_lattice_lshape.py`).

The estimate is confirmed: the ratio $27\,g^{16}\langle Q^2\rangle_0$ is 0.9808, 0.9939, 0.99961 and 0.999976 at $g^2 = 3, 4, 8, 16$. At intermediate coupling $\langle Q^2\rangle_0$ is 1.3×10^{-4}, 1.09×10^{-3} and 1.34×10^{-2} at $g^2 = 2, 1.5, 1$, growing as the coupling falls. The number at $g^2 = 0.8$ was dropped because it is converged only to about 8% in the spin cutoff, and those above to about 2%. The constant that turns Q into a volume is still not fixed, so this is a statement about the triple product, not about a physical volume.
The L-shaped cluster: gap at moderate coupling against the strip and the harmonic limit.

$$
K = \begin{pmatrix} 4 & -1 & -1 \\ -1 & 4 & 0 \\ -1 & 0 & 4 \end{pmatrix}_{(a,b,c)}, \quad \operatorname{spec} K = 4-\sqrt{2},\ 4,\ 4+\sqrt{2}\ \text{(as for the strip)}, \qquad \Delta_{g\to0} = 2\sqrt{2-\tfrac{\sqrt{2}}{2}} \approx 2.2741 \tag{15}
$$

3 checks passed (`models/su2_lattice_lshape.py`).

The lowest excitation is 2.4398 at $g^2 = 1.5$ and 2.0570 at $g^2 = 1$, against the harmonic limit 2.2741; at $g^2 = 1.5$ it equals the strip's to 10^{-3}, so the corner vertex does not show in the gap there. The weak-coupling limit itself is not reached by these numbers (the gap is below the limit at $g^2 = 1$), and no larger cluster or continuum limit was tried.

18 The ladder as a spin network

The two plaquettes sharing a link have seven links. After a spanning-tree gauge fixing, the gauge-invariant states are the spin networks of the theta graph: the three outer links of plaquette a carry one spin j_a, those of b carry j_b, and the shared link carries J, with the triangle rule at both vertices. The Hamiltonian is the Kogut–Susskind Hamiltonian in lattice units, with the plaquette operators given by squared $6j$ symbols.
Two-plaquette ladder: basis, energies and plaquette operators.

$$|j_a, j_b, J\rangle, \quad \triangle(j_a, j_b, J), \quad H = \frac{g^2}{2}\big[3C(j_a) + 3C(j_b) + C(J)\big] + \frac{1}{g^2}\big(4 - W_a - W_b\big), \quad C(j) = j(j+1)$$
$$\langle j_a{+}\epsilon, j_b, J{+}\eta | W_a | j_a, j_b, J\rangle = \sqrt{d_{j_a} d_{j'_a} d_J d_{J'}} \begin{Bmatrix} j_a & \frac{1}{2} & j'_a \\ J' & j_b & J \end{Bmatrix}^2, \quad \epsilon, \eta = \pm\tfrac{1}{2} \tag{16}$$

5 checks passed (`models/su2_lattice_ladder.py`).

A surface that separates the two plaquettes cuts one link, the shared one, so its area is the area spectrum at spin J and its mean square in the ground state is $\langle c_2(J)\rangle_0$ in units of $(8\pi\gamma_I\ell_P^2)^2$. At strong coupling the ground state is the electric vacuum $J = 0$, dressed to first order by the two single-plaquette strings; at weak coupling it spreads over large spins and the ladder is two coupled oscillators, whose virial theorem fixes the coefficient.
The mean-square area of the cut between the two plaquettes, in the ground state of the ladder.

$$\big\langle A^2_{\text{cut}}\big\rangle_0 = \big(8\pi\gamma_I\ell_P^2\big)^2 \big\langle c_2(J)\big\rangle_0, \qquad \big\langle c_2(J)\big\rangle_0 \xrightarrow{g\to\infty} \frac{\frac{2}{3}}{g^8}, \qquad g^2\big\langle c_2(J)\big\rangle_0 \xrightarrow{g\to 0} \frac{3\sqrt{10}}{10}$$
$$g^2\big\langle c_2(j_a)\big\rangle_0 \xrightarrow{g\to 0} \frac{3\sqrt{10}}{40} + \frac{\sqrt{6}}{8} = \tfrac{3}{4}\Big(\tfrac{1}{\sqrt{6}} + \tfrac{1}{\sqrt{10}}\Big) \tag{17}$$

7 checks passed (`models/qg_spin_network.py`).

The two laws meet the numerical curve at both ends (Figure 12a), and the weak-coupling approach is linear in g^2 (Figure 12b). The shared link fluctuates more than a plaquette's own links by the factor 1.746, because it carries the difference of the two loop fields. Only the leading terms are derived; the first correction in g^2 is not.

19 A third plaquette: the strip

Three plaquettes in a row have ten links. The corner vertices again force equal spins on the outer links of the end plaquettes, so the gauge-invariant states are spin networks on a graph of four trivalent vertices and six edges: j_a and j_c for the ends, t and b for the top and bottom of the middle plaquette, and J_1, J_2 for the two shared links. The electric energy is again a sum of squared areas, and the plaquette operators shift every link of the boundary of a plaquette by $\pm\frac{1}{2}$ with a product of one $6j$ symbol per vertex on the loop: two for a and c, four for b. Unlike the ladder's, these $6j$ symbols are not squared, so the signs matter.

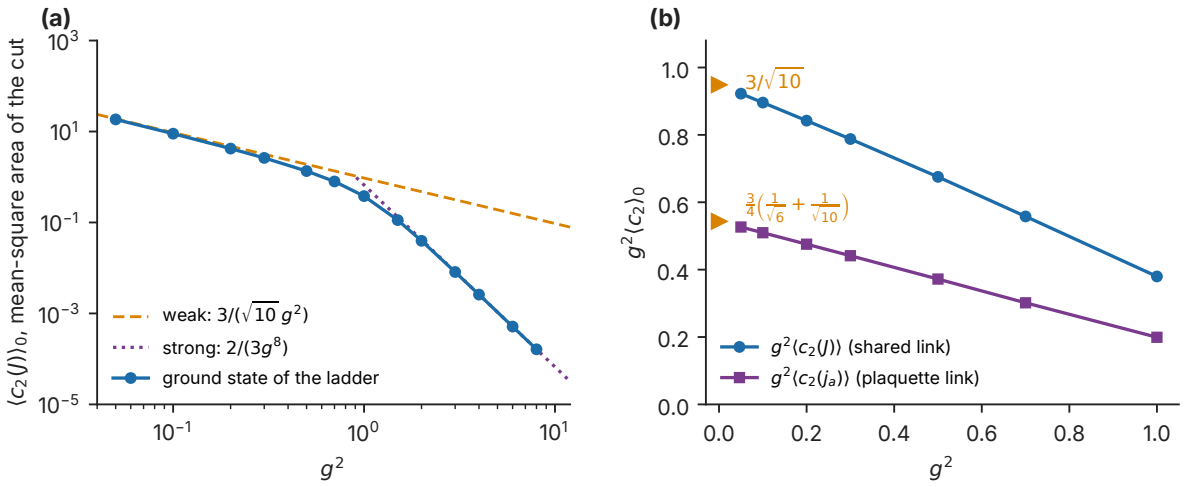


Figure 12: The surface that separates the two plaquettes of the ladder. (a) The mean-square area $\langle c_2(J)\rangle_0$ in the ground state, against g^2 (both axes logarithmic): the numerical ground states, the exact strong-coupling law $2/(3g^8)$, and the weak-coupling law $3/(\sqrt{10}\,g^2)$ from the virial theorem. (b) $g^2\langle c_2(J)\rangle_0$ for the shared link and $g^2\langle c_2(j_a)\rangle_0$ for a link of one plaquette, against g^2: both approach their exact limits (arrows, at $g^2 = 0$) linearly. The ladder's Hamiltonian is a gauge-theory Hamiltonian, not a gravitational one; the cut is a surface of its graph, not a horizon.

Three-plaquette strip: basis, energies and plaquette operators.

$$
\begin{aligned}
&|j_a, j_c, t, b, J_1, J_2\rangle, \quad \triangle(j_a, t, J_1),\ \triangle(t, j_c, J_2),\ \triangle(j_a, b, J_1),\ \triangle(b, j_c, J_2)\\
H &= \frac{g^2}{2}\left[3C(j_a) + 3C(j_c) + C(t) + C(b) + C(J_1) + C(J_2)\right] + \frac{1}{g^2}\left(6 - W_a - W_b - W_c\right)\\
\langle j_a{+}\epsilon, J_1{+}\eta|W_a|j_a, J_1\rangle &= (-1)^{(k_t-k_b)/2}\sqrt{d_{j_a} d_{j'_a} d_{J_1} d_{J'_1}}\begin{Bmatrix} j_a & \frac{1}{2} & j'_a \\ J'_1 & t & J_1 \end{Bmatrix}\begin{Bmatrix} j_a & \frac{1}{2} & j'_a \\ J'_1 & b & J_1 \end{Bmatrix}\\
W_b:\quad & s_b\sqrt{\textstyle\prod d}\begin{Bmatrix} t & \frac{1}{2} & t' \\ J'_1 & j_a & J_1 \end{Bmatrix}\begin{Bmatrix} t & \frac{1}{2} & t' \\ J'_2 & j_c & J_2 \end{Bmatrix}\begin{Bmatrix} b & \frac{1}{2} & b' \\ J'_2 & j_c & J_2 \end{Bmatrix}\begin{Bmatrix} b & \frac{1}{2} & b' \\ J'_1 & j_a & J_1 \end{Bmatrix}
\end{aligned}
\tag{18}
$$

8 checks passed (`models/su2_lattice_strip.py`).

The signs in the display were found by fitting, not derived from a convention: they were fixed against an independent construction from Wigner $3j$ symbols that shares no code with the closed forms, and agree with it on all 5047 transitions checked (states up to spin $\frac{3}{2}$ and a sample up to spin 2) to 4×10^{-16}. The three plaquette operators are symmetric and commute in the interior of the truncated space (6×10^{-16}), while the commutators reach 0.94 at the truncation boundary, as expected, and with the third plaquette trivial the operators are the ladder's, exactly, on 948 transitions.

This is the test that the strip formula of the ladder's harmonic argument lacked. The three loop variables are three-dimensional oscillators with the tridiagonal kinetic matrix of the display; gauge invariance leaves total angular momentum zero, so the lowest excitation is two quanta in the soft mode.

Weak-coupling limit of the three-plaquette strip.

$$\omega_k = \sqrt{2-\cos\tfrac{k\pi}{4}}\ \ (k=1,2,3) = \sqrt{\tfrac{4-\sqrt2}{2}},\ \sqrt2,\ \sqrt{\tfrac{4+\sqrt2}{2}}$$
$$\Delta_{g\to0} = 2\omega_1 = 2\sqrt{2-\tfrac{\sqrt2}{2}} \approx 2.2741, \quad \text{next: } \omega_1{+}\omega_2 \text{ (odd)},\ \ \omega_1{+}\omega_3,\ \ 2\omega_2 \text{ (even)} \tag{19}$$
$$\frac{\Delta_3}{\Delta_{\text{plaquette}}} \to \sqrt{\tfrac12\big(2-\tfrac{\sqrt2}{2}\big)} \approx 0.8040$$

5 checks passed (`models/su2_lattice_strip.py`).

Against this, the lowest even level of the numerical spectrum, extrapolated to $g \to 0$ from $g^2 = 0.2$–0.4 with a cubic, is 2.2749 (2.2741 predicted), and the second is 2.5520 (2.5513 predicted), with the reflection parities of the oscillator modes. The next two even levels mix at finite coupling, and only their sum could be compared (agreement to 1.3×10^{-3}). At strong coupling the first excited level is threefold degenerate, three strings of spin $\frac12$ at $3g^2/2$, and exact second-order perturbation theory splits it.

Strong-coupling splitting of the first excited level of the strip.

$$\Delta_i = \frac32\, g^2 + \frac{\mu_i}{g^6} + O(g^{-14}),\ \ \mu_i = \text{eigenvalues of } M + 2\cdot\mathbb{1},\ \ M = \begin{bmatrix} -\frac{122}{105} & -\frac{2}{21} & 0 \\ -\frac{2}{21} & -\frac{44}{35} & -\frac{2}{21} \\ 0 & -\frac{2}{21} & -\frac{122}{105} \end{bmatrix} \text{ on } (A,B,C) \tag{20}$$

3 checks passed (`models/su2_lattice_strip.py`).

The numerical levels at $g^2 = 6$ and 8 match these to 5×10^{-4} with the right parities. Between the limits the three gaps are ordered.

The strip gap against the ladder and one-plaquette gaps.

$$\Delta_3(g) \le \Delta_2(g) \le \Delta_1(g), \qquad \frac{\Delta_3}{\Delta_1} \to \sqrt{1-\tfrac{\sqrt2}{4}} \approx 0.8040\ (g\to0), \quad \to 1\ (g\to\infty) \tag{21}$$

3 checks passed (`models/su2_lattice_strip.py`).

The ordering holds on the grid $g^2 = 0.25$–8; the ratio to the one-plaquette gap is 0.835 at $g^2 = 0.25$, against the limit 0.8040, and 0.99995 at $g^2 = 8$. The three volumes say nothing about a larger or infinite system, and, like the ladder's, the gap here is that of a gauge-theory Hamiltonian on a small lattice.

Four and five plaquettes. The same graph, a ladder of squares, gives $3N-3$ labels for N plaquettes, and the display gives the Hamiltonian, with a phase σ_p that extends the three-plaquette rule by a factor $(-1)^{(k_t-k_b)/2}$ for each interior neighbour. That extension is a conjecture, and it is tested rather than derived.

N-plaquette strip: general-N spin-network Hamiltonian and its independent check.

$$H_N = \frac{g^2}{2}\sum_\ell C(j_\ell) + \frac{1}{g^2}\sum_{p=1}^{N}\big(2-W_p\big), \quad W_p = \sigma_p\sqrt{\textstyle\prod d}\prod_{v\in p}\begin{Bmatrix} x_v & \frac12 & x_v' \\ y_v' & z_v & y_v \end{Bmatrix}$$
$$\sigma_p = s_p \prod_{q\sim p,\ 1<q<N} (-1)^{(k_{t_q}-k_{b_q})/2} \tag{22}$$

5 checks passed (`models/su2_lattice_stripn.py`).

The test is an independent construction with no $6j$ symbol and no phase: wavefunctions of the N loop holonomies, with the link electric fields from the Peter–Weyl decomposition and

gauge invariance enforced by a penalty. It reproduces the ladder exactly at $N = 2$ and the strip module at $N = 3$. At $g^2 = 2$ the spin-network builder and the independent construction agree on the lowest four levels to 2×10^{-6} at $N = 4$ (ground state 3.67192) and 10^{-3} at $N = 5$ (ground state 4.5899, limited by the cut), and the plaquette operators commute away from the cut. The weak-coupling gap formula $2\sqrt{2 - \cos(\pi/(N+1))}$ (2.1827 at $N = 4$, 2.1297 at $N = 5$) is *not* tested: at $g^2 = 0.8$–0.4 the $N = 4$ gap is still falling with the energy cut (2.84 → 2.49 at $g^2 = 0.4$ between cuts 24 and 32), and $N = 5$ was too slow to converge, so no extrapolation is claimed.

20 Counting punctures and the Immirzi parameter

If a horizon of area A is pierced by N links of spin $\frac{1}{2}$, its area is N times the smallest value, and the number of invariant states is a Catalan number whose logarithm is $N \ln 2$ at leading order. Setting $S = A/4\ell_P^2$ against that count fixes γ_I. The display uses the library's own entropy and Planck length, in which G, $\hbar$ and c cancel [40].

Counting spin-1/2 punctures and the value of the Immirzi parameter [40, 41].

$$W = C_{N/2} \sim \frac{2^N}{\sqrt{\pi}\,(N/2)^{3/2}}, \qquad A = 4\sqrt{3}\,\pi N \gamma_I \ell_P^2, \qquad \frac{S}{k_B} = \frac{c^3 A}{4G\hbar} = \sqrt{3}\,\pi\gamma_I N$$
$$\frac{S}{k_B} = \ln W \simeq N \ln 2 \implies \gamma_I = \frac{\ln 2}{\pi\sqrt{3}} \approx 0.1274 \tag{23}$$

5 checks passed (`models/qg_spin_network.py`).

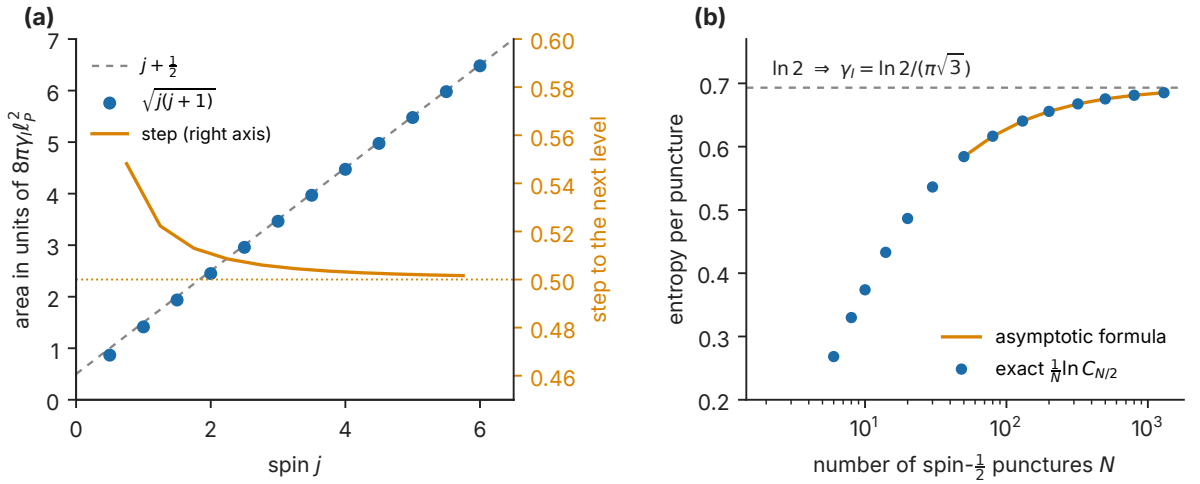


Figure 13: The area spectrum and the count of punctures. (a) The single-link area eigenvalues $\sqrt{j(j+1)}$, in units of $8\pi\gamma_I\ell_P^2$, against j, with the line $j + \frac{1}{2}$ they approach from below, and (right axis) the step between neighbouring levels, which is above $\frac{1}{2}$ and tends to it. (b) The entropy per puncture $\frac{1}{N}\ln C_{N/2}$ of N spin-$\frac{1}{2}$ punctures, from the exact Catalan numbers, with the asymptotic formula of Section 20; it tends to $\ln 2$, which fixes $\gamma_I = \ln 2/(\pi\sqrt{3})$ for this count.

The count is exact (Section 17) and the shortfall of the entropy per puncture below $\ln 2$ is $\frac{3}{2N}\ln\frac{N}{2} + \frac{\ln\pi}{2N} + \frac{9}{4N^2}$ (Figure 13b), still 0.007 at $N = 1600$. The value 0.1274 belongs to this

count: counting all spins with the area fixed gives a different number (about 0.2375 [41], quoted, not rederived here). The area spectrum does not determine γ_I, and neither does this section derive the entropy of a black hole from dynamics: the horizon area is given.

21 Einstein four-manifolds: the geometric $SU(2)$

Entry 348 of the release classifies the closed four-manifolds that are Einstein, Ric $= \lambda g$, and have positive sectional curvature: up to scale, the round S^4, Fubini–Study $\mathbb{CP}^2$ and (for the real projective quotient) $\mathbb{RP}^4$ [32]. Two companion papers carry it to the boundary case, where $S^2 \times S^2$ enters, and to an L^2 gap: a simply connected nonnegatively curved four-manifold whose trace-free Ricci tensor has small enough L^2 energy is diffeomorphic to one of the three [26, 39]. This is the one gravity entry of the release whose language is the language of this part. The frame group of an oriented four-manifold is $\mathrm{Spin}(4) = SU(2)_+ \times SU(2)_-$, the two-forms split as $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$ into the adjoints of the two factors, and the curvature operator for Ric $= 3g$ is $\mathrm{Id} + T$ on Λ^+ and $\mathrm{Id} + U$ on Λ^-, with T and U the two trace-free Weyl blocks [32]. Each block is a symmetric trace-free endomorphism of a three-dimensional space on which one $SU(2)$ acts: a spin-2 object. The link of a spin network carries a spin of one $SU(2)$; the self-dual (Ashtekar) connection of a Euclidean four-manifold is the $SU(2)_+$ part of its $\mathrm{Spin}(4)$ connection. What follows checks the algebra of that statement and the numbers that the paper of that entry [32] and the library use about the three model spaces.
so(4) = su(2) + su(2): the two SU(2)s and the Casimirs of the curvature blocks.

$$\begin{gathered}\Lambda^2 = \Lambda^+ \oplus \Lambda^- = (1,0) \oplus (0,1), \qquad (c_2^+, c_2^-)\Big|_V = \left(\tfrac{3}{4}, \tfrac{3}{4}\right), \quad \Big|_{\Lambda^+} = (2,0), \quad \Big|_{\Lambda^-} = (0,2) \\ \mathrm{Sym}^2_0\Lambda^\pm = (2,0) \oplus (0,2) \ \ (\dim 5) \text{ each}, \qquad c_2 = j(j+1)\big|_{j=2} = 6 \\ \Sigma = \sigma_+ + \sigma_- \text{ simple} \implies |\sigma_+| = |\sigma_-|\end{gathered} \tag{24}$$

5 checks passed (`models/qg_einstein4.py`).

The Casimirs are computed from explicit 4×4 generators. On the vector they are $(\frac{3}{4}, \frac{3}{4})$, on Λ^+ they are $(2,0)$, and on a Weyl block, the five-dimensional trace-free part of the symmetric square of Λ^+, the Casimir is $6 = j(j+1)$ at $j = 2$: the same function of the spin as in the area spectrum (Section 16), evaluated at spin 2. The last line of the display is the algebraic form of a decomposable two-plane having equal self-dual and anti-self-dual parts, which is also why the sectional curvature of a plane is a mean of one value from each block. In spin-foam models the corresponding statement about a bivector (the simplicity constraint) is what ties the two spins to one spin through the Immirzi parameter [46]; that is context here, and nothing in it is computed.

The curvature blocks of the three spaces are computed from their metrics in a chart, by differentiating twice, and not from a formula for the curvature of the model.
Curvature blocks of the three model Einstein four-manifolds.

$$\begin{gathered}\mathcal{R} = \begin{pmatrix} \mathrm{Id} + T & 0 \\ 0 & \mathrm{Id} + U \end{pmatrix}, \qquad \mathrm{Ric} = 3g \\ S^4: \ (T,U) = (0,0); \qquad \mathbb{CP}^2: \ T \sim \mathrm{diag}(2,-1,-1), \ U = 0; \qquad S^2 \times S^2: \ T \sim U \sim \mathrm{diag}(2,-1,-1) \\ |T|^2 = 0,\ 6,\ 6; \quad |U|^2 = 0,\ 0,\ 6; \qquad K_{\min} = 1 - \tfrac{x+y}{2} = 1,\ \tfrac{1}{2},\ 0, \qquad x = -\lambda_{\min}(T),\ y = -\lambda_{\min}(U)\end{gathered} \tag{25}$$

7 checks passed (`models/qg_einstein4.py`).

The mixed block between Λ^+ and Λ^- vanishes for the three Einstein spaces and not for the unbalanced product $S^2(K{=}3)\times S^2(K{=}1)$, whose Ricci tensor is $\mathrm{diag}(3,3,1,1)$: that block is the trace-free Ricci tensor, and its vanishing is the Einstein equation. The least sectional curvature is $1-(x+y)/2$, where x and y are the depths of the two blocks, and the condition $x+y<2$ of that paper is strict sectional positivity: S^4 (0) and $\mathbb{CP}^2$ (1) satisfy it, $S^2\times S^2$ (2) has planes of zero curvature. On every nonzero block of the three spaces $|T|^2$ or $|U|^2$ is 6, so $v^2=b^2=1$: the gap $\mathbb{E}\,v^2\geq 1$ that the entry's paper takes from [47] holds with equality on each, and $S^2\times S^2$ has $v+b=2$, exactly on the boundary of the condition $v+b<2$. The topology comes out of the same curvature.
Euler characteristic and signature of the three model spaces from their curvature.

$$\chi=\frac{1}{32\pi^2}\int\big(|\mathrm{Rm}|^2-4|\mathrm{Ric}|^2+R^2\big)\,d\mu,\qquad \tau=\frac{1}{48\pi^2}\int\big(|W^+|^2-|W^-|^2\big)\,d\mu$$
$$\mathrm{Ric}=3g:\quad \chi=\frac{3V}{4\pi^2}\big(1+s_++s_-\big),\qquad \tau=\frac{V}{2\pi^2}\big(s_+-s_-\big),\qquad s_+=\mathbb{E}\,v^2,\quad s_-=\mathbb{E}\,b^2 \tag{26}$$
$$(\chi,\tau)=(2,0),\ (3,1),\ (4,0),\qquad V=\tfrac{8\pi^2}{3},\ 2\pi^2,\ \tfrac{16\pi^2}{9}$$

6 checks passed (`models/qg_einstein4.py`).

Figure 14 puts the two characteristic identities in the library's own form. They are the ones that tie the two second moments to the volume in that paper's argument, and they are what the three rows of s_S test: the Euler number and the signature, read as equations for the same moment, agree on all three spaces. Two numbers of the papers appear as checks of the model: the Hitchin–Thorpe quantity $2\chi-3|\tau|$ is 4, 3, 8, and the condition $2\chi-3|\tau|\leq 4$ under which a Gursky–Malchiodi-type classification is quoted in that paper holds for the first two only. The energy $D(g)=\int|\mathrm{Ric}-\frac{R}{4}g|^2\,d\mu$ of the L^2 paper is invariant under rescaling in dimension four, is zero on the three Einstein spaces, and is $64\pi^2/3$ on the unbalanced product. The existence of the universal threshold ε_0 below which the classification persists is the theorem of [26]; it has no value in that paper, and none is computed here.

What corresponds to what, and what does not. (i) The $SU(2)$ of a link is, in a Euclidean four-manifold, one of the two factors of the frame group, and the electric field of a link is a section of the adjoint of that factor, which is Λ^+ when the factor is $SU(2)_+$. (ii) The two-dimensional surface whose area the spin network measures has a bivector Σ, decomposable, with $|\Sigma_+|=|\Sigma_-|$; its area is the length of Σ and the area spectrum quantises the length of the Σ_+ part, the Casimir of Section 16 (the factor $1/\sqrt{2}$ is a normalisation that is not tracked). (iii) The weak-coupling variables of the ladder are $su(2)$-valued, three components per plaquette; on the geometric side the curvature 2-form of $SU(2)_+$ evaluated on a plane is a vector of $\Lambda^+\cong\mathbb{R}^3$. That the numbers of three components agree is the whole of the correspondence at this point, and it is an interpretation. Not claimed: that the ladder, which is a two-dimensional lattice problem, or any spin network, carries an Einstein metric; that S^4, $\mathbb{CP}^2$ or $S^2\times S^2$ is a state of any quantum theory here; anything about the Lorentzian theory, in which the group of the Ashtekar–Barbero connection is $SU(2)$ for a different reason; or any statement about the classification theorem itself, which is stated in Lean in the release (`lean/docs/348.md`) and not connected to the checks above. The other gravity entries of the release are taken up in the sections that follow (the spacetime Penrose inequalities, 260, Section 22; strong cosmic censorship near Kerr, 264, Section 23; the Liouville quantum gravity limit of random planar maps, 211, Section 24); of them the Penrose inequality is the closest to what is computed here, since it bounds a mass

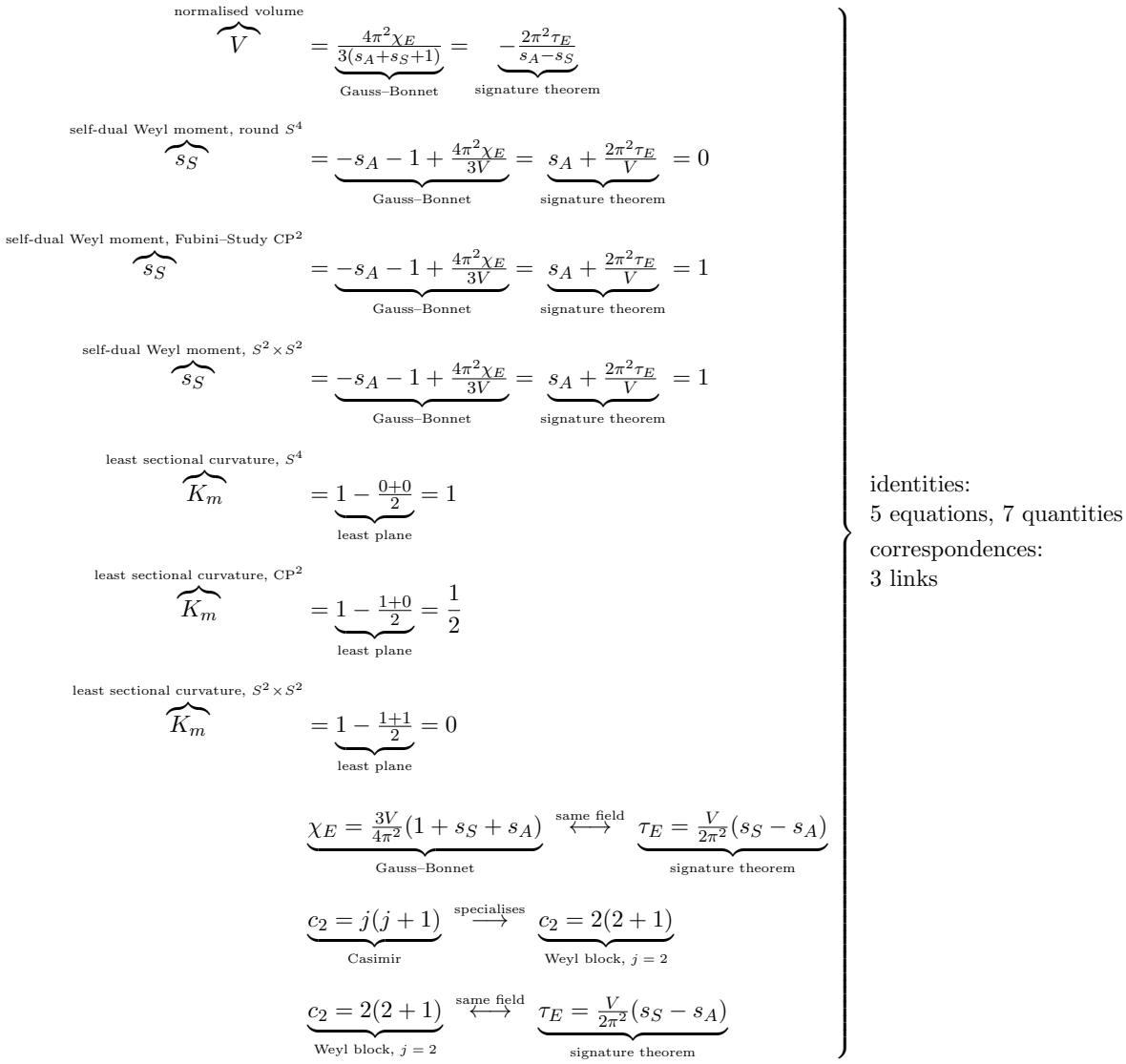


Figure 14: Einstein four-manifolds in the library, conjoined (entry 348, $\mathrm{Ric} = 3g$). The first row states the volume twice, through the Euler characteristic and through the signature, in terms of the two Weyl second moments $s_S = \mathbb{E}\, v^2$ and $s_A = \mathbb{E}\, b^2$. The next three rows state s_S both ways for the three model spaces, with χ, τ, V and s_A as computed from their metrics in Section 21: the two terms of each row are the same number, shown at the end of the row (0, 1, 1), which is the check, since the equations alone would let any s_S through. The last three rows give the least sectional curvature $1 - (x + y)/2$ of the same spaces, which is 0 for $S^2 \times S^2$, the planes of zero curvature that strict positivity excludes. The arrows are links of the library: χ and τ share the same two moments, and the Weyl block of one chirality is the spin-2 representation, whose Casimir is the $j(j+1)$ of a spin-network link at $j = 2$.

from below by an area, and the area has a spectrum in Section 16, but combining a classical inequality with a quantum area eigenvalue is a heuristic that this part does not make.

22 The Penrose inequality and the smallest area

Entry 260 proves that the invariant ADM mass m of asymptotically flat initial data satisfying the dominant energy condition is at least $\sqrt{A_{\min}/16\pi}$, where $A_{\min}$ is the minimum area of a surface enclosing a weakly trapped boundary, in three spatial dimensions and with $G = c = 1$ [35]. It states charged, four-dimensional and anti-de Sitter versions, and equality for Schwarzschild data under horizon hypotheses. The inequality is the one place in the release where a mass is bounded by an area, and the area has a spectrum in Section 16. The next display restores the units, checks that the library's Schwarzschild radius and horizon area saturate the bound, and checks the equality cases of the stated variants as algebraic identities. They are consistency checks of formulas, not tests of the theorems.
The Penrose bound: saturation by Schwarzschild, Planck units and the equality cases of its variants.

$$
\begin{gathered}
m \;\geq\; \sqrt{\frac{A_{\min}}{16\pi}}\;\; (G=c=1), \qquad M \;\geq\; \frac{c^2}{G}\sqrt{\frac{A_{\min}}{16\pi}}, \qquad \frac{M}{M_P} \;\geq\; \sqrt{\frac{A_{\min}}{16\pi\,\ell_P^2}} \\
A = 4\pi r_s^2,\; r_s = \tfrac{2GM}{c^2} \;\Longrightarrow\; M = \tfrac{c^2}{G}\sqrt{\tfrac{A}{16\pi}} \;\text{ (Schwarzschild saturates)} \\
\text{charged: } m \geq \tfrac{1}{2}\big(r + \tfrac{Q^2}{r}\big)\;\; (r \geq Q), \quad r_+ = m + \sqrt{m^2 - Q^2} \text{ saturates}; \qquad \text{4D: } m \geq \tfrac{1}{2}\big(\tfrac{A_e}{2\pi^2}\big)^{2/3}, \;\; f = 1 - \tfrac{2m}{r^2} \text{ saturates} \\
\text{AdS } (\Lambda = -3): \;\; m \geq \sqrt{\tfrac{A}{16\pi}}\big(1 + \tfrac{A}{4\pi}\big), \;\; f = 1 + r^2 - \tfrac{2m}{r} \text{ saturates}
\end{gathered}
\tag{27}
$$

6 checks passed (`models/qg_penrose.py`).

Figure 15 puts the saturated bound in the library's form. If the area in the bound is taken to be that of N spin-$\frac{1}{2}$ punctures with $\gamma_I = \ln 2/(\pi\sqrt{3})$, the bound is $M/M_P \geq \sqrt{N \ln 2/4\pi}$, which is $\sqrt{S/4\pi k_B}$ in terms of the entropy of Section 20: for one puncture, 0.2349 Planck masses.
The Penrose bound at the area of N spin-1/2 punctures (a heuristic composition).

$$
\begin{gathered}
A_N = 4\sqrt{3}\pi\gamma_I N \ell_P^2\Big|_{\gamma_I = \ln 2/(\pi\sqrt{3})} = 4N \ln 2\, \ell_P^2, \qquad \frac{M}{M_P} \;\geq\; \sqrt{\frac{N \ln 2}{4\pi}} = \sqrt{\frac{S}{4\pi k_B}}, \qquad \frac{M_{\min}}{M_P}\Big|_{N=1} = \sqrt{\frac{\ln 2}{4\pi}} \approx 0.2349 \\
\text{general punctures: } \frac{M}{M_P} \geq \sqrt{\tfrac{\gamma_I}{2}\sum_i \sqrt{j_i(j_i+1)}}, \qquad r_s \geq \frac{\hbar}{Mc} \iff \frac{M}{M_P} \geq \frac{1}{\sqrt{2}} \iff N \geq \frac{2\pi}{\ln 2} = 9.06
\end{gathered}
\tag{28}
$$

5 checks passed (`models/qg_penrose.py`).

Why this is a heuristic. The theorem is about classical initial data, and the left-hand side of the bound is a number computed from such data; an area eigenvalue belongs to an operator on a Hilbert space of spin networks. Nothing here derives a statement about a quantum state from the theorem, or the converse, and the value of γ_I used belongs to one counting (Section 20). The numbers also show where the classical reading breaks. For one puncture the area radius is $0.47\,\ell_P$, below the Planck length, and the reduced Compton wavelength of the saturating mass, $4.3\,\ell_P$, is larger than the horizon. The crude criterion that the Schwarzschild radius of the saturating mass exceeds its Compton wavelength first holds at $N = 10$ punctures. What survives is arithmetic: an exact closed form relating the smallest quantised area, the entropy and a mass in Planck units, which a reader can set beside the theorem and decide what it means.

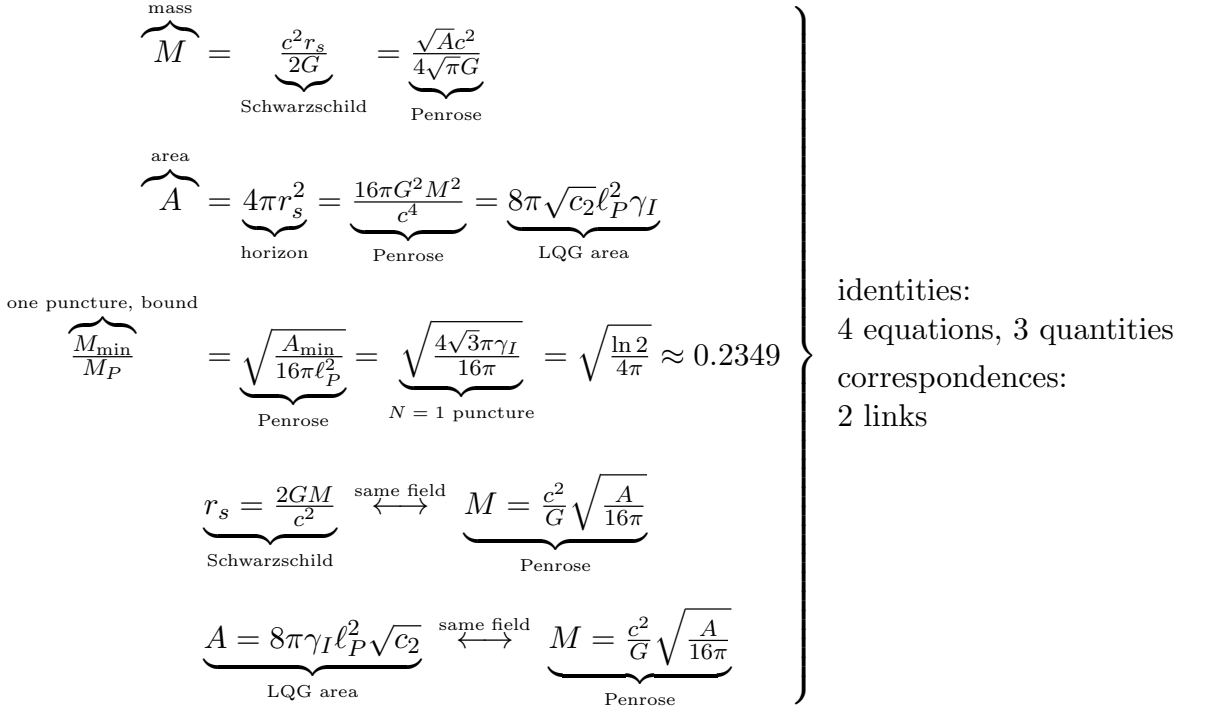


Figure 15: The Penrose inequality of entry 260 at equality, in the library. The first row states the mass twice: through the Schwarzschild radius, and as the mass that saturates the inequality for a given area. The second states the area three ways: the horizon area, the area at which the Penrose mass equals M, and the quantised area of Section 16; as in the other figures the equals sign says that each equation states the same quantity, not that the other letters share a value. The third row evaluates the bound at the area of one spin-$\frac{1}{2}$ puncture, with $\gamma_I = \ln 2/(\pi\sqrt{3})$, exactly. The arrows are library links: Schwarzschild data saturate the bound, and the area of the bound is the area that has a spectrum. Putting a quantum area eigenvalue where the theorem has a classical enclosing area is a heuristic (Section 22).

23 Strong cosmic censorship near Kerr

Entry 264 proves a local form of strong cosmic censorship near every fixed rotating subextremal Kerr bridge: in a weighted smooth topology, a dense G_δ set of nearby smooth two-ended asymptotically flat vacuum data has a maximal globally hyperbolic development with no future extension, in the classes $C^2 \subset C^1 \subset C^0 \cap W^{1,2}_{\mathrm{loc}}$ of the three companion papers (the last class, a continuous metric with a locally square-integrable weak connection, gives the strongest statement). No symmetry is imposed, and extensions need not satisfy the vacuum equations [30, 29, 33]. What the proofs use of Kerr itself is small and explicit: a reduction by scaling to $M = 1$, an inner horizon $r_- > 0$, and the absolute inner surface gravity κ, which fixes the decay of the lapse as $e^{-\kappa T}$ and the packet amplitude $A_T = e^{-\kappa T/4}$, larger than the square root of the lapse. The extension is then tested by holonomies of a family of small loops [30]. The first display computes, in exact algebra, the Kerr quantities these statements rest on; the fork read the abstract, introduction and the normalisation of the first paper and not the proofs.
Kerr horizons: radii, areas, surface gravities, Smarr formula and the first law at both horizons [30].

$$
\begin{gathered}
r_\pm = M \pm \sqrt{M^2 - a^2}, \quad r_+ r_- = a^2, \quad A_\pm = 4\pi(r_\pm^2 + a^2) = 8\pi M r_\pm, \quad \kappa_\pm = \frac{r_\pm - r_\mp}{2(r_\pm^2 + a^2)} = \pm\frac{\sqrt{M^2 - a^2}}{2M r_\pm}, \quad \Omega_\pm = \frac{a}{r_\pm^2 + a^2} \\
M = 2\Omega_\pm J + \frac{\kappa_\pm A_\pm}{4\pi}, \qquad dM = \frac{\kappa_\pm}{8\pi}\, dA_\pm + \Omega_\pm\, dJ, \qquad J = Ma \\
\frac{|\kappa_-|}{\kappa_+} = \frac{r_+}{r_-} > 1, \qquad \kappa_+ |\kappa_-| = \frac{M^2 - a^2}{4M^2 a^2}, \qquad a^2|\kappa_-| \xrightarrow{a\to 0} M, \qquad \kappa_\pm \xrightarrow{a\to M} 0
\end{gathered}
\tag{29}
$$

10 checks passed (`models/qg_kerr.py`).
The Penrose gap of Kerr, the Christodoulou-Ruffini form and the area product [35, 30].

$$
\begin{gathered}
M^2 - \frac{A_+}{16\pi} = \frac{M r_-}{2} > 0 \ \ (a \neq 0), \qquad M^2 = M_{\mathrm{irr}}^2 + \frac{J^2}{4M_{\mathrm{irr}}^2}, \quad M_{\mathrm{irr}} = \sqrt{\tfrac{A_+}{16\pi}} \\
A_+ A_- = (8\pi J)^2, \qquad A_+ - 8\pi|J| = 8\pi M(r_+ - a) \ \geq 0, \ \ = 0 \iff r_+ = r_- \ (a = M)
\end{gathered}
\tag{30}
$$

5 checks passed (`models/qg_kerr.py`).

The second display records one exact coincidence. The Penrose inequality of Section 22 is an equality for Schwarzschild and strict for every rotating Kerr, with gap $M^2 - A_+/16\pi = M r_-/2$: it vanishes exactly when the inner horizon does, which is the condition $r_- > 0$ that the near-Kerr papers need. The two theorems are different statements about different objects, and the fork can say only that they share this hypothesis. The first display also shows why the constants of the papers are tied to the fixed centre: $|\kappa_-| \sim M/a^2$ grows without bound as $a \to 0$ and both surface gravities vanish as $a \to M$.
The Kerr area bound at the area of N spin-1/2 punctures (a heuristic composition).

$$
\begin{gathered}
A_N = 4\sqrt{3}\pi\gamma_I N \ell_P^2 \ \geq \ 8\pi|J| \ (\hbar = 1) \iff N \ \geq \ \frac{2|J|}{\sqrt{3}\gamma_I} \xrightarrow{\gamma_I = \ln 2/(\pi\sqrt{3})} \frac{2\pi|J|}{\ln 2} \\
|J| = \tfrac{1}{2} : \ N \geq \frac{\pi}{\ln 2} = 4.53 \ (N \geq 5), \qquad |J| = 1 : \ N \geq \frac{2\pi}{\ln 2} = 9.06 \ (N \geq 10)
\end{gathered}
\tag{31}
$$

5 checks passed (`models/qg_kerr.py`).

What is, and is not, a connection. Two things join this entry to this part. The first is arithmetic and heuristic: composed with the area spectrum, the classical bound $A_+ \geq 8\pi|J|$

asks for at least five spin-$\frac{1}{2}$ punctures at the smallest spin $J = \hbar/2$ and ten at $J = \hbar$ (the same $2\pi/\ln 2$ as the Compton threshold of Section 22), and the Kerr mass bound $M \geq \sqrt{|J|}\, M_P$ then exceeds the Penrose bound for those punctures. It inherits every caveat of Section 22: a classical bound against an eigenvalue of an operator. It also stops at the outer horizon: the area product $A_+ A_- = (8\pi J)^2$ would make A_- a spin-network area too only if $NN' = 4\pi^2 J^2/\ln^2 2$ were an integer, which is not the case at $J = \frac{1}{2}$ (it is 20.54), and the inner horizon is not a surface of the states of Section 18 in any case. The second is structural. The proofs of entry 264 test extensions by holonomies of small loops of the Levi-Civita connection, and a spin network is built from holonomies of a connection along the links of a graph. The resemblance ends there: the connection of the proof is the Lorentzian one on the spacetime, with only L^2 coefficients in a chart; the connection of this part is an $SU(2)$ connection on a spatial slice. Nothing in the displays tests this analogy.

24 Liouville quantum gravity and the Catalan counts

Entry 211 proves that critical Fortuin–Kasteleyn planar maps converge, for $0 < q \leq 4$, to Liouville quantum gravity spheres decorated by conformal loop ensembles, and for $q > 4$ to the Brownian continuum random tree, which is the transition from surface to tree; for FK–Ising ($q = 2$, $\gamma = \sqrt{3}$) and for spanning-tree-decorated maps the graph metric and the stationary random walk converge as well, and the spectrum converges in the Ising case [27, 38, 31, 34, 36]. The parameters are linked by $q = 2 + 2\cos(\pi\gamma^2/2)$ and $\kappa' = 16/\gamma^2$ [27]. This is the entry least close to loop quantum gravity: Liouville gravity is a continuum random surface in two dimensions whose area is a multiplicative chaos measure, and the two theories share the initials and little else. The one place where the counting is the same is the one this part already uses: the Catalan numbers that give the dimension of the invariant space of N spin-$\frac{1}{2}$ legs (Section 17) and the entropy of Section 20 also count the plane trees of the tree phase and, shuffled in pairs, the tree-decorated maps. The displays check what can be recomputed.
Rooted planar maps and tree-rooted maps: brute force against Tutte and Mullin [27].

$$M_n = \frac{2\cdot 3^n (2n)!}{n!\,(n+2)!} = 2, 9, 54, 378, \ldots \;\sim\; \frac{2}{\sqrt{\pi}}\,\frac{12^n}{n^{5/2}}, \qquad T_n = \mathrm{Cat}_n \mathrm{Cat}_{n+1} = 2, 10, 70, 588, \ldots \;\sim\; \frac{4}{\pi}\,\frac{16^n}{n^3}$$
$$\mathrm{Cat}_n = \frac{1}{n+1}\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi}\, n^{3/2}} \quad \text{(plane trees, and the invariants of } 2n \text{ spin-}\tfrac{1}{2} \text{ legs)} \tag{32}$$

5 checks passed (`models/qg_planar_maps.py`).
The FK dictionary q, kappa, gamma, c and the string susceptibility [27].

$$q = 2 + 2\cos\frac{\pi\gamma^2}{2}, \quad \kappa' = \frac{16}{\gamma^2}, \quad \sqrt{q} = -2\cos\frac{4\pi}{\kappa'}, \qquad c = \frac{(3\kappa'-8)(6-\kappa')}{2\kappa'} = 25 - 6Q^2, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}$$
$$\gamma_{\mathrm{str}} = \frac{c - 1 - \sqrt{(1-c)(25-c)}}{12} = 1 - \frac{4}{\gamma^2}, \qquad \#\{\text{rooted maps}\} \sim \mu^n n^{\gamma_{\mathrm{str}} - 2} \tag{33}$$
$$q = 0, 1, 2, 3, 4: \quad \gamma^2 = 2, \tfrac{8}{3}, 3, \tfrac{10}{3}, 4; \quad c = -2, 0, \tfrac{1}{2}, \tfrac{4}{5}, 1; \quad \gamma_{\mathrm{str}} = -1, -\tfrac{1}{2}, -\tfrac{1}{3}, -\tfrac{1}{5}, 0$$

6 checks passed (`models/qg_planar_maps.py`).
Spin-1/2 invariants at level k and the FK loop weight.

$$\dim \mathrm{Inv}_k(V_{1/2}^{\otimes N}) = \#\{\text{closed walks of length } N \text{ on } A_{k+1}\} = \mathrm{Cat}_{N/2} \;\; (N \leq 2k), \qquad \# \sim \left(2\cos\frac{\pi}{k+2}\right)^N = (\sqrt{q_k})^N$$
$$q_k = 4\cos^2\frac{\pi}{k+2}; \qquad k = 1, 2, 4, \infty: \quad q_k = 1, 2, 3, 4, \qquad \tfrac{1}{N}\ln \# \;\to\; \ln 2\cos\tfrac{\pi}{k+2} = \tfrac{1}{2}\ln q_k \;\; (\ln 2 \text{ at } k = \infty) \tag{34}$$

6 checks passed (`models/qg_planar_maps.py`).

Reading the three displays. The brute-force counts reproduce Tutte's $2, 9, 54, 378$ rooted maps and Mullin's $2, 10, 70, 588$ tree-rooted maps, and Mullin's number is a shuffle of two Catalan structures, one for the tree and one for its dual. The dictionary of the second display turns the entry's parameters into central charges and exponents ($q = 0, 1, 2, 3, 4$ give $c = -2, 0, \frac{1}{2}, \frac{4}{5}, 1$), and the rule that rooted counts grow as $\mu^n n^{\gamma_{\mathrm{str}}-2}$ reproduces the exponents -3 and $-\frac{5}{2}$ found by brute force. The third display is the connection with this part. The invariants of N spin-$\frac{1}{2}$ legs are counted by closed walks on a Dynkin diagram A_{k+1} at level k of the quantum group $U_q(\mathfrak{sl}_2)$, with loop weight $2\cos(\pi/(k+2))$; setting that weight equal to $\sqrt{q}$ gives $q = 1, 2, 3$ at $k = 1, 2, 4$ (percolation, Ising, three-state Potts), and the undeformed $SU(2)$ of this part, loop weight $2 = \dim V_{1/2}$, is the critical end $q = 4$ of the range of the entry. The same Temperley–Lieb algebra acts in both, which is why the numbers agree; the fork constructs no map from a spin network to a decorated planar map, and the growth rate $\frac{1}{2}\ln q$ of the count is the growth rate of a count and not an Immirzi-type constant (the $\ln 2$ of Section 20 is the $k = \infty$ value, fixed there by a physical argument). A loop weight above 2, the $q > 4$ tree phase of the entry, has no unitary $SU(2)_k$ counterpart. Quantum-group deformations of loop quantum gravity, in which k is set by a cosmological constant, are a known direction that this part does not use and for which it cites nothing. The classical formulas used (Tutte, Mullin, the Kardar–Parisi–Zhang relation for γ_{str}) are quoted by name and checked by brute force or exact algebra where possible, not taken from a retrieved source.

25 Gauge fixing and the curvature of the link: a four-dimensional lattice result

A paper dated 30 October 2026 by Bhattacharjee, Bhattacharya, Lockwood and Romanov [43] studies the Euclidean Wilson action on $\mathbb{Z}^4$ at fixed lattice spacing. Fixing the variables on a *line field* (a set F of straight lines, one through every site, chosen so that every plaquette contains exactly one link of F) to the identity, it proves a unique infinite-volume state and an explicit transfer-matrix gap: $m \geq -\log(3\beta_W)$ for $SU(2)$ and $\beta_W < \frac{1}{3}$, plaquette blocks that reach 0.345 for $SU(2)$ and 0.76 for $SU(3)$, and for every compact simple group a range $\beta_W < 2N\kappa_G/29$ with a decay rate $m_G(\beta)$ given by the least Ricci curvature κ_G. It reduces the continuum gap to one cube condition and says that this condition, the slice ratio and the tree bounds are open. The fork read the first six pages and the appendix with the line field; it did not read the proofs, and the four-dimensional paper's own account is that these are strong-coupling results at a fixed spacing. What the fork could do is rebuild the finite statements: Table III of the four-dimensional paper (the line field), the partner counts, Table I from root-system data, and the numbers of the theorems. The next displays are those checks. They test that paper's arithmetic and combinatorics, not its theorems.

The line field of period two in four dimensions: one fixed link in every plaquette and twelve free partners.

$$\begin{gathered} \pi_x(\sigma(x)) = \sigma(x),\ \pi_x|_R \text{ a 3-cycle},\quad \pi_x(\mu) = \sigma(x + e_\mu)\ \Rightarrow\ \#\{F \cap \partial p\} = 1\ \ \forall p \\ 2(d-1) = \frac{d(d-1)}{2} \iff d = 4, \qquad \#\text{partners} = 6 \cdot 2 = 12, \qquad \#\{\text{period-2 fields}\} = 8 \end{gathered} \tag{35}$$

10 checks passed (`models/forest_gauge_4d_theorems.py`).

Partners of a free link and the Dobrushin ceiling 4/p in four dimensions.

$$p_{\text{line field}} = 12, \quad p_{\text{temporal}} = 14, \quad p_{\text{no fixing}} = 3 \cdot 6 = 18$$
$$\beta_W < \frac{4}{p}: \quad \frac{4}{12} = \frac{1}{3}, \quad \frac{4}{14} = \frac{2}{7}, \quad \frac{4}{18} = \frac{2}{9}; \qquad p = 6(d-1) \;\Rightarrow\; \frac{2}{3(d-1)} \tag{36}$$

5 checks passed (`models/forest_gauge_4d_theorems.py`).
Table I of the four-dimensional paper from root-system data, and Lichnerowicz's bound.

$$\kappa_G = \frac{h^\vee}{2\ell_\rho}, \quad \ell_\rho = \frac{\dim V_\rho\, C_2(\rho)}{\dim \mathfrak{g}}, \quad C_2(\lambda) = (\lambda, \lambda + 2\rho), \quad \beta_W < \frac{2N\kappa_G}{29}$$
$$SU(N):\ \kappa = \tfrac{N}{2},\ \tfrac{N^2}{29}; \quad SO(N):\ \tfrac{N-2}{4},\ \tfrac{N(N-2)}{58}; \quad Sp(n):\ \tfrac{n+1}{2},\ \tfrac{2n(n+1)}{29}; \quad G_2, F_4, E_6, E_7, E_8:\ \kappa = 1, \tfrac{3}{4}, 1, \tfrac{3}{4}, \tfrac{1}{4} \tag{37}$$
$$\lambda_\rho = \frac{\dim G}{N} \;\geq\; \frac{\dim G}{\dim G - 1}\,\kappa_G \quad \text{(Lichnerowicz)}, \qquad \text{equality only for } SU(2)$$

8 checks passed (`models/forest_gauge_4d_theorems.py`).
The numbers of Theorems 1-7 and Table II of the four-dimensional paper.

$$m \geq -\log(3\beta_W) \;\; (\beta_W < \tfrac{1}{3}); \qquad m \geq -\tfrac{1}{2}\log\alpha(\beta_W),\ \alpha(\tfrac{1}{3}) \leq 0.9561,\ \alpha(0.345) \leq 0.9973$$
$$m \geq -\log(1.33319\,\beta_W) \;\; (SU(3),\ \beta_W \leq \tfrac{3}{4})$$
$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \tfrac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \tfrac{29}{2}\Big)\Big) \to \log\frac{\kappa_G}{6N|\beta|}, \qquad m_G \to 0 \text{ at } \beta_W = \tfrac{2N\kappa_G}{29} \tag{38}$$
$$\tfrac{29}{28.840002} = 1.0055, \qquad SU(2):\ \tfrac{1}{12}, \tfrac{1}{8}, \tfrac{2}{9}, \tfrac{4}{29}, \tfrac{1}{3}, 0.345; \quad SU(3):\ \tfrac{3}{16}, \tfrac{9}{32}, \tfrac{9}{29}, \tfrac{3}{4}, 0.76$$

11 checks passed (`models/forest_gauge_4d_theorems.py`).

Three observations connect this to the rest of this part. None is a derivation.

The link space is the Einstein three-sphere, and the smallest spin saturates Lichnerowicz. The curvature that the four-dimensional paper uses for $SU(2)$ is the Ricci curvature of the group manifold S^3 with the bi-invariant metric $-\mathrm{Tr}(XY)$, which is $\kappa = 1$ (Einstein with constant 2 on the unit sphere, here radius $\sqrt{2}$). The electric term of the Hamiltonian of Section 18 is the Laplacian of that sphere, with eigenvalue $2j(j+1)$ in this metric. Lichnerowicz's bound $\lambda_1 \geq \frac{n}{n-1}\kappa$ is $\frac{3}{2}$ on S^3, and the spin-$\frac{1}{2}$ eigenvalue is $\frac{3}{2}$: equality, as Obata's theorem requires of the round sphere. The check above finds the same comparison strict for every other group in Table I, among them $SU(3)$ (8/3 against 12/7). The spin $\frac{1}{2}$ is also the one whose area is the smallest, $\sqrt{j(j+1)} = \sqrt{3}/2$ (Section 16), so the smallest area quantum of a link and the first electric excitation that the curvature argument of the four-dimensional paper controls are the same eigenfunction of the same Laplacian. This is a fact about S^3, not an explanation of either result.

Dimension four, twice. A field with one fixed link at every site and one in every plaquette needs the number of plaquettes through a link, $2(d-1)$, to equal the number of plaquettes at a site, $d(d-1)/2$, and the only solution is $d = 4$ (checked above). The number $d(d-1)/2$ is also the dimension of the space of two-forms, and the splitting $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$ into two copies of the adjoint of $SU(2)$, on which Section 21 rests, exists only for $d = 4$ as well. The two statements are different (one is a count of cells, the other a Hodge star), and no mechanism relates them; the coincidence is recorded because the same equation $(d-1)(d-4) = 0$ decides both.

Gauge fixing and the spin-network basis. The forest gauge solves the Gauss law by setting the holonomies on F to the identity, which leaves three free links at every site; the spin-network basis of this part solves it by coupling spins at vertices to invariants, and the ladder of Section 18 was fixed in a spanning-tree gauge to count its states. These are two ways of removing the same redundancy, and nothing is compared beyond that.

What does not follow. The gap of Section 18 is a Hamiltonian gap of a two-plaquette system in units of g^2; the gaps above are transfer-matrix gaps in lattice units at strong Euclidean coupling. The four-dimensional paper recalls that a gap uniform in lattice units forces trivial continuum two-point functions [43], so no strong-coupling bound can by itself be a statement about a continuum or about the geometry of Section 16, and none of the numbers here is compared with a number there. One printed number does not follow from the quantity that the four-dimensional paper states for it: for $SU(3)$ at $\beta_W = \frac{3}{4}$, $-\frac{1}{2}\log 0.9653 = 0.017658$ is printed as 0.0177; the model records it. The Zenodo record 10.5281/zenodo.23841532 that was proposed as a further reference could not be retrieved from this environment and is not cited.

26 Scale and blocking

The ladder has one coupling, g, and refining or coarsening the lattice flows it. Any coarse-graining of spin networks, whether block-spin or spin-foam, is a flow of couplings of this kind, so the library's account of the one-loop blocking drift is kept as context. The mass gap of a spin model and the lattice scale of a gauge theory both fall as $\exp(-\text{slope}\cdot\beta)$, but they belong to different theories. Figure 16 states the identities within each theory and draws an arrow, not an equals sign, between the two. Nothing is computed here for a gravitational flow. The equations are Spin-model transmutation slope, Spin block coupling, Gauge transmutation slope, Gauge block coupling, Lattice coupling, Euclidean correlator decay, Mass gap in physical units, Asymptotic-freedom mass scaling, Dimensional transmutation.

27 What is checked, and what is not

The dictionary holds in every case tested. The link Casimir is the squared area in units of $8\pi\gamma_I\ell_P^2$, and the ladder's electric energy is a sum of seven such squares. The area spectrum is discrete, with a smallest value $4\sqrt{3}\pi\gamma_I\ell_P^2$ and steps that fall to $\frac{1}{2}$. The invariants of a vertex are counted by Weyl's integration formula, agree with the triangle rule at trivalent and four-valent vertices, and are Catalan numbers for spin-$\frac{1}{2}$ legs. In the ground state of the ladder the surface that separates the plaquettes has a mean-square area that falls as g^{-8} at strong coupling and grows as g^{-2} at weak coupling, with exact coefficients. And counting spin-$\frac{1}{2}$ punctures against the Bekenstein–Hawking entropy gives $\gamma_I = \ln 2/(\pi\sqrt{3})$, which is a statement about that count.

Five limits apply throughout. (i) The ladder's Hamiltonian is a gauge-theory Hamiltonian, so its ground state is the ground state of that theory and not a state of quantum gravity; the geometry computed in it is the geometry that Hamiltonian induces on a fixed graph. (ii) The value 0.1274 of the Immirzi parameter comes from counting spin-$\frac{1}{2}$ punctures only; counting all spins gives a different number [41], which the fork quotes and does not rederive. (iii) Of the volume operator only the triple product on four-valent intertwiner spaces is computed, not its prefactor and not a volume operator on general vertices; the one place a four-valent lattice vertex appears (the L-shaped cluster of three plaquettes) is computed with the cluster's Hamiltonian,

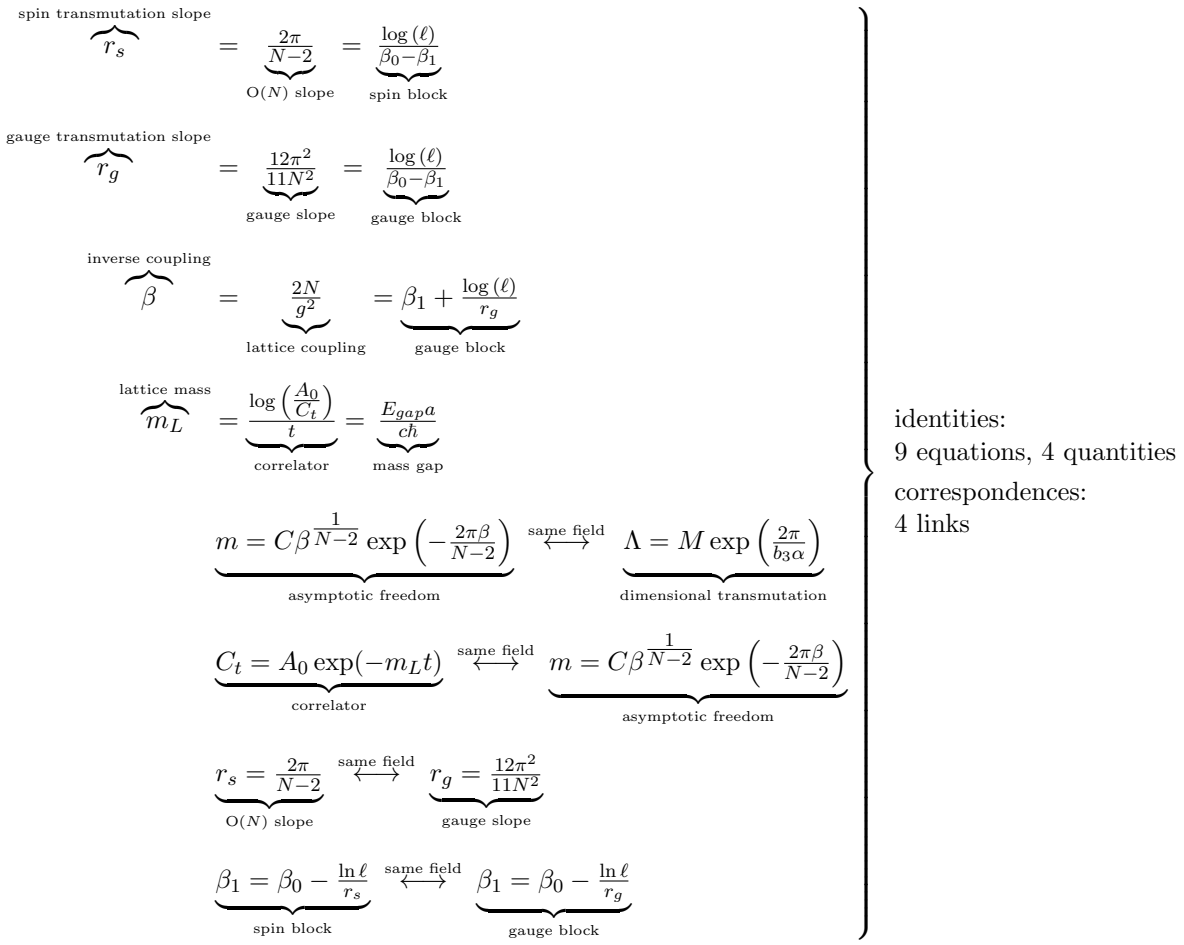


Figure 16: Dimensional transmutation across the library. The identity rows are the library's equations solved for the slopes, the coupling and the gauge-theory lattice mass. The arrows are correspondences, not identities: the $O(N)$ spin model's scaling form and the Yang–Mills lattice mass share a shape (a scale that is an exponential of an inverse coupling over a one-loop coefficient) but not coefficients, couplings or theory, so the library links them and the display does not equate them. The one exact statement across the arrow is checked: the Yang–Mills exponent $1/(2b_0g^2)$ equals $r_g\beta$, with $r_g = 12\pi^2/(11N^2)$, while the $O(N)$ exponent is $r_s\beta$ with $r_s = 2\pi/(N-2)$.

but only for the triple product, at couplings $g^2 \geq 1$. (iv) The strip is checked for $N = 1, 2, 3$ plaquettes; the strip formula for general N is a prediction, tested at three values. (v) Nothing here concerns the dynamics of gravity, diffeomorphism invariance, spin foams or a limit in which the lattice spacing disappears.

The natural next steps, in increasing distance from what is here: a volume prefactor for the triple product of Section 17, and the weak-coupling region of the L-shaped cluster; whether $\langle c_2(J) \rangle$ at an interior link of a longer strip approaches a plateau; the plaquette operators W_a and W_b, which are traces of holonomies around small loops and so of the kind a Thiemann-regularised Hamiltonian constraint is built from, composed with the volume, as the first step toward any dynamics of the geometry; and the entropy count over all spins, to see how far γ_I moves. Each is a function to add to `models/qg_spin_network.py` and, when it has an exact statement, a row to Figure 11.

Related material. The checks of this part are exact symbolic and seeded numerical computations, not machine-checked proofs. For a Lean 4 formalization of a quantum field theory (the free bosonic field in four Euclidean dimensions, against the Glimm–Jaffe axioms) see [44]. Two works of the author's collaboration are the nearest neighbours of this part, and neither is used in a computation here. The first is Part I of this paper, which computes the internal space and the boundary growth of two hierarchical tilings, and whose last section tests, with negative or generic outcomes, whether a substitution hierarchy can serve as an arrow or a clock; its discussion of gravity (Section 12) is a conjecture and a proposal and derives nothing. The second, Romanov's paper on three-dimensional lattice Yang–Mills theory in a forest gauge [50] fixes the links of a forest to the identity, as the spanning-tree gauge of the ladder does, and proves for $SU(2)$ at strong coupling and fixed lattice spacing a unique state and a transfer-matrix gap $m \geq -\log(7\beta_W/4)$ for $\beta_W < 4/7$, and a bound 9.599991 on the Hessian of the Wilson action in that gauge; by its own account some of its inputs are unrefereed. Its Euclidean couplings and transfer-matrix gaps are not those of the Hamiltonian g^2 and gaps used here, and no number of this part is compared with it; the four-dimensional paper of Section 25 extends it, and the checks of that section are those of the model `forest_gauge_4d_theorems`.

Part III
The release

28 The release behind the entries

Part II takes four entries of the OpenAI math release as subjects: Einstein four-manifolds (entry 348, Section 21), the Penrose inequality (260, Section 22), strong cosmic censorship near Kerr (264, Section 23) and the Liouville quantum gravity limit of random planar maps (211, Section 24). The fork's note `SPECTRE_CHAIR44.md` sets the Spectre and the Chair44 of Part I beside a fifth entry, 155, a tile of $\mathbb{Z}^3$ with no fully periodic tiling.

Two more groups of entries come from the fork's notes on Yang–Mills theory. Entry 215 holds the five papers on the two-dimensional $O(3)$ and $O(4)$ lattice models; the continuum-limit paper for $O(3)$ works by exact blocking, in the block renormalization tradition. Two of the fork's files tie the entry to this paper. `HIERARCHIES.md` compares those papers with the Spectre and the Chair44, as block renormalization against substitution hierarchies, and calls it the strongest

of its comparisons, one that it makes itself, since the sources say nothing about each other. And `tools/README.md` assigns to entry 215 the Monte Carlo `tools/on_model_mc.py`, the source of the library's asymptotic-freedom mass scaling in Section 26. So Part I and Part II are both tied to the same entry, by the notes and not by a computation of either part. The note `YANGMILLS.md` names two further entries whose preprints mention Yang–Mills theory, the BFSS matrix model (270) and the recovery of a metric and a connection from boundary data (365); neither is used in this paper, and the note's own account is that none of the preprints treats a gauge theory. This section draws what the release's own files say about these eight families. The graph of Figure 17 and the extra references below are read by `oai_tree.py` from `overview.tex`, `CONTENTS.md`, the `OAI:` keys in the bibliographies of the manuscripts and the Lean files; nothing is inferred from similar subject matter. The eight families hold 37 manuscripts, numbered within each family as in the figure.

Entry 155, A counterexample to periodic tiling in dimension three: (1) [25].
Entry 215, Canonical $O(3)$ continuum limit and exact $O(4)$ mass asymptotics: (1) [51]; (2) [28]; (3) [52]; (4) [53]; (5) [37].
Entry 211, The geometric phase diagram, diffusion, and spectra of random planar maps: (1) [54]; (2) [55]; (3) [27]; (4) [31]; (5) [34]; (6) [36]; (7) [38].
Entry 260, Spacetime Penrose inequalities: enclosing area, charge, rotation, and anti-de Sitter extensions: (1) [56]; (2) [57]; (3) [58]; (4) [59]; (5) [60]; (6) [61]; (7) [62]; (8) [35]; (9) [63]; (10) [64]; (11) [65]; (12) [66]; (13) [67].
Entry 264, Strong cosmic censorship near two-ended Kerr data: (1) [29]; (2) [30]; (3) [33].
Entry 348, Nonnegative-curvature Einstein classification and an L^2 topological gap: (1) [26]; (2) [32]; (3) [39].
Entry 270, Threshold and positive-energy bound states of the BFSS matrix model: (1) [68]; (2) [69].
Entry 365, Joint metric and connection recovery from one boundary patch: (1) [70]; (2) [71]; (3) [72].

An edge of the catalogue between two families is one of three kinds. A *cites* edge is a bibliography entry of one family's manuscript naming another's. A *shares* edge means that the Lean files behind the two main results import some of the same modules (modules imported by five or more families are ignored as infrastructure): evidence of related machinery, not of one result depending on the other. A *mentions* edge is a family's summary naming another result. The catalogue holds 90 edges between families, touching 82 of its 373 families, and none of them touches any of these eight. Six of the eight have a Lean scope note, which an edge of the second kind needs (the exceptions are entries 264, 270). In the release the eight are therefore separate clusters, and the connections that this paper draws between the entries are its own, or those of the fork's notes.

A Registered results of the ladder

The ladder equations on which Part II rests; each is a registered sympy result whose checks passed in the build.

The ladder: basis, Hamiltonian and plaquette operator

The seven links and two trivalent vertices of the theta graph, the Kogut–Susskind Hamiltonian in lattice units and the matrix elements of the plaquette operators, from the model

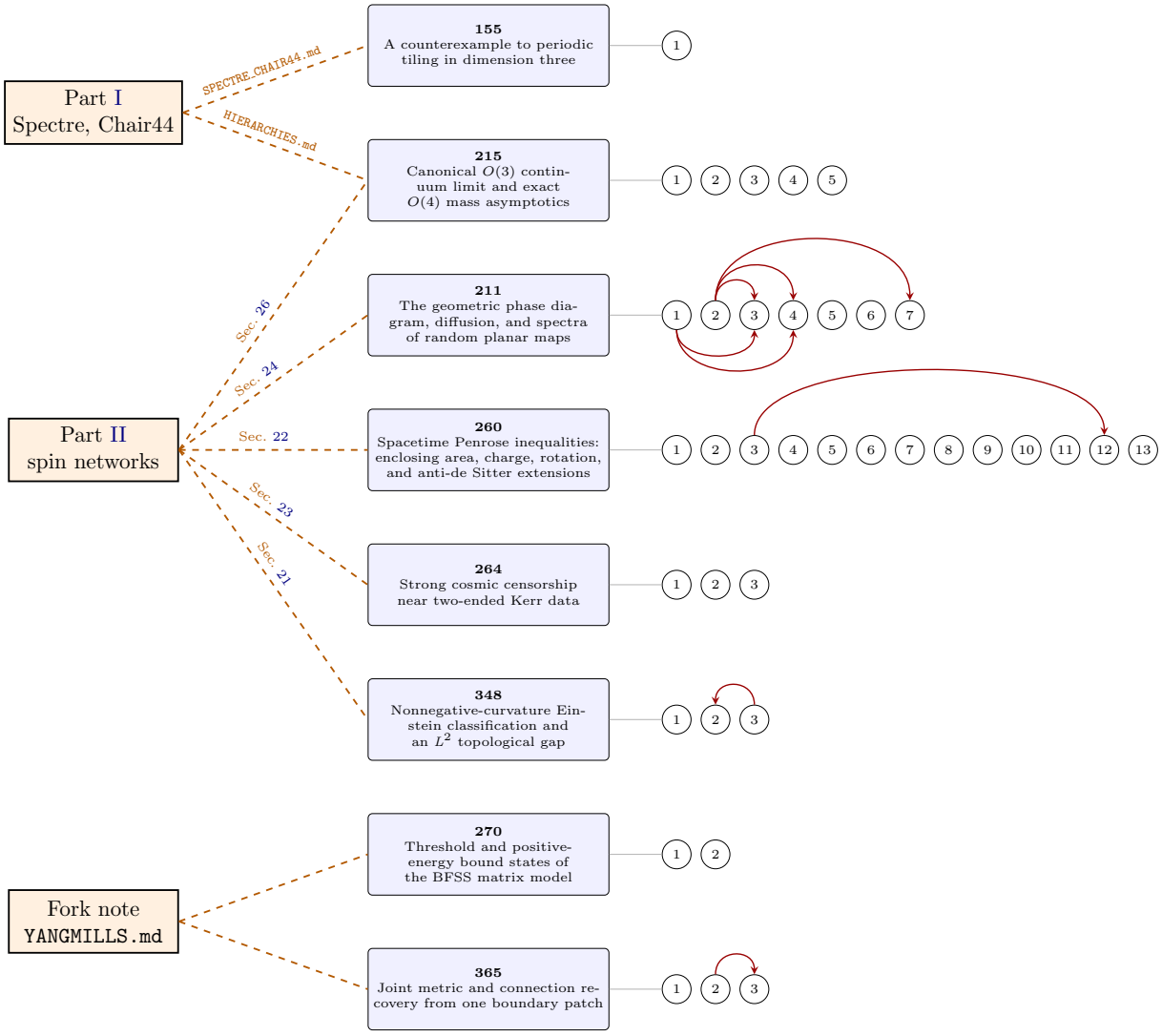


Figure 17: The eight families of the release that this paper takes up or that the fork's notes tie to it, as read by `oai_tree.py`. Each dot is a manuscript of the family on its row; a solid red arrow is a citation of one manuscript by another in its bibliography (8 in all, 8 of them between manuscripts of one family and 0 between families). The dashed lines are the places where this paper or a note of the fork meets a family, and are the only part of the figure that is not read from the release: the sections of Part II that take up an entry, the notes `SPECTRE_CHAIR44.md` and `HIERARCHIES.md` that compare entries with the tilings of Part I, and `YANGMILLS.md`, which names entries whose preprints mention Yang–Mills theory. Each is checked against the text of the note when the paper is built.

`su2_lattice_ladder`. These are the equations on which the geometry of Section 18 rests.
Two-plaquette ladder: basis, energies and plaquette operators.

$$|j_a, j_b, J\rangle, \quad \triangle(j_a, j_b, J), \quad H = \frac{g^2}{2}\big[3C(j_a) + 3C(j_b) + C(J)\big] + \frac{1}{g^2}\big(4 - W_a - W_b\big), \quad C(j) = j(j+1)$$
$$\langle j_a{+}\epsilon, j_b, J{+}\eta|W_a|j_a, j_b, J\rangle = \sqrt{d_{j_a} d_{j'_a} d_J d_{J'}} \begin{Bmatrix} j_a & \frac{1}{2} & j'_a \\ J' & j_b & J \end{Bmatrix}^2, \quad \epsilon, \eta = \pm\tfrac{1}{2} \tag{39}$$

The ladder: weak and strong coupling, and the gap

The weak-coupling limit of the spectrum from two coupled oscillators (the harmonic argument that also gives the weak-coupling geometry of Section 18), the exact strong-coupling splitting, and the gap against the one-plaquette gap. A spectral gap of a lattice gauge theory on two plaquettes: it says nothing about a gap of any field theory.
Weak-coupling limit of the ladder spectrum and the strip of N plaquettes.

$$\Delta_{g\to 0} = 2\omega_s = \sqrt{6}, \quad \omega_s = \sqrt{\tfrac{3}{2}}, \quad \omega_d = \sqrt{\tfrac{5}{2}}, \quad \text{next levels } \omega_s + \omega_d \text{ (odd)}, \quad 2\omega_d = \sqrt{10}, \quad 4\omega_s$$
$$\text{strip of } N \text{ plaquettes (prediction): } \Delta_{g\to 0} = 2\sqrt{2 - \cos\tfrac{\pi}{N+1}} \xrightarrow{N\to\infty} 2 \tag{40}$$

Strong-coupling splitting of the first excited level of the ladder.

$$\Delta_\pm = \frac{3}{2}\, g^2 + \frac{c_\pm}{g^6} + O(g^{-14}), \qquad c_+ = \frac{26}{35}, \quad c_- = \frac{14}{15} \quad (\text{one plaquette: } \tfrac{14}{15}) \tag{41}$$

The ladder gap against the one-plaquette gap.

$$\Delta_{\text{ladder}}(g) \le \Delta_{\text{plaquette}}(g), \qquad \frac{\Delta_{\text{ladder}}}{\Delta_{\text{plaquette}}} \to \frac{\sqrt{3}}{2}\ (g \to 0), \quad \to 1\ (g \to \infty) \tag{42}$$

Declarations

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The authors received no specific funding for this work, and declare no competing interests.

Code and Data Availability

All source code and data supporting this study are available at:
https://github.com/brentharts/spectre/
https://github.com/brentharts/spectre-monotile-py/
https://github.com/brentharts/six-birds-tiles/

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fork read pages 1-6 and Appendix A and rebuilt the finite statements in models/forest_gauge_4d_theorems.py; it did not read the proofs. Strong-coupling lattice results at fixed spacing; the paper lists the cube condition, the slice ratio and the tree bounds as open. *Unrefereed as far as the fork knows; the paper reports computer-assisted inputs and AI-assisted drafting.* https://github.com/creelie/4Y-M

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